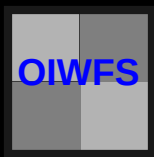


Near IR On-Instrument Wavefront Sensor (OIWFS)—Contents

Hubert Yamada

oiwfs_h ty_ TOC / 05-00 / 2001-07-02

This report contains a collection of tables of contents for the documentation for the Gemini Near Infra-Red On-Instrument Wavefront Sensor (OIWFS).



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Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRA) and Wavefront Sensor (OIWS)— Definitions, Abbreviations and References

Hubert Yamada

niri_h ty_009 / 05 / 1999-10-23

This report lists all abbreviations and references which are specific to the Gemini “Near Infra-Red Imager” and “On Instrument Wavefront Sensor.”

1 Introduction

1.1 Purpose of the document

This is a master list of all abbreviations and acronyms that are used by the software documentation for the Near Infra-Red Imager (NIRA) and On Instrument Wavefront Sensor (OIWS).

1.2 Scope

The software product to be developed from this document will be the control system for the Near Infra-Red Imager (NIRA) and the On Instrument Wavefront Sensor (OIWS).

1.3 Abbreviations and Acronyms

This list is incomplete. For a more complete listing, see the Abbreviations and Acronyms list in [16].

A&G	Acquisition and Guiding
A/D	Analog to Digital Converter
CICS	Core Instrument Control System
CC	Components Controller
DC	Detector Controller
GNAAC	Gemini NOAO Aladdin Array Controller
GNIRS	Gemini Near Infra-Red Spectrograph
HS	Hall-effect Sensor
ICD	Interface Control Document
IS	Instrument Sequencer

N/A	Not Applicable
NIRI	Near Infra-Red Imager
OCS	Observatory Control System
OIWFS	On Instrument Wave-Front Sensor
OMS	Oregon Microsystems
TBD	To Be Determined
TCS	Telescope Control System
WCS	World Coordinate System
UH IfA	University of Hawaii, Institute for Astronomy

1.3.1 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 are given like this [1].

1.4 Definitions

This list is incomplete. For a more complete listing, see the CICS definitions list in [17].

Hall-effect Sensor. An electronic component which is used to detect and to measure magnetic field strength.

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Gemini Near Infra-Red Imager and Wavefront Sensor



OIWFS — Mechanism Testing

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niri_h ty_015 / 1.00 / 2001-07-02

This document describes the procedure for verifying that the OIWFS mechanical components are properly working.

1 Introduction

1.1 Purpose

This document describes the procedure for insuring that the OIWFS mechanical components are working properly. It assumes that all wiring has already been verified for correctness, and that the default channels are used for all input and output boards. It assumes that all of the boards have been correctly configured¹.

Note that although the Digital Input / Output board is intended to permit the OIWFS to interface to the interlock bus, there is no requirement that it actually must connect to the interlock bus. Support for the interlock bus is minimal, and is provided so that the user may define instrument-specific support, as required.

The current release of the OIWFS software requires a Bancomm board to interface to the A&G system. It can, however, be tested without the Bancomm board. The only operations that will fail are those that relate to the “follow” command.

Note that the OIWFS was initially designed to work with NIRI. Since delivery of NIRI, changes have been made to the NIRI software (including the OIWFS software). Later changes have not been propagated into the NIRI developers’ software, so some changes may need to be performed by the Gemini staff in order to integrate the OIWFS with the Gemini control system.

1. Note: For GNIRS, all boards except the Digital Input/Output board were correctly configured at the time that the system was shipped. The Digital Input/Output board should be configured according to Gemini standards, as described in [30].

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Precautions

2.1 Temperature

The mechanisms should never be operated while the instrument is being cooled or warmed. This is particularly important if the mechanism is being tested in a small liquid-nitrogen cooled dewar. Severe temperature gradients may occur, which can cause the mechanisms to bind. Operating under those circumstances can break the mechanism.

It is strongly recommended that a temperature sensor be placed on a part of the mechanism which is as far away from the source of heating or cooling, in order to verify that the entire mechanism has reached thermal equilibrium before operation. Note that for the focus stage, the moving part of the mechanism has very poor thermal contact with the rest of the mechanism. Also, the motor shaft has very poor thermal contact with the motor housing.

2.2 Motor Heating

The mechanisms are not intended to be operated continuously for extended periods of time (i.e., longer than five minutes.) During prolonged, continuous operation, the motors can become hot. This should be avoided. It is recommended that any tests which require continuous motion of the mechanisms should be performed intermittently, with adequate time to permit the motor to cool.

2.3 Wear

The mechanisms are not intended to be operated continuously. Avoid day-long tests, as this will reduce the lifetime of the mechanisms.

3 Procedure for Testing Mechanisms

The procedure described here assumes that the OIWFS control software (at least the Engineering subsystem) has already been configured and installed, as described in the release

notes. It assumes that the Engineering subsystem has been verified to work correctly in simulation mode.

3.1 Preparation

1. Source your local configuration file
2. Run niriEngStart. This will bring up the top level engineering screen.
3. Press the Interlocks button. This will bring up the interlocks screen.
4. Press the “WARM” button, to set the system temperature. (For cold tests, the “COLD” button should be used instead.) Wait for one minute. You must set the system temperature each time that the IOC is rebooted. You must set the temperature each time that the system completes warming or cooling. It is strongly recommended that the system temperature be set to “CHANGING” while the system is being warmed or cooled; this will set an interlock that will protect the mechanisms by keeping them from moving.
5. Close the Interlocks screen.
6. Return to the top level engineering screen.
7. Press the “Wavefront Sensor CC” button. This will bring up the top-level Wavefront Sensor Components screen.

4 Filter Wheel

1. From the top-level Wavefront Sensor Components screen, press the “Filter wheel” button. This will bring up the filter wheel engineering screen.
2. Identify the motor driver module which will be used to drive the filter wheel.
3. Enter a number in the new position (“New Pos”) field that is 40000 more than the current position (“Cur Pos”).
4. You should see all of the following things happen. If any of these does not, then there is a problem with the mechanism.
5. The LED on the motor driver module should turn yellow.
If it is off, then the motor is not enabled
If it is green, then the motor is enabled, but not receiving pulses
If it is red, then the driver is in a fault state. Turn the motor driver box off for 20 seconds, then turn it back on again. If the LED again turns red, then there is a problem with the electronics, motor, or wiring.
6. The motor should immediately start moving.
7. The value of the sensors marked 1P and 1B should change slightly, but should be less than 0.100 V most of the time. Occasionally, the sensor values will rise up or drop down to a much larger value, and then immediately return back to a low value. It will take constant inspection of the meters to see this happen.
If all of these happen, then the motor is turning, the system is working at a minimal level, but is not necessarily performing to specification.
8. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.

9. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Twenty or thirty cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

4.1 Focus Stage

1. From the top-level Wavefront Sensor Components screen, press the “Focus stage” button. This will bring up the filter wheel engineering screen.
2. Locate the motor which is to drive the focus stage
3. By hand, with motor off, turn the main gear of the mechanism near the center of travel
4. Enter a number in the new position (“New Pos”) field that is 500 more than the current position (“Cur Pos”).
5. Verify that the motor turns and then stops.
6. Verify that the hall-effect sensor meters (labelled 2P and 2B) move.
7. Press the Datum/Diag button.
8. Watch and listen to the mechanism very carefully. If you can see or hear the mechanism hit a limit, stop it immediately. If this happens, there is a serious problem with the mechanism or its sensors. Low-level calibration may be required.
9. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.
10. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Twenty or thirty cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

4.2 Gimbal

Only the x-axis will be described here. The y-axis is tested identically.

1. From the top-level Wavefront Sensor Components screen, press the “Gimbal X” button. This will bring up the gimbal x-axis engineering screen.
2. Locate the motor which is to drive the filter wheel
3. Move the mechanism away from the limits of travel. (May require mechanical adjustment).
4. Enter a number in the new position (“New Pos”) field that is 70000 more than the current position (“Cur Pos”).
5. Verify that the motor turns and then stops.
6. Verify that the mirror support moves. Note that because of the high gear ratio, this motion is very difficult to see.
7. Verify that the two right hall-effect sensor meters (marked 2P and 2B) move.
8. Verify that the two left hall-effect sensor meters (marked 1P and 1B) move. Note that these meters will normally have a value that is close to 0. They will move off-zero very briefly. It takes careful observation to see this motion. If no motion is seen, repeat from step 4, with a value that is 70000 less than the starting position. If motion is still not seen, then the hall sensors are not being read properly or the entire gear train is not turning.

9. Press the Datum/Diag button. This will find the home position for the mechanism and immediately perform a quick mechanism check. Check the Status and Error/Info boxes. If all goes well, the system should return to the Idle state. There should be no messages in the Error/Info box at this time. If there are any messages, then the system has failed the test. See the calibration procedure.
10. If all goes well, press the “cycle” button. This will perform a continuous mechanism test. Because of the very large range of travel of this mechanism, two or three cycles should be sufficient to verify correct operation. End the cycle test by pressing the stop button.

**Gemini
Near
Infra-Red
Imager
and
Wavefront
Sensor**



Summary of the GNIRS OIWFS Mechanism Testing

Hubert Yamada

niri_h ty_016 / 1.00 / 2001-07-02

This document describes the tests which were performed to insure that the GNIRS OIWFS mechanisms were working correctly.

1 Introduction

The GNIRS OIWFS mechanisms were built using the same designs as the NIRI OIWFS mechanisms. The NIRI mechanisms had received extensive testing, both in the lab and on the telescope, so it was felt that the design was sound. For this reason, it was deemed unnecessary to perform extensive and time-consuming mechanism tests. It appeared sufficient to perform a basic test of all mechanism functions, with particular attention being paid to those elements which had proven to be problematic with the NIRI OIWFS.

1.1 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.2 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.3 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Filter Wheel

The filter wheel built from a design that has proven to be very reliable. For this reason, it was felt that it was acceptable to test the mechanism at room temperature, without cooling it to its operating temperature.

Testing was done using the actual interface boards and stepper-motor driver modules that will be used in GNIRS. Custom interface cables were used in place of the cables which will be used in the dewar; the other cables, which will be external to the dewar, were the cables which are to be used with GNIRS. The mechanism was configured as it was expected to be used, including the Hall-effect sensors.

The standard OIWFS datum / diagnose function was then used, with the NIRI calibration constants. It was expected that the datum function would work correctly, but that the diagnose function would fail until the system had been calibrated. It was found that the direction of rotation of the motor was the reverse of NIRI. It is believed that the NIRI motors were inconsistently wired. Since all of the GNIRS motors were wired identically, it was felt that it was preferable not to reverse the direction of motor motion to match NIRI. Instead, all calibration constants were reversed.

With the corrected calibration constants, the filter wheel was able to datum properly, but failed the diagnosis phase. This was expected, as the normal calibration constants are sensitive, and need to be regenerated for each mechanism. The low-level calibration constants are well-defined by the mechanism. They were not changed, and were not expected to need to be changed.

The standard mechanism calibration procedure was then carried out, and the new constants were verified to be reasonable, and then were loaded into the database. The datum / diagnose function was again performed, and was completed with no problems. The datum / diagnose function was then executed a few times, with no failures detected.

The mechanism was then moved to each filter position in sequence. The mechanism reached each position and the Hall-effect sensor reading confirmed that the mechanism was in the correct position.

Finally, the OIWFS cycle test was started. The mechanism was run through 30 test cycles. No errors were detected during the cycle test.

3 Focus Stage

The filter wheel built from a design that has proven to be reliable. For this reason, it was felt that extensive testing was unnecessary, but that cryogenic operation needed to be verified.

Testing began with the mechanism sitting in an upright position. The mechanism was moved to the center of travel. Some slight misalignment of the mechanism were discovered and corrected at this time. The home position was found to be slightly closer to the center of travel of the mechanism than NIRI. This has the possibility of introducing slightly more inconsistency to the home position, because the mechanism is not pre-loaded at the home position. It

was felt that the home position was sufficiently distant from the central position to insure a repeatable home position.

The mechanism was centered, and moved back and forth from limit to limit several times. It was found that the mechanism moved freely, with no apparent problems. No evidence of motor stalling — a common problem with this type of mechanism, particularly near the limits of travel — was detected.

The low-level calibration constants were then adjusted to match this mechanism. This was expected to be necessary, since the constants are dependent on very fine positioning of the Hall-effect sensors. The limits of travel were adjusted, also expected, as the limits depend on the orientation of the main screw. The mechanism was datumed, and performed as expected. The normal calibration was then carried out. The datum / diagnose test was then performed several times, with no failure. Finally, a cycle test was carried out for about 30 cycles. The mechanism passed with no problems.

The mechanism was then cooled to its operating temperature. The mechanism was unable to datum itself. Further inspection showed that the home position had moved slightly as the system had cooled. This was not a problem with Niri, though it was a problem with a similar Niri mechanism. At that time, it was decided to resolve the problem by changing some of the low-level calibration constants, depending on the temperature. This was found to be necessary with the Niri science-channel focus stage, which has a similar design. There is no explanation of why this is necessary with the GNIRS OIWFS focus stage, but not with the Niri OIWFS focus stage. With this change to the software, the mechanism passed the datum / diagnose test several times. Finally, a cycle test was performed for about 30 cycles. No problems were discovered during the cycle test.

Finally, to verify that nothing had changed with the mechanism was warmed again. It was able to pass the datum / diagnose test and a cycle test of approximately 30 cycles.

At this time, it is possible to say that the GNIRS OIWFS mechanism passes its tests as well as the corresponding Niri mechanism did. However, optical testing is still required to insure that it is behaving as well as the Niri mechanism.

4 Gimbal Mirror

The gimbal mirror was built from a working design, so it was felt that it was unnecessary to perform the detailed testing that was required for the Niri mechanism. This mechanism is designed to very high precision, so optical testing was required.

The mirror was first calibrated while it was warm. The mechanism was found to move smoothly. The mechanism was datumed a few times, and found to return to the same home position (as determined using the Hall-effect sensors). The mechanism was then calibrated and passed a small number of cycle tests. Slight mechanical adjustments were necessary in order to provide the full range of travel. The calibration and cycle tests were then repeated.

The system was then analyzed using an alignment telescope. The image of a cross-hair was projected onto the gimbal mirror, and the reflected image was examined as the gimbal was tipped. The reticle was able to be read to an accuracy of about 0.1 reticle unit, which corresponded to a mirror move of approximately 150 steps. (The exact scale was different for both axes, because of geometric factors.)

The mirror was offset by 100000 steps and then returned to its home position. This was repeated 20 times. There appeared to be a slight drift in the position, that was approximately 0.1 or 0.2 reticle units.

The mechanism was tested at cryogenic temperatures. One axis failed to move at all at cryogenic temperatures. The other axis failed also after further cold testing. The problem was eventually isolated, and repaired.

The previous calibration and test procedure was repeated, but the slight drift in position was not observed.

The mechanism was then cooled. The mirror was offset by 100000 steps and then returned to the home position. The cross-hair returned exactly to the zero position to the accuracy of observation.

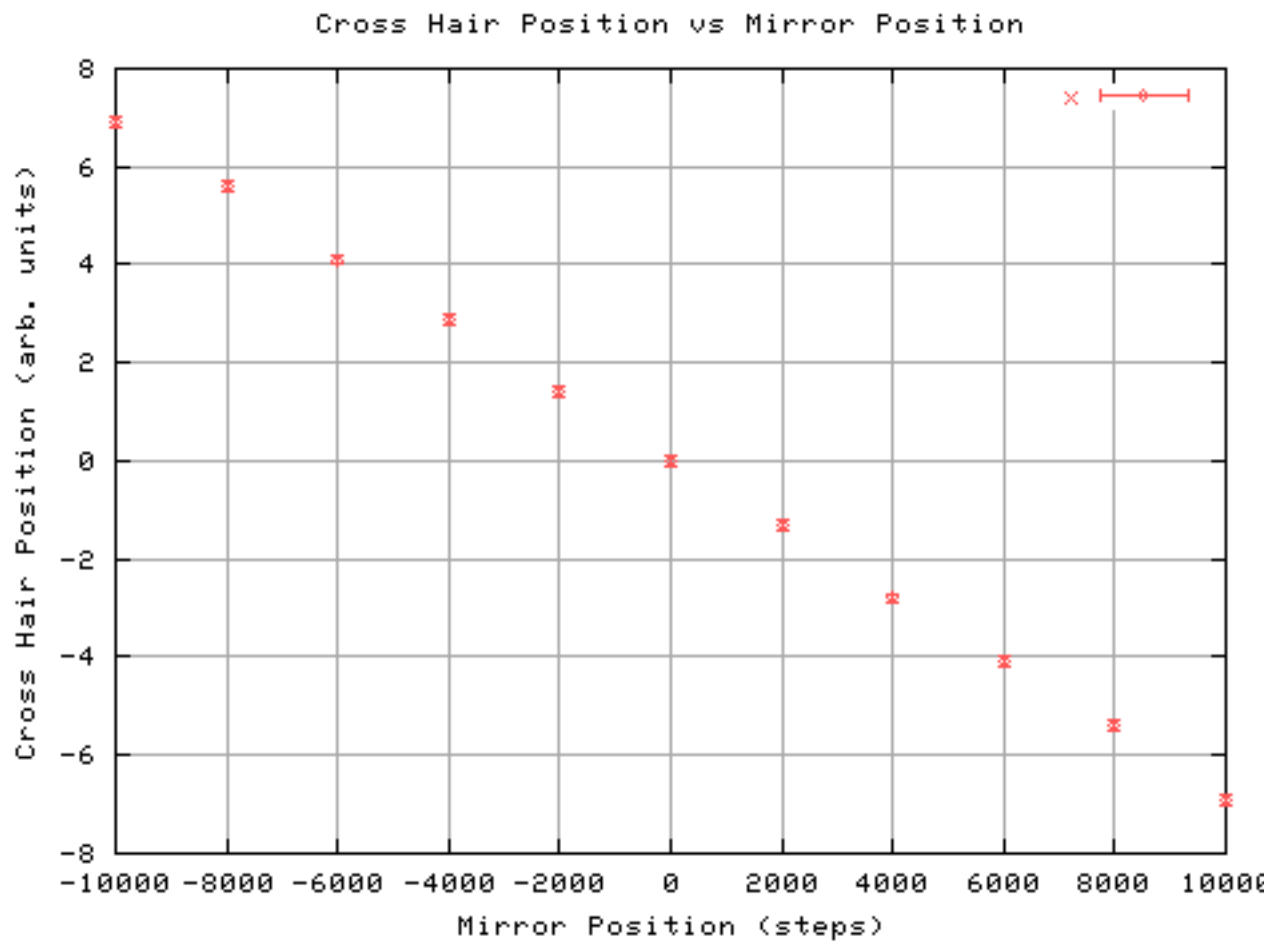
The mechanism was calibrated, and the datum / diagnose function was repeated three times for each axis. Each axis was cycle tested for five cycles.

The mechanism was then moved to a series of points, and the reticle position of the cross hair was then read. The following data was recorded. Note that no corrections were made for any slight distortions which might have been caused by the optics of the test system. The data was found to be linear, with any discrepancies easily explained by the limitations of the test setup.

TABLE 1.

x Axis Cross Hair Position as a Function of Mirror Position

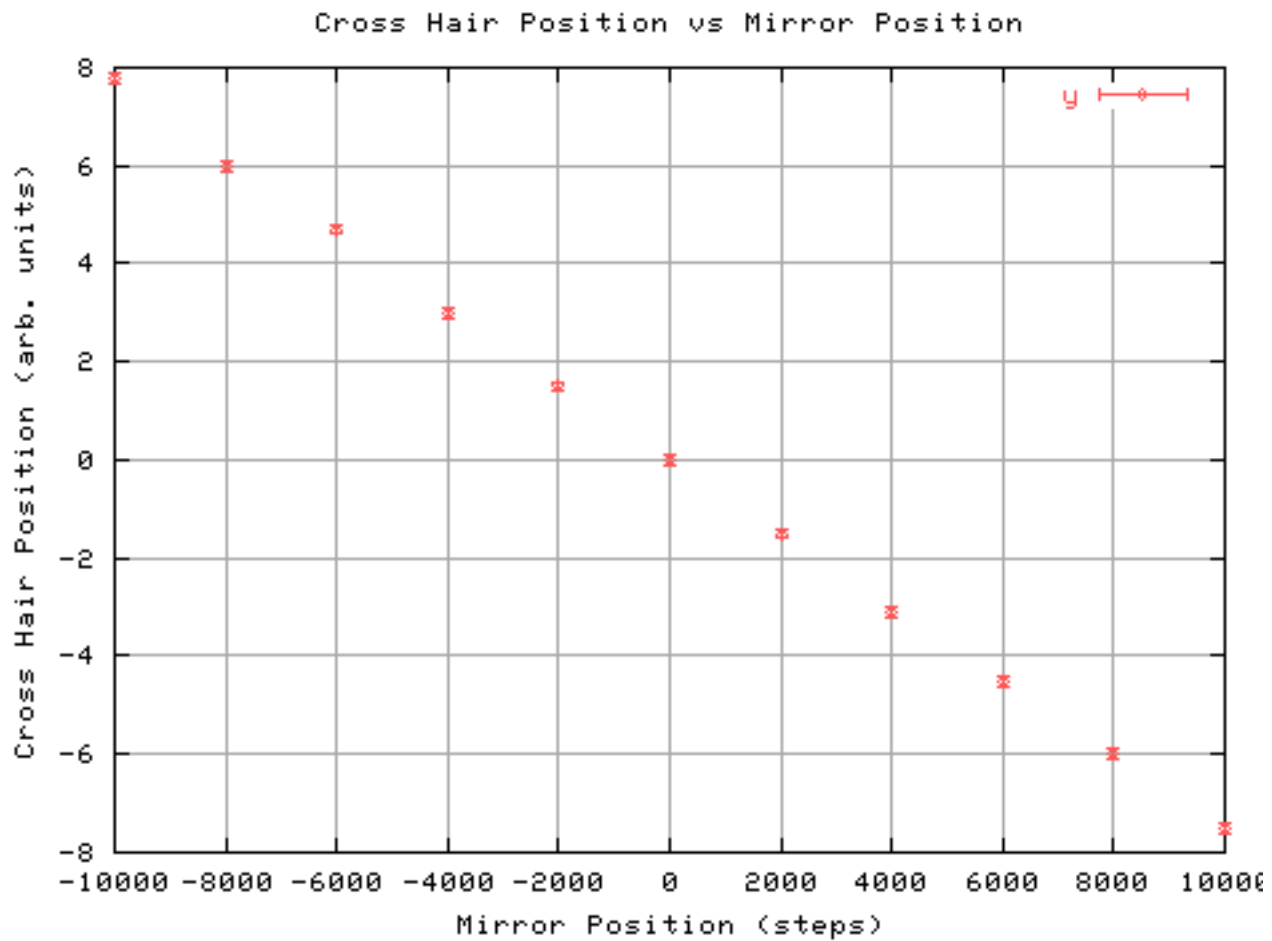
Mirror position (ministeps)	Cross hair position (Reticle units)	Cross hair uncertainty (Reticle units)
-10000	6.9	0.1
-8000	5.6	0.1
-6000	4.1	0.1
-4000	2.9	0.1
-2000	1.4	0.1
0	0.0	0.1
2000	-1.3	0.1
4000	-2.8	0.1
6000	-4.1	0.1
8000	-5.4	0.1
10000	-6.9	0.1

FIGURE 1. x Axis Cross Hair Position vs Function of Mirror Position**TABLE 2.** y Axis Cross Hair Position as a Function of Mirror Position

Mirror position (ministeps)	Cross hair position (Reticle units)	Cross hair uncertainty (Reticle units)
-10000	7.8	0.1
-8000	6.0	0.1
-6000	4.7	0.1
-4000	3.0	0.1
-2000	1.5	0.1
0	0.0	0.1
2000	-1.5	0.1
4000	-3.1	0.1
6000	-4.5	0.1

Mirror position (ministeps)	Cross hair position (Reticle units)	Cross hair uncertainty (Reticle units)
8000	-6.0	0.1
10000	-7.5	0.1

FIGURE 2. y Axis Cross Hair Position as a Function of Mirror Position



Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Software Installation and Release Notes

Hubert Yamada and Steven Beard

niri_h ty_006 / 1.00-00 / 2001-06-27

This document describes the procedure for building and installing the NIRI/OIWFS software, the NIRI engineering software, and the Near Infra-Red OIWFS software, release 1.00.

1 Introduction

1.1 Purpose

This document describes the procedure for building and installing the NIRI CC and OIWFS software. It also describes version-specific features of this release.

This package provides all of the software that is necessary to control NIRI or the OIWFS according to gemini specifications. The higher level CAD/CAR/APPLY interface is provided by niri_1.00 which is based on the CICS. The CICS and the higher-level interface software was written by Steven Beard.

The low-level engineering package is part of the software distribution. It provides the functions that are necessary to operate the NIRI and OIWFS mechanisms. It may also be used as a stand-alone package which provides all the tools that are necessary to interface to the NIRI CC and OIWFS hardware, to verify that the hardware is working correctly, and to calibrate the hardware. When used as a standalone package, it does not satisfy all of the gemini interface requirements.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the instruments which will be using the NIRI OIWFS.

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Release Notes

This is NIRI software 1.00, which is intended to be the final software release of the University of Hawaii software.

The following features are missing or broken and are expected to be fixed by the Gemini telescope developers.

- Some hardware subsystems (window cover, air ionizer) are not supported;
- The World Coordinate System is not supported;
- Communications with the detector controller need further work;
- Health problems cannot all be traced to alarms;

2.1 The 1.00 release (2000-06-27)

Currently under development.

- Feature Freeze;
- WCS code now added (though WCS support still not present);
- A&G Interface now conforms fully to the ICD’s;
- Some of the A&G interface and time-handling software was changed, to conform to the A&G system, as implemented;
- The documentation has been finalized.

2.2 The 1.00 alpha 6 release (1999-11-11)

The alpha 6 release includes the following changes:

- Fixed many small problems with the Engineering User interface;
- Fixed some problems with the OMS driver which occurred when the requested step count was too large;
- Added start-up scripts for OIWFS only mode;
- The “INVALID” position warning and the move-verification errors are now triggered by the same tolerances;
- The A&G interface now conforms better to the ICD’s;

- The directory structure has been significantly rearranged, to be more consistent;
 - The CICS_{CC} directory is renamed to CC;
 - The CICS_{WFS} directory has been renamed to WFS;
 - The gnaac-interface directory is now a complete EPICS-application tree (the DC directory);
 - The sequencer functions have been moved into an IS directory;
 - The GNAAC simulator directory has been removed;
- The startup scripts have been rewritten to be easier to modify;
- WfsBeam is now implemented;
- The directory structure and configuration files are now finalized.

2.3 The 1.00 alpha 5 release (1999-10-23)

The alpha 5 release includes the following changes:

- Restructured CAD files to make the logic clearer;
- The A&G interface is now functional.

2.4 Release V003, Releases 1.00 alpha 1 - 1.00 alpha 4

These were not publicly released. They involved extensive restructuring of the CICS-derived code to make it easier to modify. They also included changes to support communications with the detector controller; better support for the Hall effect sensors; temperature control; interlocks; temperature-dependent records; support for the full range of NIRI mechanisms; and many other smaller changes.

2.5 Release V002

Mostly bug fixes and changes in the way that configuration is handled. There is now a sequencer control screen.

2.6 Release V001

This is the initial release and was not widely distributed.

3 Hardware configuration

3.1 VME Boards

All boards should be set as specified in [30], except as described below.

The address of the Oregon Microsystems boards must be set to 0xFC80 (first board) and 0xFC90 (second board). The address of the XYCOM 490 board must be set to 0xE000. (See your Oregon Microsystems and Xycom documentation). Interrupt vector 0xE0 must be available for use by the Xycom board.

3.2 Omega Temperature controllers

The Omega temperature controllers should be set to 1200 baud, using the front panel commands.

3.3 MV167 Real time clock

Unfortunately, some of the EPICS device support (e.g., the stepper motor record) assumes a 60 Hz clock, which can cause occasional, intermittent problems; since the NIRI/OIWFS software no longer requires the stepper motor record, it may no longer be necessary to set the real time clock, but it is recommended as a precaution

The following instructions will fix the problem. Warning: These instructions are only used for the MV167, which is the gemini standard CPU for instrument systems. *Do not follow these instructions on any other system!* At the VxWorks prompt:

```
m 0xffffc00fe
```

You will see 0xffffc00fe:, followed by a 4 digit hexadecimal number, followed by a dash. Record it for later reference. Enter:

```
21ee
```

You will see 0xffffc0100, followed by a four digit hexadecimal number, followed by a dash. Enter a period:

```
.
```

To undo these changes, repeat the procedure, except that you must enter the number that you recorded earlier, instead of 21ee.

4 Prerequisites

This software was developed under Solaris 2.5.1 and epics3.12.2GEM5 / VxWorks 5.2. I do not guarantee that it will work correctly with any other operating system release or EPICS release. At this time no customization to the operating system is necessary. Minor changes to EPICS may be necessary, as described below.

In order to run the CICS/NIRI interface to the GNAAC detector controller, a current version of the GNAAC detector controller software must have already been installed and running on a separate IOC. (The GNAAC simulator, is no longer included with the software, because it is very out of date.)

In order to build and to run the CICS/NIRI software, the Gemini slalib, Gemini astlib, and Gemini timelib are all required.¹

Also required:

- gunzip.

1. These packages are available via ftp from <ftp://ftp.gemini.edu/pub/epics/gemLibs.tar.gz>

4.1 VxWorks routing bug

There is a bug in VxWorks that sometimes causes communications problems between the instrument sequencer and the detector controller software (running on a separate CPU).

Execute a `routeShow` command, from the VxWorks prompt. This will show your routing table. Typically, you will see two entries

destination	gateway
0.0.0.0	xxx.xxx.xxx.rrr ^a
xxx.xxx.xxx.0 ^b	xxx.xxx.xxx.yyy ^c

- a. The address of your router.
- b. Your local subnetwork address.
- c. Your host address.

On some systems, you may see the following:

destination	gateway
0.0.0.0	xxx.xxx.xxx.rrr ^a
xxx.xxx.xxx.0 ^b	xxx.xxx.xxx.yyy ^c
xxx.xxx.xxx.0	xxx.xxx.xxx.rrr

- a. The address of your router.
- b. Your local subnetwork address.
- c. Your host address.

If you see this, you should edit your `$local.vws` file, located in `$EPICS/extensions/src/uae/templates/startup`. There will normally be a line in this file of the form:

```
routeAdd "0", "xxx.xxx.xxx.rrr"
```

where `xxx.xxx.xxx.rrr` is the address of your router.

After this line, add the command:

```
routeDelete "xxx.xxx.xxx.0", "xxx.xxx.xxx.rrr"
```

where `xxx.xxx.xxx.0` is the address of your local subnetwork and `xxx.xxx.xxx.rrr` is the address of your router.

If you are unable to modify your epics distribution, you may add this command to your VxWorks startup script instead.

4.2 Bug fix for e2sr (pre-GEM5)

Important: There is a bug in the `e2sr` command, which is part of the EPICS distribution (fixed in GEM5). This bug will cause `e2sr` to crash, if a variable is set to an empty value. It is not possible to compile the software, unless the bug is patched.

If an version of EPICS preceding GEM5 is used², the following patch should be applied:

Edit `edifParser()`, in `$EPICS/extensions/src/edif/edifLib.c`, by replacing

```
while (*in != EOS) {  
  
    ...  
  
}
```

with

```
if (in != NULL) {  
    while (*in != EOS) {  
  
        ...  
  
    }  
  
}
```

5 Building, Loading, and Using the NIRI Low-Level Engineering Software

The `niriEng` package is a minimal package which is designed to run the NIRI and OIWFS hardware, and to simulate the hardware. It does not satisfy the Gemini interface requirements. It does provide all functions which are necessary to calibrate and debug the NIRI hardware.

Note: the full NIRI/CICS package, described below in “Building, Loading, and Using the CICS/NIRI/OIWFS software” on page 10, uses the `niriEng` package³ for its low-level functions. All functions which are provided by the `niriEng` package are available in the CICS/NIRI package; there are three reasons to use the `niriEng` package directly:

- It loads a *lot* faster
- It compiles a *lot* faster
- It uses less memory (probably not a real problem)

5.1 Obtaining the Software

Ftp to `gemftp.ifa.hawaii.edu`, and download

```
/pub/gemini_IFA/niri/source/niri_1.00.tar.gz
```

2. This is not to imply that releases of EPICS other than GEM5 are supported!

3. The previous release of this package was called `niriCC`

For instruments other than NIRI, there will also be a file:

```
/pub/gemini_IFA/niri/source/XXXX_1.00.gz
```

Where XXXX will be the name of the instrument, e.g., “gnirs.” Get this file in addition to the standard niri file. This file contains hardware parameters which are unique for each mechanism that is built.

5.2 Unpacking the Software

Execute the following commands:

```
mkdir niri_1.00
cd niri_1.00
cp (...) /niri_1.00.tar.gz .
gunzip < niri_1.00.tar.gz | tar -xvf -
```

For instrument XXXX (other than niri), after executing the previous commands, execute:

```
cp (...) /niri_1.00_XXXX_00.tar.gz .
gunzip < niri_1.00_XXXX_00.tar.gz | tar -xvf -
```

You only need the ENG subdirectory. The use of the remaining contents of this directory is described below, in “Building, Loading, and Using the CICS/NIRI/OIWFS software” on page 10.

5.3 Configuring a standard EPICS system

Previous releases of this software required several changes to the EPICS distribution. This package now uses a modified version of the motor record, and does not need to replace the stepper motor record.

5.4 Configuration

Create a `niriLogin` file; the file `niriLogin_SAMPLE` is provided as an example. You will need to edit this file to set the path to your installed NIRI software, and to set your prefix (or prefixes).⁴

Warning: The instrument prefix, e.g., `niri:`, must contain at most five characters, including the final colon. The status-and-alarm-database prefix, e.g., `niri:` or `niri::`, must contain at most six characters, including the final colon. The instrument name, e.g., `NIRI`, must contain at most four characters.

You should not need to change any other files. If you do need to do so, it should be reported as a bug.

The `csh/tcsh` configuration file “`niriLogin`” sets up various environment variables and your path. You must source `niriLogin` before building or running any of the `niri` software. It might be wise to source “`niriLogin`” in your `.login` or `.cshrc` file:

```
source niriLogin
```

4. `NiriLogin` assumes that you are using `csh`, `tcsh`, or some other shell that is compatible with the `csh/tcsh` syntax. If not, you will need to make an equivalent file for your shell, using `niriLogin` as a starting point.

The variables which must be defined in niriLogin are:

TABLE 1.

Variable	Default value	Purpose	Change? ^a
NIRI_ENGDIR	/gemini/niri/ENG	The path to the NIRI engineering source directory.	Always
NIRI_PREFIX	niri:	This prefix is prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon.	If needed
NIRI_NAME	NIRI	This is used to generate a description of various records. It should be four or fewer alphanumeric characters.	If needed
NIRI_TMPSCAN	“Passive”	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a scan rate, e.g., “1 second”.	If needed
NIRI_TMPVAR	0	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a record field, e.g., niri:cc:tmpvar.	If needed
NIRI_TMPHB	0	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a record field, e.g., niri:cc:heartBeat.	If needed
NIRI_TARGETS	mv167	Which EPICS-supported hardware platform ^b .	Not recommended

- a. Items which are marked “Not recommended” will have unexpected results, unless the niri software is modified accordingly. Details are outside the scope of this document.
- b. The mv167 is the only supported architecture at this time.

5.4.1 Initial Setup

In the \$NIRI_ENGDIR directory, type:

```
./niriSetup
```

You will get a few warning messages, e.g., that “.cvsignore” does not exist. These are harmless and can be ignored.

5.4.2 Environment subsystem

The environment simulator is now working. *It is no longer necessary to disable it if the appropriate supporting hardware is not available.*

5.5 Building the software

In the \$NIRI_ENGDIR directory, type

```
gmake
```

This should build the system. The software should compile without warnings, *except*

- All C modules will warn “‘rcsid’ defined but not used.”
- The SNL code compiler (*.stpp files) generates a lot of warnings about undefined variables.

5.6 Making dependency list

Switch to the source directory

```
cd $NIRI_ENGDIR/src
```

Make the dependency list

```
gmake depends
```

This will insure that all files will be remade if the record configuration files or C header files are changed.

5.7 Loading the NIRI CC and OIWFS control software or simulator

Connect to your IOC and set the start-up script to:

- (...) /bin/mv167/startupSim (simulator),
- (...) /bin/mv167/startup (control software),
- (...) /bin/mv167/startupWfsSim (OIWFS simulator), or
- (...) /bin/mv167/startupWfs (OIWFS control software).

where (...) represents the full, absolute path of the niri engineering directory. Then reboot the IOC. The IOC should reboot without any errors. At this time, you should be running the software that you selected.

5.8 Running the diagnostic software

- Make sure that NIRI_ENGDIR is defined.
- Make sure that \$NIRI_ENGDIR/bin/solaris is in your path

Type:

```
niriEng
```

This should start up the niri low-level engineering screen. Note: This screen is designed to control the whole NIRI system. If it is used to control the OIWFS system, non-OIWFS control widgets will be white. Separate screens to control the OIWFS only are not provided.

The low level engineering screens are documented in [5].

5.9 Cleaning and rebuilding

The set up and installation process generates a number of files automatically. To insure that the old files are removed before building the system with a new configuration, type:

```
gmake reset
```

Do not use

```
gmake clean
```

because this fails to remove some files which should be removed, and remove some files which should not be removed. To restore the system to its original state, type:

```
gmake resetall.
```

6 Building, Loading, and Using the CICS/NIRI/OIWFS software

The niri package is designed to run the NIRI and OIWFS hardware, and to simulate the hardware. This package is intended to satisfy all Gemini interface requirements for a science instrument. It provides all functions which are necessary to calibrate and debug the NIRI hardware.

Note: the full NIRI/CICS package uses the niriEng package (described above in “Building, Loading, and Using the NIRI Low-Level Engineering Software” on page 6) for its low-level functions.

6.1 Obtaining the Software

Ftp to gemftp.ifa.hawaii.edu, and download

```
/pub/gemini_IFA/niri/source/niri_1.00.tar.gz
```

6.2 Unpacking the Software

Execute the following commands:

```
mkdir niri_1.00
cd niri_1.00
cp (...)niri_1.00.tar.gz .
gunzip < niri_1.00.tar.gz | tar -xvf -
```

For instrument XXXX (other than niri), after executing the previous commands, execute:

```
cp (...)niri_1.00_XXXX_00.tar.gz .
gunzip < niri_1.00_XXXX.tar.gz | tar -xvf -
```

6.3 Configuring a standard EPICS system

It is no longer necessary to modify your EPICS distribution. This package now uses a modified version of the motor record, and does not need to replace the stepper motor record.

However, the address of the Oregon Microsystems board must be set to 0xFC80 and the address of the XYCOM 490 board must be set to 0xE000. (See your Oregon Microsystems and Xycom documentation). Note also that interrupt vector 0xE0 must be available for use by the Xycom board.

6.4 Configuration

Create a niriLogin file; the file niriLogin_SAMPLE is provided as an example. You will need to edit this file to set the path to your installed NIRI software, and to set your prefix (or prefixes).⁵

Warning: The instrument prefix and detector controller prefix, e.g., niri:, must contain at most five characters, including the final colon. The status-and-alarm-database prefix and detector-controller status-and-alarm-database prefix, e.g., niri: or niri::, must contain

5. NiriLogin assumes that you are using csh, tcsh, or some other shell that is compatible with the csh/tcsh syntax. If not, you will need to make an equivalent file for your shell, using niriLogin as a starting point.

at most five characters, including the final colon. The instrument name, e.g., **NIRI**, should contain at most four characters.

The variables which must be defined in niriLogin are:

TABLE 2. Variables

Variable	Default value	Purpose	Change ^a
NIRI_DIR	/gemini/niri	The path to the NIRI software source directory.	Always
NIRI_DCPKGDIR	/gemini/gnaac/gnaacSrc	The path to the NIRI detector-controller (GNAAC) source directory.	Always
NIRI_SLALIBDIR	/gemini/slalib	This is the path to the installed Gemini slalib package.	Always
NIRI_TIMELIBDIR	/gemini/timelib	This is the path to the installed Gemini timelib package.	Always
NIRI_ASTLIBDIR	/gemini/astlib	This is the path to the installed Gemini astlib package.	Always
NIRI_PREFIX	niri:	This prefix is prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon.	If needed
NIRI_DCPREFIX	niri:	When accessing records in the detector controller, this prefix will be prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon. It should not contain the dc: prefix, which is normally added to detector-controller records.	If needed
NIRI_SADPREFIX	\${NIRI_PREFIX}	This prefix is prepended to all records in the status and alarm database. It should be four or fewer alphanumeric characters, followed by one or two colons.	If needed
NIRI_DCSADPREFIX	\${NIRI_DCPREFIX}	When accessing records in the detector controller status and alarm database, this prefix will be prepended to all records. It should be four or fewer alphanumeric characters, followed by a colon. It should not contain the dc: prefix, which is normally added to detector-controller records.	If needed
NIRI_NAME	NIRI	This is used to generate a description of various records. It should be four or fewer alphanumeric characters.	If needed
NIRI_TMPSCAN	“Passive”	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, “Monitoring an external record.”, on page 10. Set this to a scan rate, e.g., “1 second”.	If needed

TABLE 2. Variables

Variable	Default value	Purpose	Change ^a
NIRI_TMPVAR	0	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, "Monitoring an external record.", on page 10. Set this to a record field, e.g., niri:cc:tmpvar.	If needed
NIRI_TMPHB	0	This is used by the temperature interlock, which is described in [6], Section 2–4.2.1, "Monitoring an external record.", on page 10. Set this to a record field, e.g., niri:cc:heartBeat.	If needed
NIRI_USEBANCOMM	0	If this is set to 1, then the Bancomm clock board is not initialized. This is useful for debugging the system when the timebus is not available ^b .	For debugging only
NIRI_USEVXTIME	1	If this is set to 0, then the internal system clock is used, instead of the timebus. This is useful for debugging the system when the timebus is not available ^c .	For debugging only
NIRI_AGPREFIX	ag:	This is the top-level prefix of the A&G system. The default value points to a simulation record, to avoid writing to the real A&G inadvertently. This may be set to aid in \${NIRI_PREFIX} debugging the system ^d .	For debugging only
NIRI_LIBDIR	\${NIRI_DIR}/LIB	The path to the NIRI shared source directory.	Not recommended
NIRI_ENGDIR	\${NIRI_DIR}/ENG	The path to the NIRI engineering source directory.	Not recommended
NIRI_CCDIR	\${NIRI_DIR}/CC	The path to the NIRI public components controller interface directory	Not recommended
NIRI_WFSDIR	\${NIRI_DIR}/WFS	The path to the NIRI public wavefront sensor components source directory	Not recommended
NIRI_WCSDIR	\${NIRI_DIR}/WCS	The path to the NIRI world coordinate system software directory.	Not recommended
NIRI_ISDIR	\${NIRI_DIR}/IS	The path to the NIRI instrument sequencer source directory	Not recommended
NIRI_DCDIR	\${NIRI_DIR}/DC	The path to the NIRI to GNAAC interface directory	Not recommended
NIRI_TARGETS	mv167	Which EPICS-supported hardware platform ^e .	Not recommended

- Items which are marked “Not recommended” will have unexpected results, unless the niri software is modified accordingly. Items which are marked “For debugging only” should be returned to the default value before running a working system. Details are outside the scope of this document.
- Note that this causes incorrect behavior, because local time is different from Gemini time.
- Note that this causes incorrect behavior, because local time is different from Gemini time.
- Note that this causes incorrect behavior, because the A&G status record will not be updated.
- The mv167 is the only supported architecture at this time.

You should not need to change any other files. If you do need to do so, it should be reported as a bug.

The csh/tcsh configuration file “niriLogin” sets up various environment variables and your path. You must source niriLogin before building or running any of the niri software. It would probably be wise to source “niriLogin” in your .login or .cshrc file:

```
source niriLogin
```

Run niriSetup to configure the software for your EPICS system, and copy in some standard EPICS files.

```
./niriSetup
```

6.5 Building the system

Execute the following command:

```
gmake
```

This should build the system. The software should compile without warnings, *except*

- All C modules will warn “‘rcsid’ defined but not used.”
- The SNL code compiler (*.stpp files) generates a lot of warnings about undefined variables.

6.6 Making dependency list

Switch to the source directory

```
cd $NIRI_DIR/src
```

Make the dependency list

```
gmake depends
```

Repeat with \$NIRI_LIBDIR/src, \$NIRI_ENGDIR/src, and \$NIRI_DCDIR/src.

This will insure that all files will be remade if the record configuration files or C header files are changed.

6.7 Loading the NIRI/CICS OIWFS/CICS control software or simulator

Connect to your IOC and set the start-up script to

- (...) /bin/mv167/startupSim (simulator),
- (...) /bin/mv167/startup (control software),
- (...) /WFS/bin/mv167/startupSim (OIWFS simulator), or
- (...) /WFS/bin/mv167/startup (OIWFS control software),

where (...) represents the full, absolute path to the niri_1.00 directory. Then reboot the IOC. The IOC should reboot without any errors. At this time, you should be running the software that you selected.

6.8 Running the user interface

The system may be started by running the command:

```
niriStart
```

This will start a top level control screen from which you can start the NIRI CC, WFS, DC, and low-level engineering screens. Note: This screen is designed to control the whole NIRI system. If it is used to control the OIWFS system, non-OIWFS control widgets will be white. Separate screens to control the OIWFS only are not provided.

6.9 Cleaning and rebuilding

The set up and installation process generates a number of files automatically. To insure that the old files are removed before building the system with a new configuration, type:

```
gmake reset
```

Do not use

```
gmake clean
```

because this fails to remove some files which should be removed, and remove some files which should not be removed. To restore the system to its original state, type:

```
gmake resetall.
```

7 Multiple copies of the software

The NIRI/OIWFS software is designed to make it easy for multiple copies of the software to be built and in use simultaneously on separate IOC's. Multiple copies may be kept in synch by using CVS. Each copy should be built in a separate directory and should have its own version of niriLogin, with the program prefix and name defined differently. In order to make it easy to have two copies simultaneously (i.e., a database which is intended for general use, and a version with experimental changes, or a simulator running simultaneously on another IOC, the following features are provided:

- All symbolic links are created by the configuration script, for ease of use with CVS.
- All configuration information is stored in niriLogin, and in files that are automatically generated by the configuration process.
- The only time that it is actually necessary to source niriLogin is before running niriSetup and the user interface scripts (niriEng, niriStart, etc.).
- It is not necessary to source niriLogin before running gmake.

8 Limitations of the simulation

The simulation is a reasonably accurate representation of the working niri system except:

- Mechanisms are idealized, i.e., all Hall-effect sensors lie on top of each other, there is no noise in the signals, there is no backlash, and the mechanisms cannot fail.
- The Hall effect sensors do not read the same values as the real mechanisms.
- The "FAST" simulation mode is not really very fast for datuming.

- The environment subsystem is a very poor simulation.

9 Unfinished tasks

This section describes tasks which remain to be done. These tasks, for the most part, are tasks that are fairly easy to do at the telescope, and very difficult to do with a standalone system.

9.1 WCS

1. Calibration constants for the WCS have not been computed.

This is easily done empirically, with far less effort than it would take to compute the values.

9.2 Networking and communications

1. The time bus has not been tested. Time functions have not been tested.

This requires equipment which is not available to the NIRI developers. The necessary subroutines are in place, and if they function as expected, this should be a simple operation.

2. The Interlock bus has not been tested.

NIRI currently does not need to generate an interlock event, nor will it need to respond to an interlock demand. The software and hardware are provided so that it will be possible to interface with the interlock bus if needed. This should be tested.

9.3 wfsBeam

1. The filter positions which block the light path to the wfs have yet to be entered in the appropriate .lut file.

This cannot be done until the final arrangements of filters is known.

10 Known bugs and limitations

This section describes problems with the NIRI software. All bugs listed here are believed to be non-critical or have a reasonable work-around. In general, bugs in the EPICS or GEM EPICS have not been fixed, if there is an adequate work-around, in the interest of portability. The problems here are generally of the following types:

- Non-fatal bugs which have proven difficult to isolate;
- Bugs which would require extensive redesign to repair;
- Bugs which would require significant time to repair, and which do not have a significant effect on the quality of the system;
- Bugs in the underlying EPICS or GEM EPICS system.

10.1 Problems with the simulation

1. The simulation starts in an invalid location.

When the system is first rebooted, its initial position is 0, and its initial Hall-effect sensor values are also 0.0. The Hall sensors will be computed correctly after the first motion command.

2. It is not possible to switch into simulation mode using the sequencer commands.

This was discussed with gemini, and it was decided that it was too difficult to implement with the existing tools. Simulation mode requires a separate startup script.

3. The GNAAC simulator is not supported.

The GNAAC simulator is very out of date and is not supported. The changes which are required to make it work with the NIRI software are probably small. The GNAAC simulator should be able to run on the same IOC as the NIRI software.

4. FAST Simulation mode is not really “FAST”

The FAST simulation mode takes a fair amount of time to complete some routine tasks (notably datuming). This could be improved with a small amount of effort, if the time and resources were made available.

5. VSM Simulation modes is not supported at all

The requirement for VSM was relaxed in discussions with Gemini. “FAST” mode is probably adequate for most purposes.

10.2 Problems with edd / dm

1. When the IOC is rebooted, some of the menus in the edd/dm user interface do not get loaded properly. The contents of the menu will be listed as “no menu.” Furthermore, some of the text / numeric displays show incorrect values (usually, obviously meaningless values).

This is a problem with edd/dm. A work-around is to close (not just iconify) the affected windows, and restart them.

2. The user interface dies unexpectedly.

This is a known bug with edd/dm. If you open a window (call it window A), and use a button on window A to open another window (call it window B), then close window A, then the user interface may crash unexpectedly at some time in the future. This is a known problem with edd/dm. There is no known work-around, other than to use caution when closing windows.

3. The user interface corrupts the color map of an X (including Openwindows) display.

This is a known bug with edd/dm. There is no known work-around. You will have to restart your windowing system.

4. The user interface screens cannot be edited with edd or displayed with dm.

All user interface tools for NIRI have been built using edd2.4 and dm2.4, which uses an incompatible file format. The following commands should be used in place of the edd and dm commands.

```
env EPICS_DISPLAY_PATH=`pwd`/O.solaris:`pwd` edd2.4
env EPICS_DISPLAY_PATH=`pwd`/O.solaris:`pwd` dm2.4
```

10.3 Stability problems

1. The system sometimes stops responding to EPICS channel-access commands (once per month).

This problem occurs randomly, and only when the cooling motors are in use. This may have been caused by a bug in the motor record, which has now been fixed.

10.4 Compiler Problems, Code, and Coding Standards

1. All C modules warn “‘rcsid’ defined but not used” during compilation.

The Gemini Programming Standards Document [22] requires “rcsid” to be defined and requires code to be compiled with a high warning level. These two requirements cannot be met simultaneously.

2. The SNL code compiler (*.stpp files) generates a lot of warnings about undefined variables.

These warnings are very difficult to suppress, because many constants are defined in C include files, but the SNL compiler cannot understand most C include files. Suppressing this error would require significant changes to header files, including system files, EPICS files, and Gemini-provided files or would require constants to be defined in more than one place, which is poor programming practice.

3. Boolean variables are not used consistently. Sometimes TRUE is 0, and sometimes it is any non-zero value. Sometimes FALSE is 1, and sometimes it is 0.

The EPICS Boolean convention (0 True, 1 False) is the reverse of the C convention (0 False, non-zero True). The EPICS convention is being used, the definitions EPICS_TRUE and EPICS_FALSE are always used. When using the C convention definitions are not always used. The engineering subsystem uses the C convention exclusively. All other systems should use the EPICS convention exclusively, except when interfacing to the Engineering subsystem, though there may be a few places where the C convention is used (these need to be corrected when found.)

10.5 Mechanical problems

1. The cooling motors do not stay synchronized.

There does not seem to be any way to fix this with the current hardware.

2. The software assumes that the number of motor-steps for a full rotation of one of the filter wheels and other wheel-like mechanisms is an integer.

Unfortunately, some of the wheels are a non-integer number of motor steps. This can cause slight positioning problems when the wheels have rotated a large number of turns from their home position. This is not believed to be a serious problem.

It may be necessary to datum the system occasionally or to use the diagnose command to return it to 0.

10.6 Datuming and diagnostic commands

1. The diagnose commands and redatum commands can be very slow for the wheels.

These commands move the wheels back to the zero point, without taking advantage of the coordinate wraparound. This could be improved, but the diagnose and redatum commands are intended for infrequent, engineering use.

2. If a mechanism is put into motion by a low-level engineering command, and if it is in the not-datumed state, then a warning message appears: “Mechanism position lost spontaneously.”

It is not trivial to get rid of this warning message, because of the way that the SNL code is written. The warning message should just be ignored.

3. There is no warning if the pupil viewer is not properly seated.

The pupil viewer is an unusual mechanism. It seats itself by overrunning its limits and forcing itself into a hard stop. Handling this case would require a significant rewrite of the control code. In general, if the mechanism is ever handled manually, it is necessary to move it from one limit to the other or to datum it. (The high-level datum command automatically does a diagnose).

4. The wfsBeam record is of limited accuracy.

It would be difficult and time-consuming to figure out which orientations of the gimbal mirror would permit light to fall on the wave-front sensor with different pick-off mirrors. This and similar effects are completely neglected. The only things which are considered are mirror positioning and filter transparency.

10.7 Calibration tools

1. The scan tool times out a lot, especially for the focussing stage. When it times out, the user interface leaves most of the controls disabled. The interface must be exited and restarted to fix the problem.

The scan tool is used very infrequently, and never by the end users. This really *should* be fixed, but is a low priority.

2. The scan tool, the position tool, and the cycle tool handle don’t handle text input very well.

The version of tcl / tk which is shipped with the GEM5 release is quite old. I do have a newer version of tcl / tk and have ported etwish to that release, but I don’t use it with the

NIRI software, because that would make my utilities incompatible with the unmodified GEM5 release.

10.8 Other Problems

1. The number of frames for the observe command should really be part of the setup command.

The frames parameter is likely to change frequently, and shouldn't really require a complete set up of the detector controller in order to change its value. There could be a separate command, just to set the frame count.

2. Using caget or caput to access elements of type "double" in the HallStep record causes a fault, which renders the record non-functional.

There is probably some EPICS requirement that is not being satisfied by the HallStep record. Oddly enough, this problem only occurs with caget and caput. Edd / dm, cau, pvload, and etwish all have no problem.

3. There is no easy way, short of rebooting, to restore all parameters that can be set using the low-level engineering interface to their initial values. Using the low-level engineering interface, it is easy to put the system into a state in which the usual commands do not function properly, even though the system may appear normal. The low-level engineering interface is capable of putting the system into a state from which rebooting will not recover and can even put the system into a state from which software recovery is not possible.

There are just too many things that can be done with the low-level engineering interface to safeguard against all possibilities. There is no way to avoid this, except by disabling the low-level engineering commands completely. The low-level engineering commands are on screens which have a distinct black title bar with bright yellow letters. Be sure that you know what you are doing, before using the low-level engineering commands!

4. The current position for the guide probe can show "INVALID" when it should show "Undefined."

This is because of a limitation on the number of entries in a lutin/lutout record. In practice, the named positions are not used for the guide probe.

11 Possible Enhancements

The following modifications are candidates for release 1.01.

1. Datuming is too slow.

The datuming algorithm is very conservative, in order to work reliably with no modification under all conditions, both at room temperature and cryogenic temperatures.

2. The temperature control system is not as well automated as it should be.

Scheduling will make it impossible to complete the front end to the temperature control system. The existing code is sufficiently usable that simplifying the task is considered a low priority.

3. There are problems with the motor record.

Janet Tvedt discovered several problems with the motor record, and provided patches to fix these problems. These patches have not been incorporated into the hmotor record (NIRI’s customized motor record), and some of them should probably be applied. Currently, the hmotor record is working fairly reliably in the mode in which NIRI requires it, so there seems to be little need to apply these changes. If the hmotor record is used in other projects, however, there may be problems.

4. Utility scripts need to be rewritten.

The utility scripts in “scripts” should be rewritten in another language. They are really too complicated to be shell scripts. In particular, there are a number of global variables which are reused everywhere. It is not a serious problem with the existing scripts, but requires careful attention to which variables are reused.

5. Health problems cannot be traced to alarms

This is a Gemini requirement which is not properly met. In fact, although the health tree is quite extensive, alarms are practically non-existent.

6. Issuing multiple movement commands to a single mechanism can cause problems. In particular, it is not possible to issue a datum command and a motion command at the same time.

There is a race condition, which can occur. It would be possible to add an interlock to keep this from happening, but for now, nothing is done. This should never occur in normal operation anyway; however, it should really be fixed.

12 Design Decisions (“It’s not a bug, it’s a feature!”)

1. Mechanisms do not datum themselves automatically.

Mechanisms that datum themselves automatically can be a real pain to work with during hardware and software testing and integration. Furthermore, the mechanisms retain knowledge of their current positions, even through a reboot (though not a power cycle), which can be a significant time saver, allowing the entire time-consuming datuming procedure to be bypassed. Since NIRI does not appear to need frequent rebooting under normal conditions, this is not a serious problem.

2. The Health state is not set to the WARNING mode if an operation fails.

The CICS sets its health to WARNING if an operation fails. This seemed to be rather unintuitive and was misleading to the people who tested the NIRI software, so it was removed. In general, any failure which can cause an operation to fail will trigger some warning, most commonly, the backlash warning or the not-datumed warning.

3. If a mechanism times out, the CAR record is not set to the ERR state.

Timeout is now handed via a warning SIR record, instead of via the CAR record. This is because certain operations can take a very long time under certain conditions. For example, datuming all mechanisms is much more time consuming than might be expected, because all mechanisms cannot be operated simultaneously.

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Architectural Design Document

Hubert Yamada

niri_hyt_001 / 05 / 1999-10-23

This report gives a architectural software design for the Gemini “Near Infra-Red Imager and On-Instrument Wavefront Sensor software.”

1 Introduction

This report gives an architectural software design for the Near Infra-Red Imager (NIRI) and provides description of its software decomposition.

1.1 Purpose of the document

This document describes the difference between the NIRI design and the template which is provided by the CICS, described in [15]. It is assumed that the reader of this document is familiar with that document.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS);
- the developers of the Gemini Acquisition and Guidance units
- the developers of the GNIRS OIWFS components controller and other groups which will be using the design of the NIRI OIWFS.

1.2 Scope

The software product to be developed from this document will be control software for the Near Infra-Red Imager (NIRI).

1.3 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.4 Stylistic Conventions

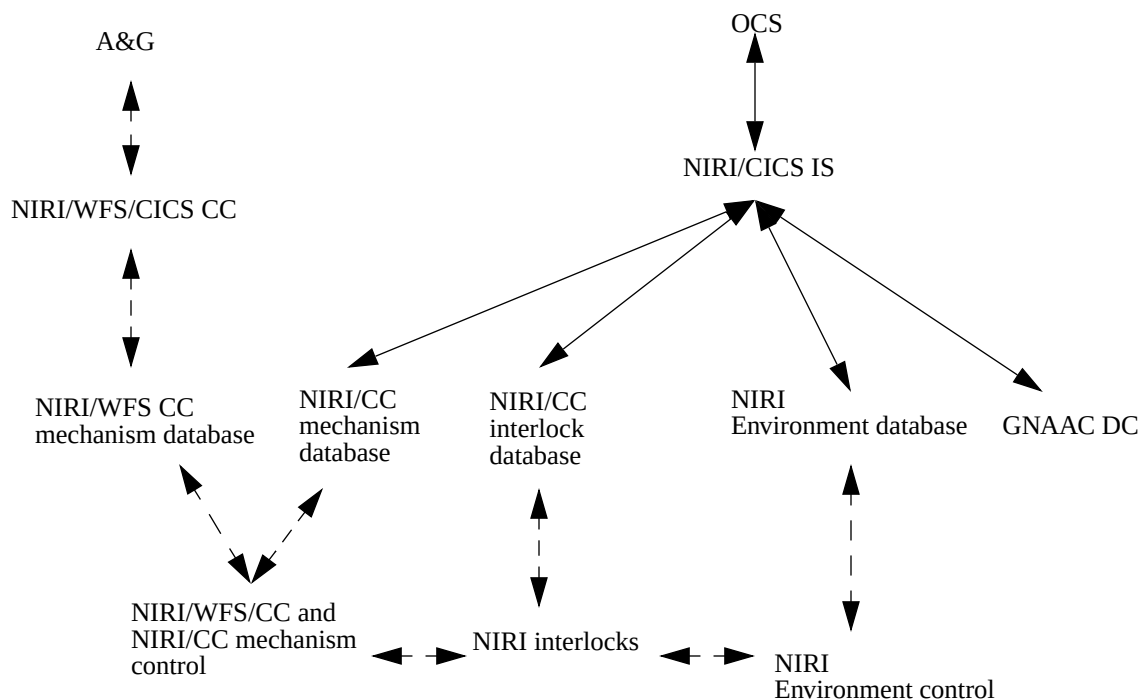
References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.5 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Implementation

At this time, the NIRI software is expected to follow the CICS architectural design very closely. The differences will be detailed in a later version of this document.



Solid lines represent commands that are linked by apply records

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Detailed Design Document

Hubert Yamada

niri_h ty_002 / 09 / 1999-10-23

This report gives a detailed software design for the Gemini “Near Infra-Red Imager” and On Instrument Wavefront Sensor.

1 Introduction

This report presents a the software design for the control software for the Gemini Near Infra-Red Imager.

This report supersedes the preliminary document, niri_h ty_000.fm, “Optical-Components Controller: Implementation Issues,” which was released as part of the NIRI PDR documents.

Note that the design of many of the hardware elements is not finalized at this time. Some control algorithms may need to be modified.

1.1 Purpose of the document

This report gives a describes the differences between the design of the software for the Near Infra-Red Imager (NIRI) and the prototype instrument which is described in the CICS documents [14]. This document includes a description of the software for both NIRI and the NIRI OIWFS¹ components controllers. At some time, the OIWFS documents will be placed into a separate document.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS);
- the developers of the Gemini Acquisition and Guidance units;

1. For purposes of this document, the terms NIRI OIWFS and OIWFS will be considered synonymous, though the NIRI OIWFS will be used by other systems.

- the developers of the GNIRS OIWFS components controller and other groups which will be using the design of the NIRI OIWFS.

It is assumed that readers of this document are familiar with the CICS prototype instrument system which is described in [14], [15], [16]. Throughout this document, unless otherwise stated, it is assumed that the software design for the NIRI control software is the same as the software design for the CICS prototype instrument

1.2 Scope

The software product to be developed from this document will be the control system for the Near Infra-Red Imager (NIRI).

1.3 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.4 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.5 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Implementation

2.1 Overview

The NIRI and NIRI OIWFS control software will be based on the Core Instrument Control System (CICS). Whenever possible, CICS commands will be duplicated. New commands and their corresponding records will be as similar to existing CICS commands as possible. If there is nothing in the CICS which can be adapted to our purposes, a new command will be designed and implemented, using the specifications in the ICD’s (especially ICD 7a and ICD 1b).

The instrument sequencer (IS) will forward sequence commands to the components controller (CC) and to the detector controller (DC); it will get the World Coordinate System (WCS) information from the TCS; it will modify the WCS information to include instrument-specific information; it will forward the modified coordinates to the DC; and it will set the TCS pointing origin.

The detector controller is a separate work package, and is described in [18].

FIGURE 1. Module Dependencies

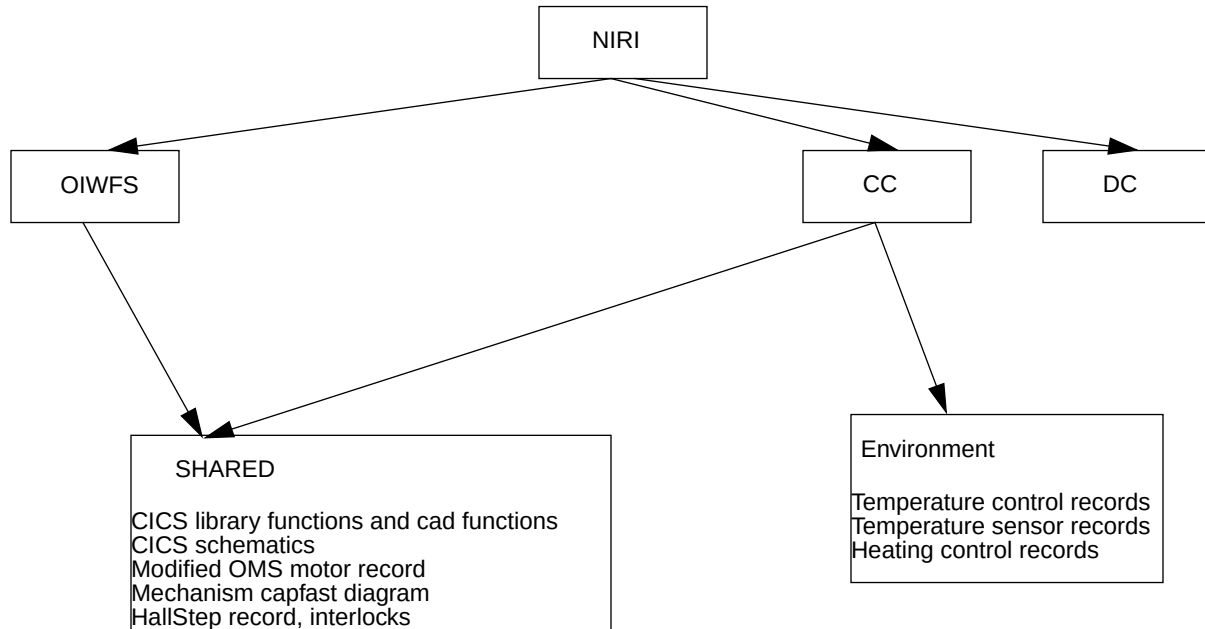
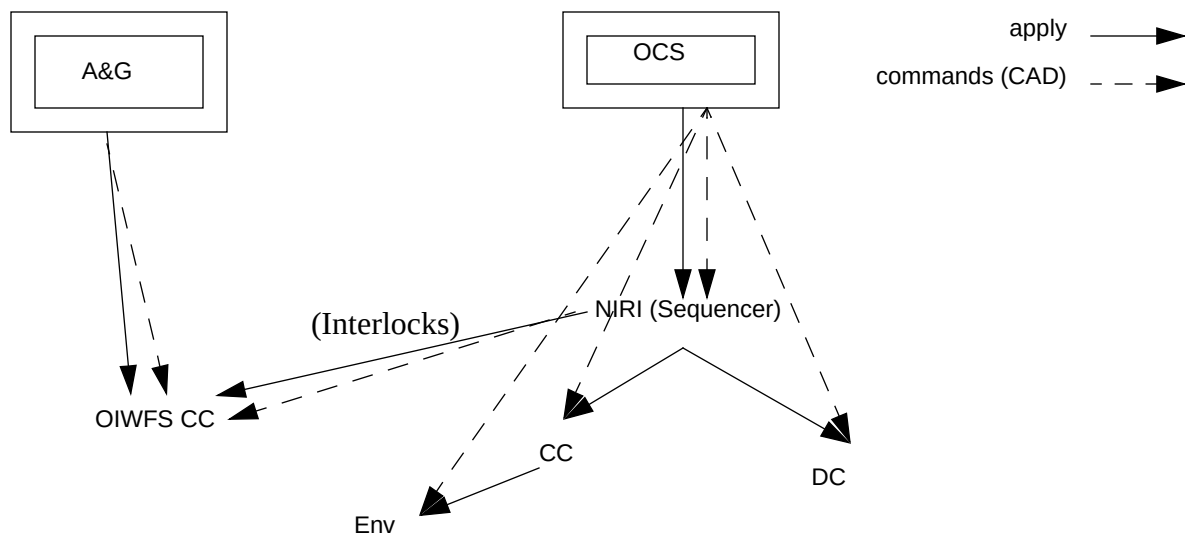


FIGURE 2. Command Flow



2.2 Static Storage

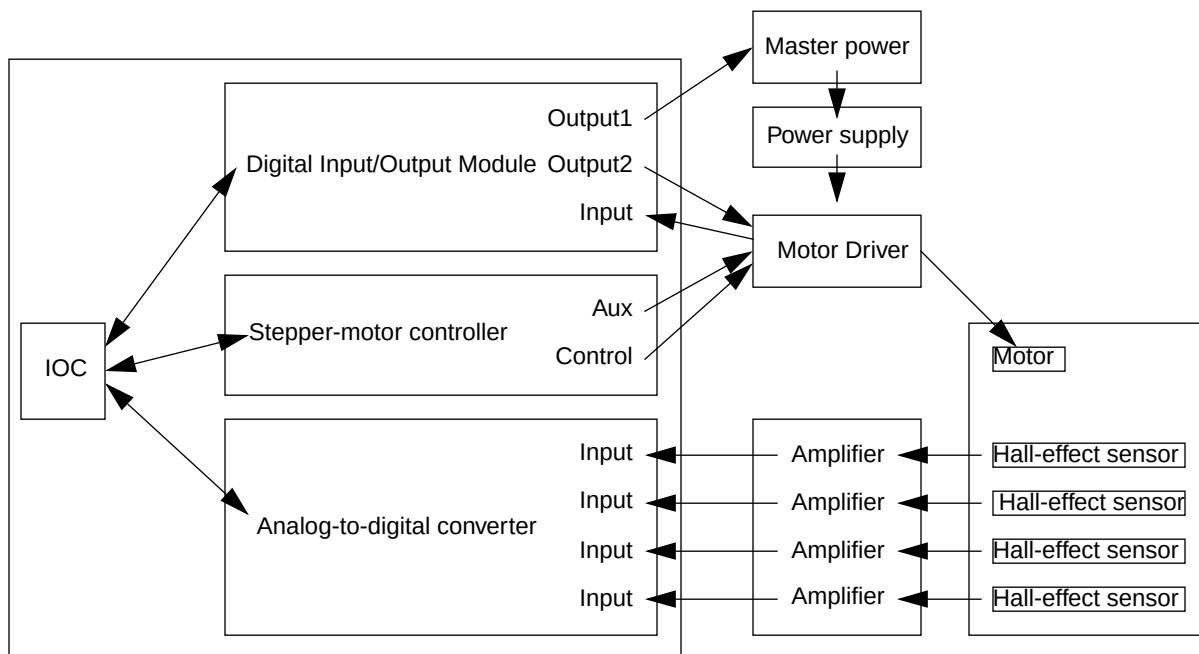
2.2.1 Static Look-Up Tables

Upon initialization of the system, calibration information will be loaded into appropriate EPICS process variables using pload. This information will be necessary to calibrate the Hall-effect sensors. The calibration data will be generated by a separate program, which will be described in [5].

3 Stepper Motor / Hall-Effect Sensor Control

All mechanisms will be controlled using a combination of stepper motors and Hall-effect sensors.

FIGURE 3. Control system overview (single mechanism)



3.0.1 Notes on Figure 3

- All inputs in Figure 3 lead in to the same amplifier and A/D boards. This will probably not be true. The inputs may be distributed among several boards, so that the system will remain partially functional if a single board fails.
- Although the master power control is shown in this single-component diagram, this will actually be shared among all mechanical components. The master power will be shut off only when all components are idle.
- The Digital Input/Output, “Output2” and “Input” are used to check the status of the stepper motor driver, and to reset it if there is a fault.

- The Stepper-motor controller Aux output will be used to shut down the power to the motor when it is not in motion.

3.1 Hall-Effect Sensors and the Home Position

The stepper motors are not encoded. Instead, each component will have a magnet at its datum (home) position. A Hall-effect sensor will be used to detect this magnet to locate the datum position. For some components there may also be other magnets giving physical limits and other reference positions. These secondary magnets may be distinguished from the home-position magnet by using the opposite polarity, and may be distinguished from each other by a different peak field strengths at the sensor².

One sensor is designated as the “primary” sensor. This sensor will be connected to the “home” input of the Oregon Microsystems stepper-motor controller, which will speed up the datuming process. If this sensor fails, then the datum position may still be found, but will require a slower search process.

3.1.1 A discrete positioning mechanism (4-position filter wheel)

Figure 4 describes a simple, 4-position filter wheel. For mechanisms with discrete positions (e.g., filter wheels) there will be reference magnets for each distinct position, as well as the magnets for the home position. For the filter wheel, the reference magnets (and their corresponding sensors) are mounted on two concentric rings. By using a combination of magnets with different field strengths at the sensor, each position may be uniquely encoded. Note that if either sensor fails, the system will continue to operate, though the encoding scheme may no longer be unique.

2. Different field strengths at the sensor may be produced either by varying the depth of the magnet or by using magnets with different strengths.

FIGURE 4.

A simple mechanism (wheel)

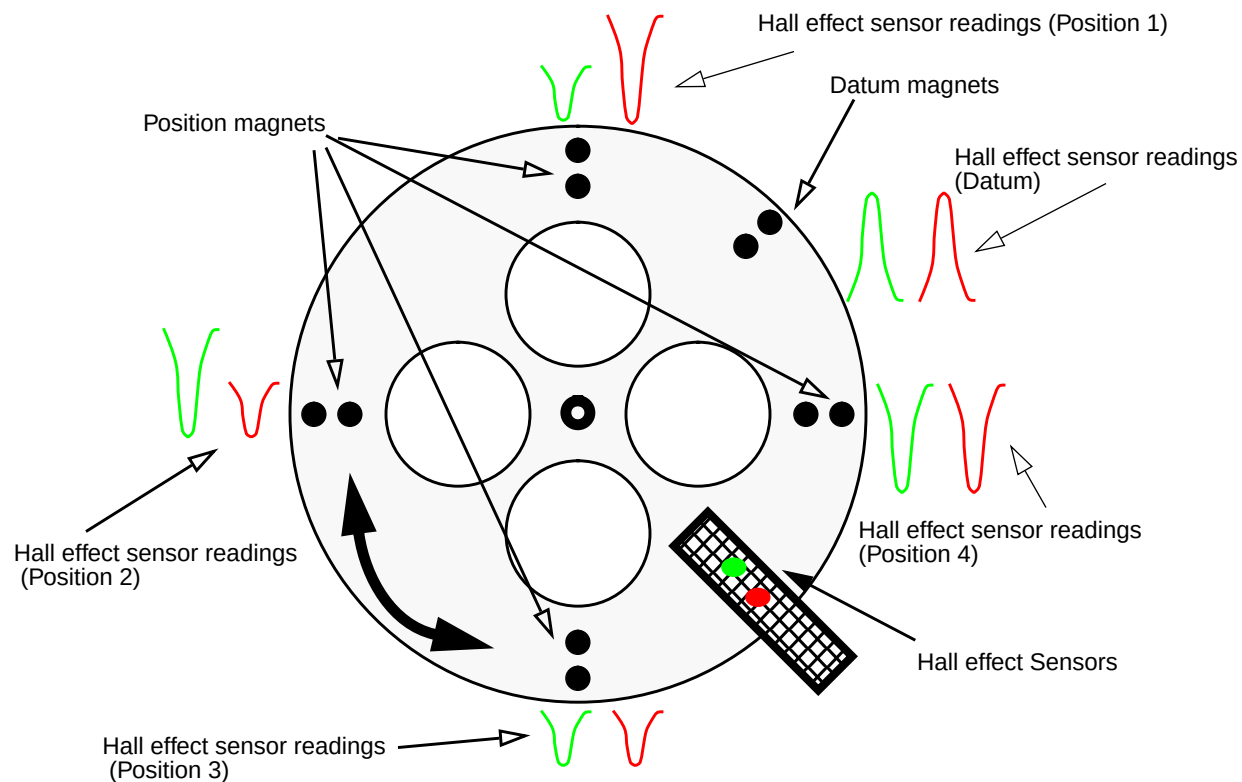


Table 1 describes the sensor values corresponding to Figure 4.

TABLE 1.

Sensor Values for Figure 4

Position	Sensor 1 (V)	Sensor 2 (V)
1	-2.0	-4.0
2	-4.0	-2.0
3	-2.0	-2.0
4	-4.0	-4.0
Datum	+4.0	+4.0

Table 2 describes typical sensor values for a twelve-position wheel. Note that it is assumed that it is only possible to distinguish four different magnet strengths. In practice, it may be possible to distinguish more values.

TABLE 2.

Typical Sensor Values (12 position wheel)

Position	Sensor 1 (V)	Sensor 2 (V)
1	-4.0	-4.0
2	-4.0	-3.0
3	-4.0	-2.0
4	-4.0	-1.0
5	-3.0	-4.0
6	-3.0	-3.0
7	-3.0	-2.0
8	-3.0	-1.0
9	-2.0	-4.0
10	-2.0	-3.0
11	-2.0	-2.0
12	-2.0	-1.0
Datum	+4.0	+4.0

For smaller mechanisms with a small number of positions, the two-sensor arrangement will be modified to use a single back-to-back sensor pair.

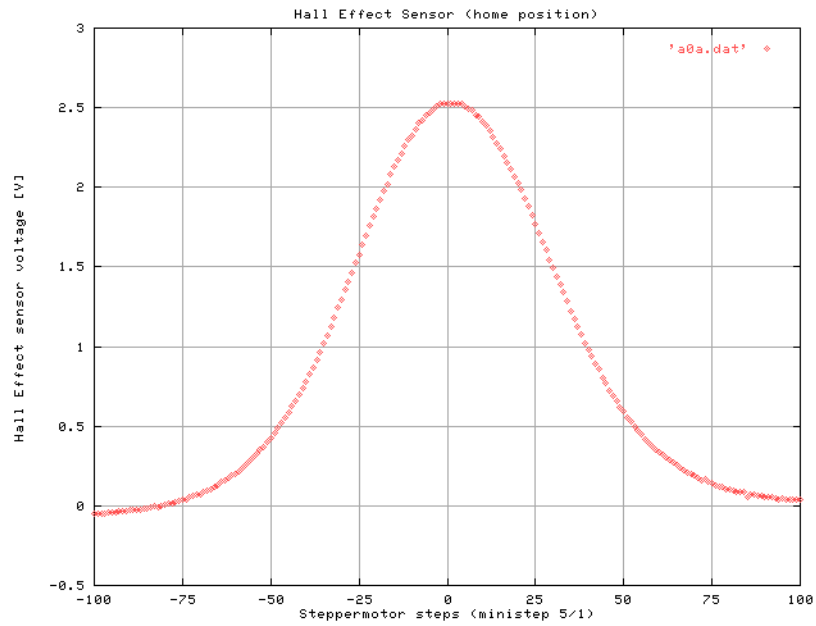
3.1.1.1 Sensor data plots (home magnet)

Figure 5 is a data plot of the Hall-Effect sensor values near a datum magnet. Note that the peak is sharply defined and nearly symmetric, though careful inspection shows a slight asymmetry. The data values become negative in the wings of the peak. This is probably a real effect, due to the reversal of the direction of the magnetic field near a magnet.

This data was taken using a 5/1 mini-step stepper motor controller. In practice, the mini-steps may prove to be useless, because the motor will tend to fall into a detente when power to the motor is removed.

FIGURE 5.

Hall effect sensor readings (Datum magnet)



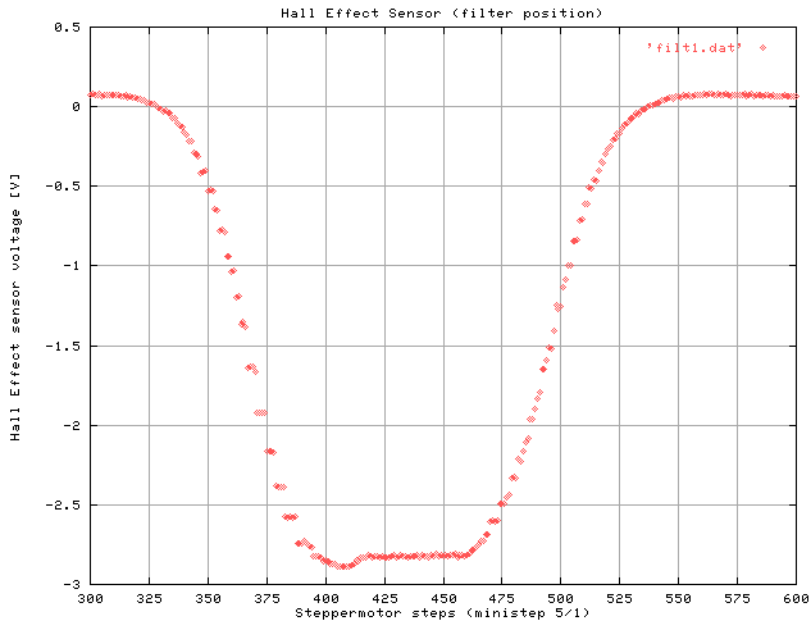
3.1.1.2 Sensor data plots (Confirmation magnet)

Figure 5 is a data plot of the Hall-Effect sensor values near a datum magnet. Note that the (negative) peak is flattened. This demonstrates the effectiveness of the Geneva mechanism which is used to turn the filter wheel. (The Geneva mechanism causes the motion of the wheel to be non-uniform. The motion of the wheel is slowed in the vicinity of the filters, making the hardware relatively insensitive to positioning errors.) The data values become positive in the wings of the peak. This is probably a real effect, due to the reversal of the direction of the magnetic field near a magnet.

This data was taken using a 5/1 mini-step stepper motor controller. In practice, the mini-steps may prove to be useless, because the motor will tend to fall into a detente when power to the motor is removed.

FIGURE 6.

Hall effect sensor readings (Filter magnet)



3.1.2 Continuous-positioning mechanisms (OIWFS focusing stage)

In some cases, a single sensor will be unable to provide adequate resolution. Higher resolution will be provided by using a combination of magnets and sensors on different parts of the mechanism (e.g., for the focusing stage, one magnet/sensor pair on the moving part of the stage, and a second pair on the main gear).

Figure 7 describes an (over)simplified model of a focusing stage, which requires two sensor pairs to define the datum position. The coarse sensor is able to position the stage in the vicinity of the datum, unambiguously, but with poor resolution. The gear sensor is able to determine the position very accurately, but the true position can be incorrect by an integral number of gear rotations.

A dual-magnet linear sensor arrangement is shown. The coarse-positioning or “linear” magnets are placed with opposite polarities, producing two adjacent peaks, one positive, and one negative. Between the peaks, there is a ramp, which is (almost) linear. The stage is designed so that it always operates in the linear region, providing a useful (though not extremely accurate) encoding of the position of the stage. Note that any changes in the sensitivity of the sensor electronics or constant offset can affect this encoding scheme. This is sufficient, because the encoding scheme needs only to position the stage at the datum, to within one turn of the main gear.

FIGURE 7. A two-sensor arrangement (OIWFS Focusing stage)

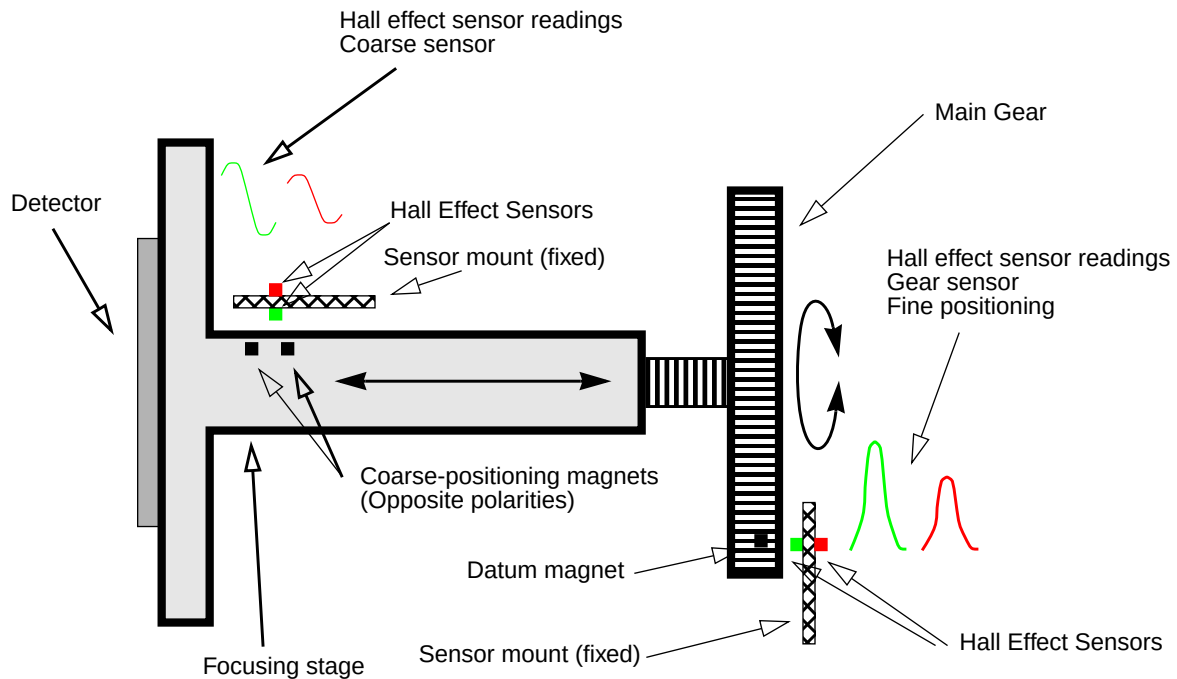
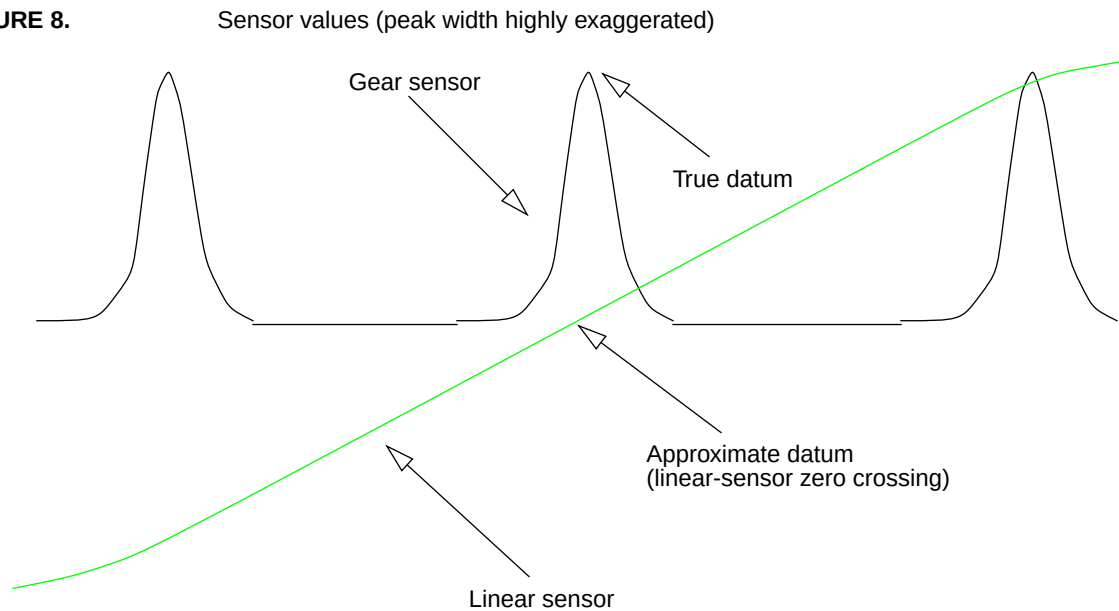


Figure 8 shows the expected reading from the Hall-effect sensors.

FIGURE 8.



Unlike the filter wheel, the Hall-effect sensors here are mounted on top of each other. This arrangement is simpler than for the filter wheel, because it requires only a single magnet, not a pair of magnets. Unfortunately, because it is not possible to insure that the sensors are precisely concentric with each other, there is a slight difference in the peak position. This must be compensated for via an offset value, which must be determined empirically. Furthermore, the signal strength for the sensor which is further away from the magnets is noticeably weaker.

One sensor is designated as the “primary” sensor. This sensor will be connected to the “home” input of the Oregon Microsystems stepper motor controller, which will speed up the datuming process. If this sensor fails, then the datum position may still be found, but will require a slower search process.

3.1.2.1 Reference points

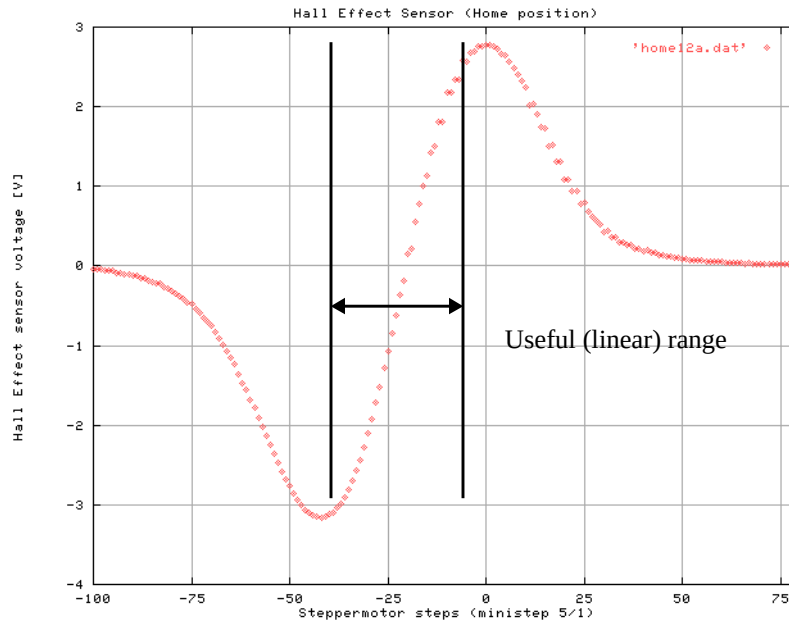
Although there are no predefined locations, a list of (arbitrarily chosen) reference points will be defined. To insure proper operation of the system, a command (“DIAGNOSE”) will be provided to move the mechanism to each reference point, and the expected value of the hall-effect sensors will be compared the actual value.

3.1.2.2 Sensor data plots (“linear” arrangement)

Figure 9 shows measured values for the Hall effect sensors in a “linear” arrangement. Note that the data values are almost linear in the region between the peaks. This provides a useful partial encoding scheme; however it is too susceptible to instabilities in the electronics to be used as a precise encoding scheme.

FIGURE 9.

Hall effect sensor readings (“linear” magnet arrangement)



3.1.3 Other continuous-positioning mechanisms (science focus and gimbals)

Unfortunately, not all of the continuous-positioning mechanisms are identical. The focussing stage for the science detector and the OIWFS gimbal mirrors are similar to the focussing stage for the OIWFS detector, but are not identical.

3.1.3.1 The focussing stage for the science detector

The larger range of travel of the focussing stage for the science detector means that the full range of travel of the stage cannot be uniquely encoded, as it is for the OIWFS focussing stage. To get around this problem, additional magnets are installed on the translation stage to insure that there is a region in which the signal value is a one-to-one function of the position. The ends do not provide a unique signal, but the sign of the signal is sufficient to indicate the location of the linear region.

FIGURE 10. A two-sensor arrangement (Science Focusing stage)

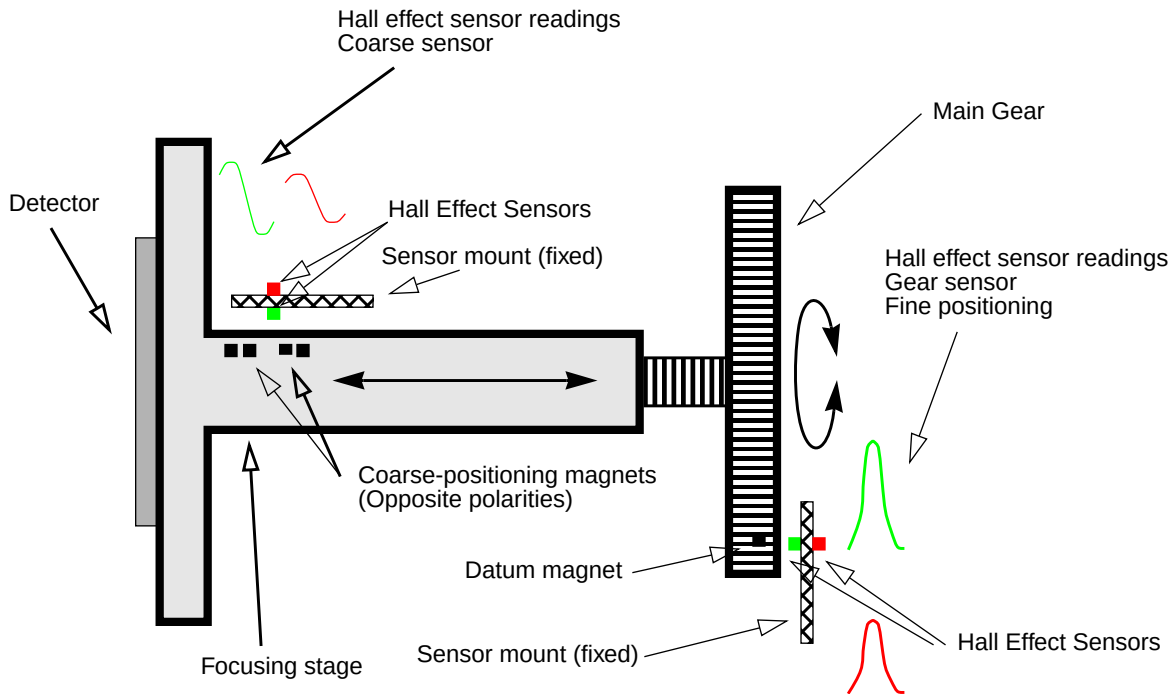
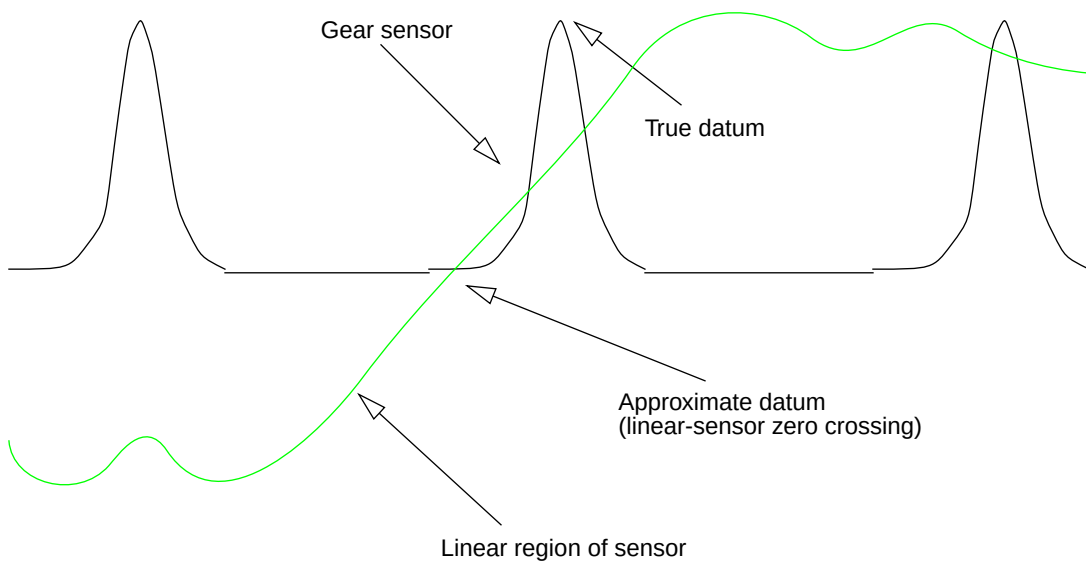


Figure 11 shows the expected reading from the Hall-effect sensors.

FIGURE 11. Sensor values (peak width highly exaggerated)



3.1.3.2 The OIWFS gimbal mirrors

The larger range of travel of the gimbal mirrors stage and the precise positioning requirements require still another variation for the gimbal mirrors. Instead of a linear positioning sensor, a single magnet is used, and the location of the peak is used to determine the true home position.

FIGURE 12. A two-sensor arrangement (gimbal mirror stage)

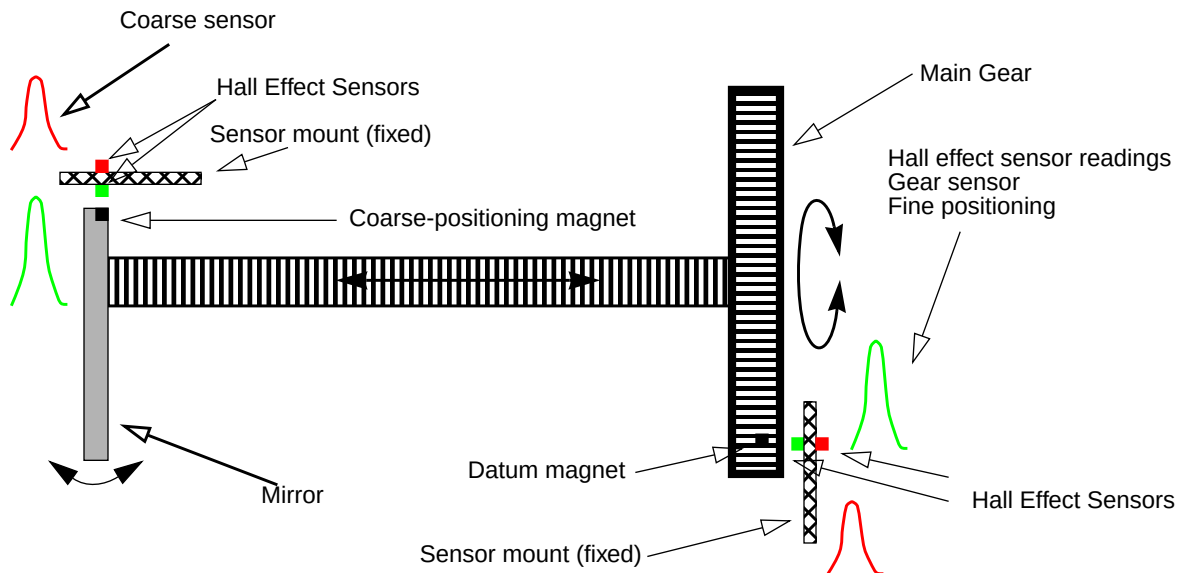
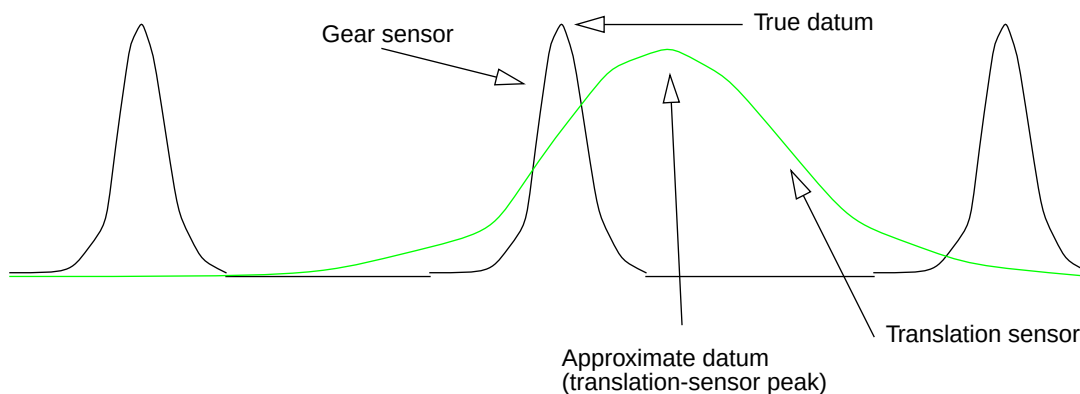


Figure 13 shows the expected reading from the Hall-effect sensors.

FIGURE 13. Sensor values (peak width highly exaggerated)



3.1.4 Two position mechanisms (the pupil viewer)

The pupil viewer has two defined positions. Each position is located at a hard stop, at an end of travel.

FIGURE 14. A one-sensor arrangement (pupil viewer))

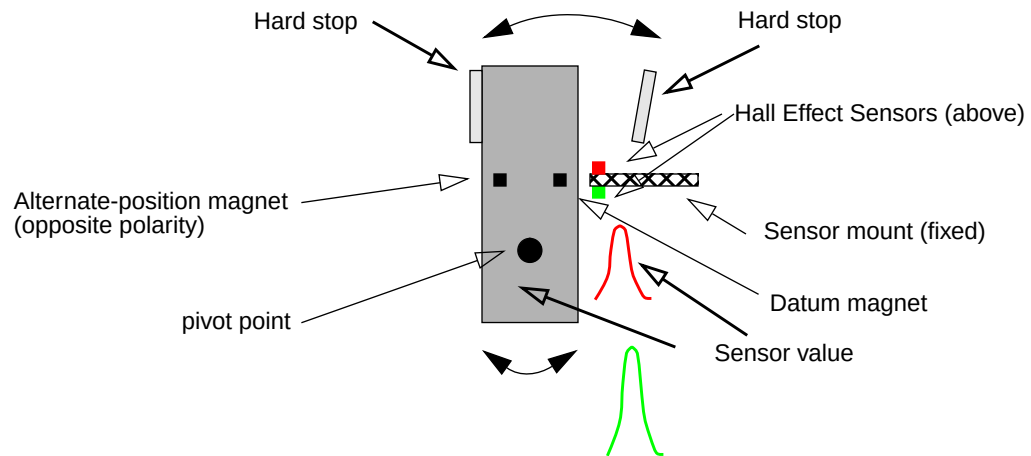
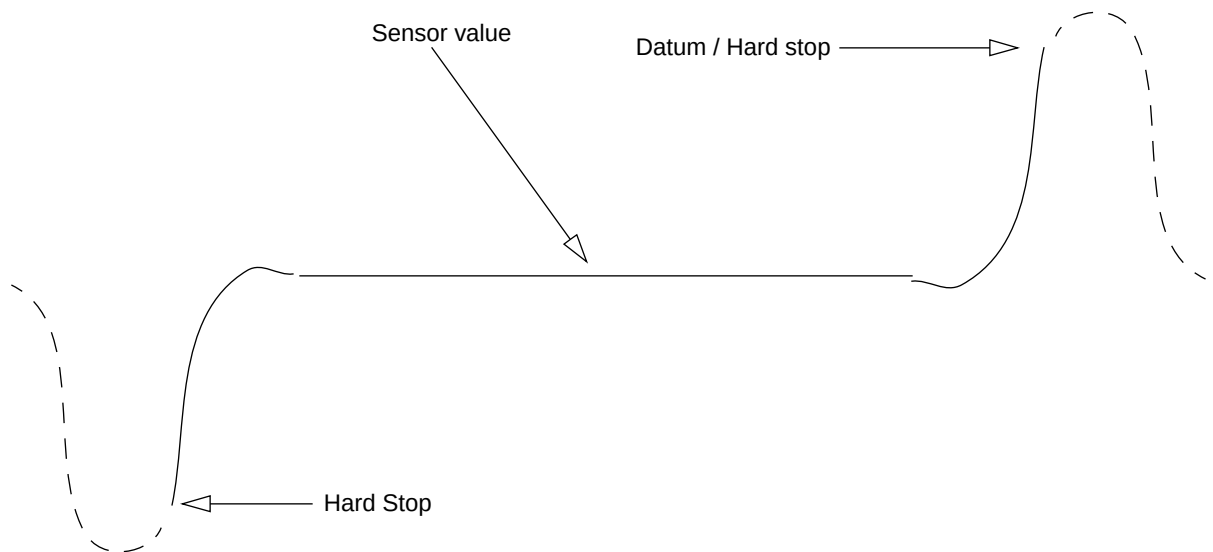


Figure 13 shows the expected reading from the Hall-effect sensors.

FIGURE 15. Sensor values (peak width highly exaggerated)



3.2 Peak Finding Algorithm³

For the rest of this document, we will refer to “finding the peak value” associated with each magnet/sensor pair, using the following algorithm

Locate the largest sensor value with a brute-force search.
Sample the sensors values near the maximum sensor value.
Fit a parabola to the sensor values.
Use the peak of the parabola as the sensor value.

This symmetric algorithm should be relatively insensitive to slight changes in the gain of the electronics, and to any biases in the electronics.

3.3 Stepper-motor errors

The hardware will be designed so that the stepper motors will never lose counts if all mechanisms are working properly. If any counts are lost, a problem exists which must be detected and corrected. The Hall-effect sensors may be able to detect the problem. If the sensors are able to detect a problem, an ERROR will be generated so that appropriate error recovery may be taken.

3.4 Backup sensors

All of the Hall-effect sensors have backup sensors. The system will be designed to use a primary sensor to datum the corresponding mechanism and to confirm the instrument configuration, and a secondary sensor for confirmation purposes only. Either sensor may be disabled if it fails. If either sensor is disabled, it may not be possible to confirm the configuration of the corresponding mechanism. If the primary sensor fails a slower datuming algorithm will be used. If both sensors are disabled, it will not be possible to confirm the current position nor will it be possible to datum the mechanism. Disabling either sensor will generate a WARNING.

3.5 Hard Limits and Limit Switches

Most of the mechanisms do not have hard limits and can be driven continuously indefinitely. The exceptions are described in the following section.

3.5.1 The focusing stages

The wavefront-sensor focusing stage and the science-detector focusing stage both have hard stops. Neither mechanism has a limit switch or a passive way of detecting that the mechanism has hit a limit. Both mechanisms are designed so that they will not be damaged by *occasionally* hitting a hard stop. The software is designed with soft limits and it should be impossible to hit a hard stop unless there is a hardware problem. If a hard stop is encountered, it will be necessary to redatum the mechanism.

3.5.2 The Pupil Viewer

3. The peak-finding algorithm is implemented within the HallStep record, documented in [35] in Section 4 on page 6.

The pupil viewer is designed so that it always hits a hard stop whenever it hits the limits of its travel.

3.5.3 The OIWFS Gimbal Mirror

The gimbal mirror can be damaged by overrunning its limits. There are limit switches to prevent this from happening. These switches are not intended to be used in normal operation. The software will be designed with soft limits so that the mechanism will never hit a limit.

If a motion command causes the hardware to hit a limit switch, the mechanism will stop and the motion command will complete with an error message. It will then be possible only to back the mechanism out of the limit. It should not be necessary to redatum the mechanism after hitting a limit.

Warning: The limit switches for both axes are tied together. If either axis hit the positive limit, then both axes will stop, and it will only be possible to move the axes in a negative direction. The same holds true for the negative limits. It is possible to reach a state in which it is not possible to move the mechanism: If one axis actuates the positive limit, and the other axis actuates the negative limit, then the mechanism will not be able to move. In this case, it will be necessary to determine which mechanism is in which limit, and to carefully back out of each limit in the proper direction. The procedure for determining which limit switch has been actuated by which axis is described in the hardware reference manual.

3.6 Simultaneous motor motion

It will not be possible to move all motors in the system simultaneously, because of power restrictions. (This will be handled by the lower level software, and should be transparent to the upper level interface.)

Whenever a motor becomes active, a counter will be incremented. When the maximum number of motors is running, new operations will be delayed until the motor count drops below the maximum.

3.7 Stepper Motor Control Module Failure

If the motor draws too much current, the stepper-motor control modules can enter into a fault state. The motor is unable to operate if a fault occurs. This fault generates an immediate ERROR until the stepper-motor control module is reset. A command will be provided to reset the fault, which should be a very infrequent event.

Note: A fault condition also exists if power is not applied to the control module. This is normal. Obviously, it is impossible to reset this fault. It is not possible to distinguish this from the over-current fault condition.

3.8 Diagnose

A DIAGNOSE command will be provided for each component (This should not be confused with the standard TEST and VERIFY commands.) This command will move the hardware and monitor the Hall-effect sensors in order to determine if the hardware is functioning. It should be possible to discover a single, failed, sensor. It will not be possible to distinguish a

double sensor failure from a failed motor. Unlike the TEST command, the DIAGNOSE command *will* move the hardware.

3.9 Redatum

A REDATUM command will be provided for each component. (This should not be confused with the standard TEST and VERIFY commands.) This command will move to the expected datum position and confirm that it really is at the datum. If the verification fails, then an error will be returned. This command will not be used as part of the normal configuration-change procedure. It will only be used if a possible hardware problem is suspected.

4 Environment Control

4.1 Cooling Motor

The closed-cycle coolers are driven by the same stepper-motor controller as the niri mechanisms.

4.2 Temperature regulation

The NIRI and NIRI OIWFS temperature control is handled with three off-the-shelf Omega temperature controllers. These controllers accept ascii commands via their serial ports. A custom EPICS record is used to pass commands to these temperature controllers.

4.3 Heating Resistors

Warm-up is handled by a combination of heating resistors, which can be disabled using the digital input/output board (controlled using an EPICS binary output record) and the Omega temperature controller.

It will be necessary to coordinate the warming of the components with the warming of the detector, to insure that the detector is always slightly warmer than the surrounding instrument. Overall control of the cooling motors and the temperature regulation is handled by a custom record.

4.4 Temperature Sensors

An undetermined number of temperature sensors will monitor the behavior of the system if the temperature is out of the valid range an ERROR will be generated.

The temperature sensors will be read using an off-the-shelf Omega digital thermometer, which accepts commands via a serial port. A custom record has been provided to read the temperature.

5 Special Calibration Programs

There will probably be a need for special purpose software for engineering purposes, e.g., to generate the look-up tables of filter positions. These programs will never be used in normal operation. After delivery of the instrument to gemini, the programs will only need to be run after daytime engineering in which significant changes are made to the NIRI hardware.

These programs will be documented in [5].

6 The Gemini Interlock System

The appropriate hardware and software support will be provided for the Gemini Interlock System, but NIRI will never generate an interlock event, and will need to carry out no actions in response to an interlock demand.

7 Special Epics Records

7.1 Modified records

7.1.1 Motor record (OMS VME8, VME44 device/driver)

The EPICS motor and OMS VME8, VME44 device provides support for the Oregon Micro-systems VME8-8 board. We have produced a modified motor record and device support. (If necessary to avoid conflicts, this record will be called the “nirimotor” record.

- The record is modified to allow direct control over the AUX output. (The AUX output is used to control the power to the stepper motors from the stepper motor driver modules.) The PWR field has three possible values: “ON”, “OFF”, “AUTO”. “AUTO” is the default and is backwards compatible with the existing record.
- Constants and global variables have been renamed, to have a consistent “MOTOR_” or “motor” prefix.

7.2 Custom Records, Device Support, and Subroutine Records

The following will probably be necessary. They will be some combination of custom records, new device support for existing records, subroutine records, gensub records, and SNL code:

- Records/code to control the cooling and warming of the instrument;
- Records/code to optimize the motions of the filter wheels;
- Records/code to control the cooling motor;
- Records/code to control the temperature controller.

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Implementation Notes and Record Reference

Hubert Yamada

niri_h ty_011 / 03 / 1999-10-23

This report contains a reference manual for several custom records and capfast symbols which are central to the implementation of The Gemini Near Infra-Red Imager and On Instrument Wavefront Sensor.

1 Introduction

This report presents a brief overview of the implementation of the control software for the Near Infra-Red Imager and On Instrument Wavefront Sensor. It also contains reference manuals for several custom EPICS records and capfast symbols.

1.1 Purpose of the document

This report provides a description of the interfaces and behavior of various NIRI software components.

The intended audience of this document is:

- the maintainers of the NIRI software;
- the reviewers of the NIRI system;
- the reviewers of the NIRI software;
- anyone who may wish to understand the details of NIRI.

2 Overview

The NIRI top-level directory (niri_VXYY, where XYY is a release number in the form major release, minor release) is made up of the following subdirectories

LIB. Shared utilities and schematics

CICSCC. NIRI CC code, adapted from the CICS

CICSWFS. NIRI WFS code, adapted from the CICS

ENG. NIRI mechanisms, WFS mechanisms, and NIRI environment control

DC. The GNAAC DC (Disabled)

Standalone subsystems of this system can be built by modifying the appropriate Makefiles and startup scripts in the root directory to compile only the subdirectories containing the desired subsystem, the LIB subdirectory, and the ENG subdirectory. For example, a stand-alone wavefront sensor can be built by including just the “LIB”, “CICSWFS”, and “ENG” directory. The ENG subdirectory is designed to build a product which is capable of functioning by itself.

The NIRI top-level directory, the LIB directory, and the CICSCC directories are adapted from the corresponding CICS directories, with the following changes:

- The niri_VXYY/ascii directory includes files from the ENG/ascii directory.
- The CICSCC directory has additional mechanisms, to simulate the entire NIRI instrument
- The channel-access links to the CICS motor simulator has been replaced with channel-access links to the ENG databases.

In addition, the CICSWFS directory is essentially a copy of the CICSCC directory, customized with the NIRI components.

2.1 The ENG directory

Note: The ENG directory is designed to be run as a standalone package for low-level engineering of the niri mechanisms, or as a part of the complete niri system.

The ENG directory contains three independent databases for controlling different mechanisms: The CC database, the WFS database, and the environment-control database. It contains a single database, the GBL database, which provides low-level interlocks. It contains databases which may be loaded to simulate the corresponding subsystems. There are also some special records, device support, and other C code which are required to support the databases.

It is possible to simulate the NIRI mechanisms, the WFS mechanisms, and the NIRI environment controls by loading the appropriate simulation database(s), which are also included in the ENG directory. The top-level user interface should be have identically with the simulation or with the actual hardware. Some slight differences may be noted in the lower-level engineering interfaces.

2.1.1 The CC Database

The components-controller database is made up of several instances of the same capfast hierarchical symbol, the mech symbol.

2.1.2 The WFS Database

The wavefront-sensor database is structured identically to the CC database, with a different set of mechanisms.

2.1.3 The GBL Database

The global parameters database provides access to interlocks, to keep the system from moving when it is unsafe (e.g., the mechanisms are not at an equilibrium temperature) or unacceptable (e.g., during an observation) for mechanisms to move. It also restricts the number of mechanisms that are in motion. (The power supply is not able to provide power for all motors simultaneously.)

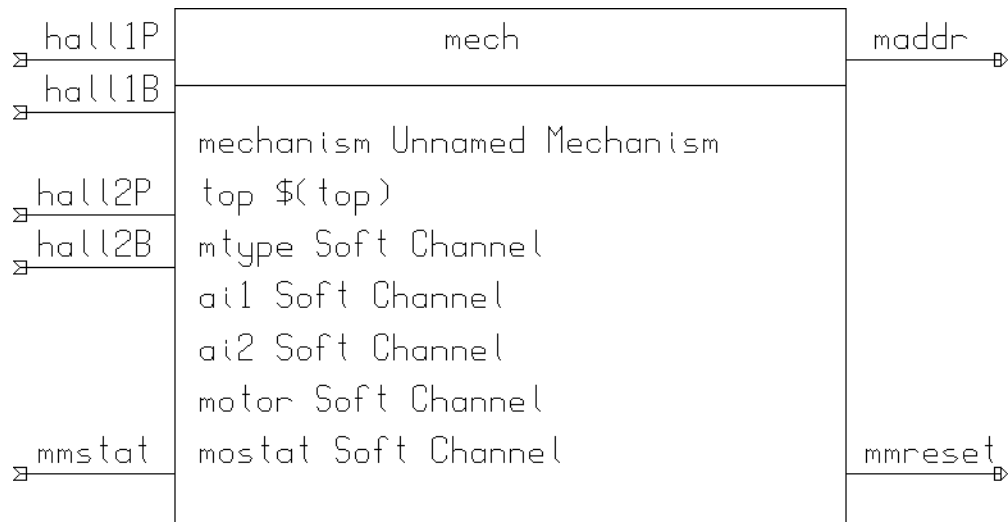
2.1.4 The ENV Database

The environment-control database is very simple, containing only a few records to monitor and to control the temperature sensors and temperature controllers.

3 The Mech Symbol and Diagram

All mechanical components are controlled by instances of a single capfast hierarchal symbol and the associated capfast diagram. Custom parameters are passed in via input and output ports (mostly hardware addresses) and via macro substitution.

3.0.1 The Mech Symbol



The following table describes the macro variables, inputs, and outputs of the mech hierarchal symbol.

TABLE 1. Mech.sym macro variables and ports

Macro/Port Name	Type	Value
hall1P	Input record hardware address	Primary Hall Effect sensor (wheel or main gear)
hall1B	Input record hardware address	Backup Hall Effect sensor (wheel or main gear)

TABLE 1. Mech.sym macro variables and ports

Macro/Port Name	Type	Value
hall2P	Input record hardware address	Primary Hall Effect sensor (linear sensor or unused)
hall2B	Input record hardware address	Secondary Hall Effect sensor (linear sensor or unused)
mmstat	Input record hardware address	Motor control module status (working or fault)
maddr	Motor control record hardware address	Motor controller board
mmreset	Output record hardware address	Motor control module reset (resets fault state)
mechanism	Macro variable	Text string (name of mechanism)
top	Macro variable	Prefix for generating record names
mttype	Macro variable	Type of motor controller (EPICS Device or Soft HS [simulator])
ai1	Macro variable	Type of input sensor (EPICS Device or Soft HS [simulator])
ai2	Macro variable	Type of input sensor (EPICS Device, Soft Channel [unused], or Soft HS [simulator])
motor	Macro variable	Type of motor (EPICS Device or Soft HS [simulator])
mostat	Macro variable	Type of motor status input (EPICS Device or Soft Channel [unused])

3.1 External interfaces (the NIRI/CICS interface)

This section describes the fields that the NIRI/CICS interface uses to control the mechanisms in the mech hierarchal symbol. These are the only fields that should be considered “public”¹. All other fields are to be considered a private to the implementation, and may be changed at any time.

TABLE 2. External Interfaces

Field	Action	Read or Write	Action	Comments
\$(top)MoveTo.VAL	Demand Position	Write	Move to new position	If an operation is already in progress, a new operation will fail.
\$(top)Op.VAL	Operation Code	Write	Perform operation.	See Table 3 on page 5 for a list of operation codes If an operation is already in progress, a new operation will fail.
\$(top)Stop.VAL	Stop	Write	Interrupt operation	Write a 1 to this field to interrupt an operation
\$(top)Busy.VAL	Busy	Read	Operation in progress?	Non-zero means operation in progress

1. The low-level engineering interface interacts with other fields; do not use it as an example of the correct way to interface with the mech hierarchal symbol.

The Mech Symbol and Diagram

TABLE 2. External Interfaces

Field	Action	Read or Write	Action	Comments
\$(top)Done.VAL	Done	Read	Status after operation	Non-zero means error
\$(top)Hallstep.MESS	Completion message	Read	Error/Status message after an operation	Cleared by the next operation
\$(top)Hallstep.DATM	Datumed?	Read or Write	Mechanism datumed?	Can be set or cleared externally or internally at any time. Set after successful datum, cleared if datum failed, cleared upon reboot. For informational purposes only; does not affect workings of record.
\$(top)Hallstep.HSWN	Hall effect sensor disabled?	Read	Hall-effect sensor disabled?	Can be set or cleared externally or internally at any time.
\$(top)Moving.VAL	Moving?	Read	Mechanism moving?	Non-zero means that it is moving. This is primarily for informational displays. Monitor \$(top)Busy.VAL instead, because the mechanism may briefly stop moving while an operation is still in progress.
\$(top)Mpos.VAL	Current Position?	Read	Mechanism current position	This field takes approximately 0.1 seconds to update. It is intended to be used for informational displays. For precise positioning, wait at least 0.1 second after Moving.VAL becomes 0 before reading this field.
\$(top)Speed.VAL	Desired speed?	Read	Mechanism speed	This is the desired speed after the mechanism has accelerated, <i>not</i> the current speed.

The following table describes all operation codes. *These codes might be out of date. Check values in hsConsts.h.*

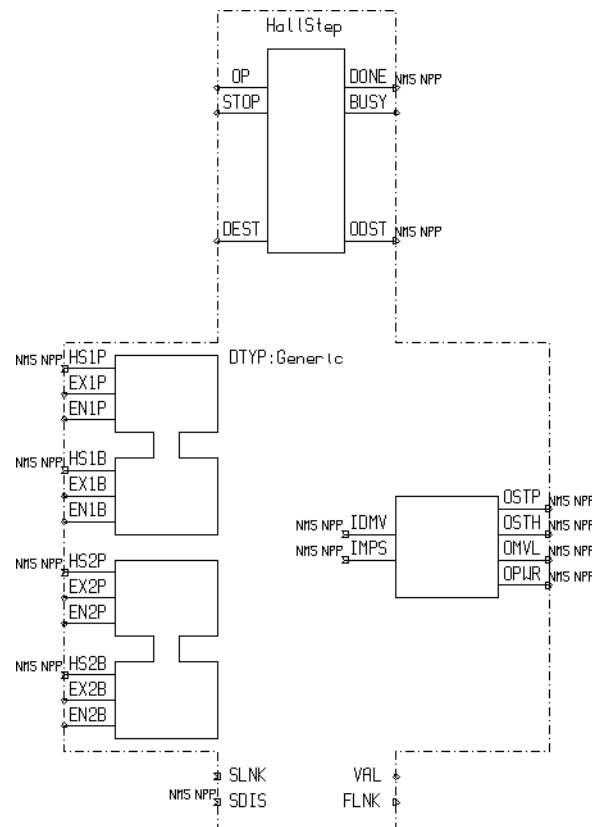
TABLE 3. Mech Hierarchical Symbol Operation codes

Code	Action	Public?	Comments
0	No Op	No	Reserved for internal use
1	Move	No	Use \$(top)MoveTo.VAL instead
2	Datum mechanism	No	For engineering. Normal programs should use Datum + Diagnose
3	Redatum mechanism	Yes	
4	Cycle mechanism	No	For engineering
5	Diagnose mechanism	Yes	
6	Datum + Diagnose	Yes	

4 The HallStep Record and Symbol

The “mech” diagram controls the hardware via the (custom) “hallStep” record. All commands to the hardware will pass through this record. (Hall effect sensor values may be read directly from the ai record.) Nothing should interface directly to the motor record, except for the records in the mech symbol.

In general, it should never be necessary to interface to the HallStep Record. Use the public fields in the mech symbol instead.



The following table describes the fields of the HallStep record.

TABLE 4. HallStep fields

Name	Summary	Description
VAL	Subr. return value	Error code returned if operation is rejected
OVAL	Old return value	
PREC	Display Precision	
STOP	Stop operation	Setting this field to non-zero cancels the current operation
OP	Operation	See Table 5 on page 9
DEST	Destination	Destination for Move operation

The HallStep Record and Symbol

TABLE 4. HallStep fields

Name	Summary	Description
OPST	Operation status	Text description of current action
MESS	Operation error msg	Text description of error or other important information
HOPR	High Operating Range	Upper limit of motion (Ignored during datum)
LOPR	Low Operating Range	Lower limit of motion (Ignored during datum)
RAW	Raw mode	Device dependant (e.g. ignore limits)
DONE	Operation error code	Output link, writes zero [no error] or non-zero [error] after operation completes
BUSY	Currently processing?	Output link, writes 1 when operation begins, 0 when operation ends
DSL P	“Datum slope P”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DSL B	“Datum slope B”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DOFP	“Datum offset, rough P”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DOFB	“Datum offset, rough B”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DBSH	“Datum backlash”	Exact meaning depends on device support (Wheel or Stage), used for datuming
DOF1	“Sensor offset, 1P-1B”	Exact meaning depends on device support (Wheel or Stage), used for datuming
WTHP	“Peak width 1P”	Estimated width of sensor peak (gear or wheel)
PKP	“Peak height 1P”	Estimated height of sensor peak (gear or wheel)
WTHB	“Peak width 1B”	Exact meaning depends on device support (Wheel or Stage)
PKB	“Peak height 1B”	Exact meaning depends on device support (Wheel or Stage)
CSTP	“Cycle stop”	Stop cycle test
CTOL	“Cycle tolerance”	Sensor error for automatic stop
CSET	“Cycle set values”	(Flag) On next cycle, set expected values for cycle test from observed value
CDSA	“Cycle A dest.”	First position for cycle test
CDSB	“Cycle B dest.”	Second position for cycle test
CDSC	“Cycle C dest.”	Third position for cycle test (normally 0)
E1PA	“Cycle A expect”	Expected primary (gear or wheel) sensor value at first test position
E1PB	“Cycle B expect”	Expected primary (gear or wheel) sensor value at second test position
E1PC	“Cycle C expect”	Expected primary (gear or wheel) sensor value at third test position
V1PA	“Cycle A read”	Actual primary (gear or wheel) sensor value at first test position
V1PB	“Cycle B read”	Actual primary (gear or wheel) sensor value at first test position
V1PC	“Cycle C read”	Actual primary (gear or wheel) sensor value at first test position
E1BA	“Cycle A expect”	Expected secondary (gear or wheel) sensor value at first test position
E1BB	“Cycle B expect”	Expected secondary (gear or wheel) sensor value at second test position
E1BC	“Cycle C expect”	Expected secondary (gear or wheel) sensor value at third test position
V1BA	“Cycle A read”	Actual secondary (gear or wheel) sensor value at first test position
V1BB	“Cycle B read”	Actual secondary (gear or wheel) sensor value at second test position
V1BC	“Cycle C read”	Actual secondary (gear or wheel) sensor value at third test position
E2PA	“Cycle A expect”	Expected primary (linear) sensor value at first test position
E2PB	“Cycle B expect”	Expected primary (linear) sensor value at second test position

The HallStep Record and Symbol

TABLE 4. HallStep fields

Name	Summary	Description
E2PC	“Cycle C expect”	Expected primary (linear) sensor value at third test position
V2PA	“Cycle A read”	Actual primary (linear) sensor value at first test position
V2PB	“Cycle B read”	Actual primary (linear) sensor value at second test position
V2PC	“Cycle C read”	Actual primary (linear) sensor value at third test position
E2BA	“Cycle A expect”	Expected secondary (linear) sensor value at first test position
E2BB	“Cycle B expect”	Expected secondary (linear) sensor value at second test position
E2BC	“Cycle C expect”	Expected secondary (linear) sensor value at third test position
V2BA	“Cycle A read”	Actual secondary (linear) sensor value at first test position
V2BB	“Cycle B read”	Actual secondary (linear) sensor value at second test position
V2BC	“Cycle C read”	Actual secondary (linear) sensor value at third test position
HSWN	“Warn HS”	Flag set when any hall effect sensor is disabled
HS1P	“Input Link HS1P”	Link to Hall-effect sensor input record
EX1P	“Expect HS1P”	Expected value of sensor after next move command
EN1P	“Enable HS1P”	Enable/Disable Hall-effect sensor
HS1B	“Input Link HS1B”	Link to Hall-effect sensor input record
EX1B	“Expect HS1B”	Expected value of sensor after next move command
EN1B	“Enable HS1B”	Enable/Disable Hall-effect sensor
HS2P	“Input Link HS2P”	Link to Hall-effect sensor input record
EX2P	“Expect HS2P”	Expected value of sensor after next move command
EN2P	“Enable HS2P”	Enable/Disable Hall-effect sensor
HS2B	“Input Link HS2B”	Link to Hall-effect sensor input record
EX2B	“Expect HS2B”	Expected value of sensor after next move command
EN2B	“Enable HS2B”	Enable/Disable Hall-effect sensor
IDMV ^a	“Motor finished moving”	Input link, used to monitor motion flag on motor-control record
IMPS ^a	“Motor Cur. position”	Input link, used to monitor motion flag on motor-control record
OMVL ^a	“Motor New position”	Output link, used to write new position on motor-control record
OSTH ^a	“Motor Set home”	Output link, used to trigger “set home” operation on motor-control record
OSTP ^a	“Motor Stop motion”	Output link, used to write “stop” on motor-control record
OPWR ^a	“Motor Power”	Output link, used to control power to on motor-control record
ODST	“Destination Link”	Output link, after a move, writes the current position

a. IDMV, IMPS, OMVL, OSTH, OSTP, and OPWR are very closely tied with the behavior of the motor record. If the motor record is modified, the behavior of these fields may need to change.

The following table describes all operation codes. *These codes might be out of date. Check values in `hsConsts.h`.*

TABLE 5.**HallStep Record Operation Codes**

Code	Action
0	No Op
1	Move
2	Datum mechanism
3	Redatum mechanism
4	Cycle mechanism
5	Diagnose mechanism
6	Datum + Diagnose

4.1 Record processing

This record is processed only when an operation code is written to the OP field.

4.1.1 When an operation code is processed:

If the record is processing another operation, the new operation fails, and the VAL field is set to a non-zero value.

If the operation code is invalid, the new operation fails and the VAL field is set to a non-zero value.

If the operation code is valid, the operation code is passed to the completion task, and the completion task is awakened.

Record processing completes, and 0 is returned in the VAL field.

4.1.2 When the completion task is given a new operation

The task writes a non-zero value to the BUSY link.

The task performs the desired operation (described below), reading sensor values from the sensor input records, writing control codes to the motor-control record. If there is an error, the task sets the MESS field to a descriptive message.

The task writes an error code (or 0 if there is no error) to the OPST field.

The task writes a 0 to the BUSY link.

For the move command:

The DEST field is read.

The hardware is moved to the new location.

The sensor values are compared to the expected sensor values (not yet implemented).

For the datum command:

The DATM field is set to 0

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

If the operation succeeded, the new datum is set.

If the operation succeeded, the DATM field is set to 1

For the redatum operation

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

The new datum is *not* set; the MESS field indicates how different the new datum is from the last datum

For the cycle operation

The hardware moves to the first test position (set by the CDSA, field. At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field.

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field.

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The process begins again, starting with the first test position, until the operation fails, or until the CSTP field is set to a non-zero value.

For the diagnose operation

The hardware moves to the first test position (set by the CDSA, field.

At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field.

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field.

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

For the datum + diagnose operation

The DATM field is set to 0

The hardware performs whatever operations are necessary to datum the hardware (see Section 3.1 on page 5 of [3]).

If the operation succeeded, the new datum is set.

The hardware moves to the first test position (set by the CDSA, field).

At the first test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the second test position (set by the CDSB, field).

At the second test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

The hardware moves to the third test position (set by the CDSC, field).

At the third test position, it reads the Hall effect sensors and compares them to the expected values.

If the values are not within the stated tolerance, the operation fails and processing terminates.

If the operation succeeded, the DATM field is set to 1

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and On-Instrument Wavefront Sensor (OIWFS)—Calibration and Maintenance Tools

Hubert Yamada

niri_h ty_004 / 08 / 2000-01-25

This document describes the software calibration programs and procedure for NIRI and the Near-Infrared OIWFS.

1 Introduction

1.1 Purpose

This document describes the software calibration programs and procedure for NIRI.

The intended audience of this document is:

- The NIRI reviewers;
- The developers of the Gemini Observatory Control System (OCS);
- Groups which will be using the design of the NIRI OIWFS (e.g. GNIRS);
- The maintainers of NIRI after it is delivered to Gemini.

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 When is it necessary to recalibrate a mechanism?

2.1 What is the difference between normal calibration and low-level calibration.

Normal calibration is a procedure which must be carried out for many different reasons (listed below). It is often necessary: if the zero point of the analog input board drifts, if wires are replaced, if the mechanism is partially disassembled.

2.2 When is normal calibration necessary?

Normal calibration is necessary if a mechanism appears to be working and datums itself at the correct location, but the final verification phase (diagnose) fails. It is also necessary if the mechanism is moving to the correct positions, but returns an error when it moves.

The procedure for normal calibration is described in Section 5.3 and Section 6.4.

2.3 When is low-level calibration necessary?

Low-level calibration is necessary only if a mechanism has been disassembled and reassembled; appears to be operating correctly for the most part; and cannot datum itself or datums itself at an insane position. Low-level calibration should be considered only after all alternatives have been excluded. Note that only the focus stages and the gimbal mirror require recalibration. For all other mechanisms, all constants are well-defined by mechanical means.

2.4 When should the mechanism not be calibrated?

If the mechanism is not functioning properly, recalibrating it will mask problems or cause it to return false error messages.

3 Running the tools

3.1 The Engineering User Interface

You must know how to operate the NIRI / OIWFS Engineering user interface before you begin the calibration. See the software release notes [7] and the User's Guide for the Engineering User Interface [8] for more details.

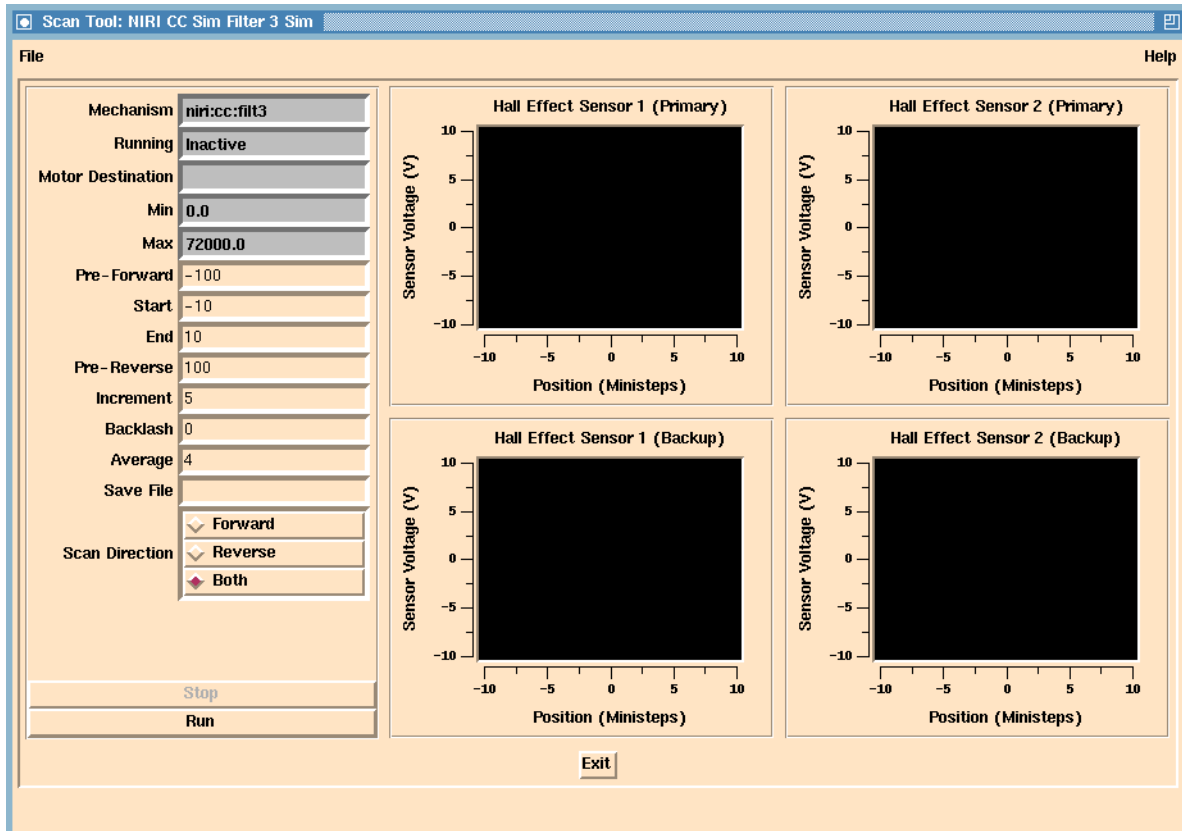
4 Scan Tool

The "scan tool" is used for characterizing hardware, for measuring backlash, and for diagnosing problems. It is a useful tool for diagnosing problems with mechanisms, particularly with mechanisms that are not moving freely. It can be helpful in low-level calibration of mechanisms. It is not used in normal operation.

4.1 Starting the “Scan Tool”

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Scan Tool.”

4.2 Using The “Scan Tool”



The normal mode of operation is as follows:

Forward Pass:

Move to “Pre-forward”.
 Move to “Start”, and read the Hall effect sensors
 Move to “Start” + “Increment”, and read the Hall effect sensors
 (Repeat until passes “End”)

Reverse Pass:

Move to “Pre-reverse”
 Move to “End”, and read the Hall effect sensors
 Move to “End” - “Increment”, and read the Hall effect sensors
 (Repeat until passes “Start”)

The difference between the values on the forward and reverse pass provides an indication of the amount of backlash in the system. Significant differences between the forward and reverse passes indicate a serious hardware problem, i.e., something is slipping or sticking.

5 The “Cycle Tool”

NIRI and the OIWFS are designed to provide quick verification that a mechanism is operating properly. This is done by placing analog Hall-effect sensors and magnets at various locations. When the mechanism is in a known, good state, the mechanism is moved to three test positions. The Hall-effect sensor values are read at each location.

The diagnose command verifies that the mechanism is operating correctly by moving it in sequence to each of these test positions and verifying that the sensor values are close to the which were recorded when the system was known to be operating correctly. The Cycle command is identical to the diagnose command, except that it repeats the test, until told to stop.

The “Cycle Tool” is a program that aids in this part of the calibration procedure. It moves the system to each of the test positions, and then records the values, which may then be written to a file. The file is read whenever the system is booted.

Warning: A side effect of running the “Cycle Tool” is that the in-memory table of expected values is changed. This is generally desirable, but can be misleading.

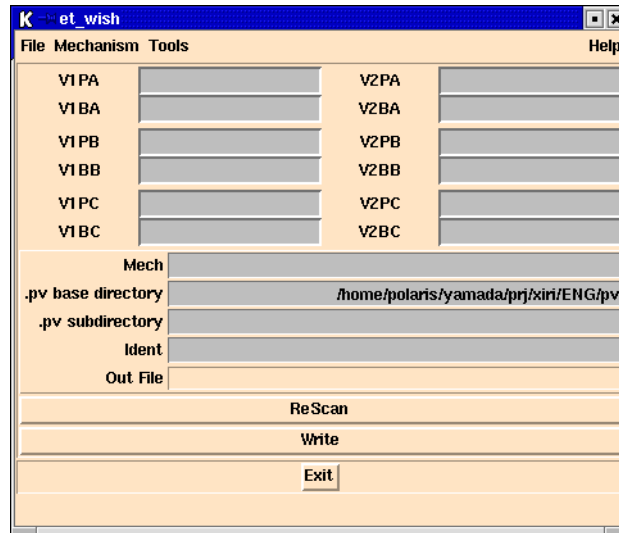
Warning: If the “Cycle Tool” is run while the mechanism is not working correctly then the results returned by the Cycle Test and Diagnose Test are suspect.

Note: The “Diagnose Test” is automatically performed when the “Datum” command is called from the high-level, interface. This is equivalent to the Datum/Diag command in the low-level engineering interface.

5.1 Starting the “Cycle Tool”

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Cycle Tool.”

5.2 Using the “Cycle Tool”



The fields / buttons of the cycle tool are:

V1PA. Voltage at test position A, sensor pair 1, primary sensor.

V1BA. Voltage at test position A, sensor pair 1, backup sensor

V2PA. Voltage at test position A, sensor pair 2, primary sensor.

V1BA. Voltage at test position A, sensor pair 2, backup sensor

V1PB. Voltage at test position B, sensor pair 1, primary sensor.

V1BB. Voltage at test position B, sensor pair 1, backup sensor

V2PB. Voltage at test position B, sensor pair 2, primary sensor.

V1BB. Voltage at test position B, sensor pair 2, backup sensor

V1PC. Voltage at test position C, sensor pair 1, primary sensor.

V1BC. Voltage at test position C, sensor pair 1, backup sensor

V2PC. Voltage at test position C, sensor pair 2, primary sensor.

V1BC. Voltage at test position C, sensor pair 2, backup sensor

Mech. The EPICS prefix for all fields related to this mechanism

.lut base directory. Where output files are created.

.lut subdirectory. Subdirectory of .lut base directory where files are created

Ident. The name of the default output file.

Out File. Name of the output file (user editable).

ReScan. Collect data.

Write. Write data file.

Exit. Exit from the “Cycle Tool.”

5.3 Normal Calibration (Phase 1)

1. Press the “ReScan” button to move the mechanism to the three test positions.
2. Examine the values. All values should be reasonable. (See Section 5.4.)
3. Examine the “Out File” field. Verify that the correct file will be created. Change the “Out File” field if you do not want to overwrite the normal file.
4. Press the “Write” button to write the data.
5. Press the “Exit” button to exit the “Cycle Tool.”
6. From a shell on the software server, change into the “.pv base directory” (from the “Cycle Tool”).
7. Run “gmake” in the “.pv base directory.”

Note: If you forget to press “Write” before pressing “Exit,” you will not be warned, but the file on disk will not be updated.

5.4 Expected Values (Phase 1)

The following tables describe the expected sensor values for each mechanism at the test positions.

The window cover has no sensors, and never needs to be calibrated. All sensor values read as 1 or 2 at the limits and 0 between limits.

TABLE 1.

Filter Wheels, The Pupil Mask Wheel, The Focal Plane Mask Wheel, The Entrance Beam Splitter, the Steering mirrors

Abbreviation	Sensor	Position	Significance	Value (V)
V1PA	1 (Primary)	A	First Position	< -0.5
V1BA	1 (Backup)			< -0.5
V2PA	2 (Primary)			0
V2BA	2 (Backup)			0
V1PB	1 (Primary)	B	Last Position	< -0.5
V1BB	1 (Backup)			< -0.5
V2PB	2 (Primary)			0
V2BB	2 (Backup)			0

The “Cycle Tool”

Abbreviation	Sensor	Position	Significance	Value (V)
V1PC	1 (Primary)	C	Home	> 1
V1BC	1 (Backup)			> 1
V2PC	2 (Primary)			0
V2BC	2 (Backup)			0

TABLE 2.

The Pupil Viewer

Abbreviation	Sensor	Position	Significance	Value (V)
V1PA	1 (Primary)	A	Alternate	< -1
V1BA	1 (Backup)			< -1
V2PA	2 (Primary)			0
V2BA	2 (Backup)			0
V1PB	1 (Primary)	B	Home	> 1
V1BB	1 (Backup)			> 1
V2PB	2 (Primary)			0
V2BB	2 (Backup)			0
V1PC	1 (Primary)	C	Home ^a	> 1
V1BC	1 (Backup)			> 1
V2PC	2 (Primary)			0
V2BC	2 (Backup)			0

a. If the corresponding values for positions B and C are not nearly identical, then the mechanism may have moved unexpectedly.

TABLE 3.

The Focus Stage

Abbreviation	Sensor	Position	Significance	Value (V)
V1PA	1 (Primary)	A	Large Positive	Near 0
V1BA	1 (Backup)			Near 0
V2PA	2 (Primary)			Any ^a
V2BA	2 (Backup)			Any
V1PB	1 (Primary)	B	Large Negative	Near 0
V1BB	1 (Backup)			Near 0
V2PB	2 (Primary)			Any
V2BB	2 (Backup)			Any
V1PC	1 (Primary)	C	Home	> 1
V1BC	1 (Backup)			> 1
V2PC	2 (Primary)			Any
V2BC	2 (Backup)			Any

a. The #2 sensor values should change significantly between the A, B, and C positions. Either $A > C > B$ or $B > C > A$.

TABLE 4.

The Gimbal Mirror

Abbreviation	Sensor	Position	Significance	Value (V)
V1PA	1 (Primary)	A	Large positive	Any
V1BA	1 (Backup)			Any
V2PA	2 (Primary)			Any ^a
V2BA	2 (Backup)			Any
V1PB	1 (Primary)	B	Large Negative	Any
V1BB	1 (Backup)			Any
V2PB	2 (Primary)			Any
V2BB	2 (Backup)			Any
V1PC	1 (Primary)	C	Home	Any
V1BC	1 (Backup)			Any
V2PC	2 (Primary)			Any
V2BC	2 (Backup)			Any

a. The #2 sensors should change significantly between the A, B, and C positions.

5.5 Configuration file

The test positions are defined in ENG/pv/cc.lut and ENG/pv/wfs.lut.

6 The “Position Tool”

6.1 Starting the “Position Tool”

From the low-level engineering screen for the mechanism, press the “External Tools” button using the right mouse button, then select “Position Tool.”

6.2 Using The “Position Tool”

NIRI and the OIWFS are designed to provide verification that a mechanism is in the proper configuration. This is done by placing analog Hall-effect sensors and magnets at various locations. When the mechanism is in a known, good state, the mechanism is moved to all standard positions. (For mechanisms with a continuous range of travel, the standard positions are at positions that are convenient for checking mechanism operation.)The Hall-effect sensor values are read at each position.

Each time that the mechanism is moved to a standard position, the expected sensor values are compared to the actual sensor values. If the values do not agree, then an error is returned.

Furthermore, any time that the mechanism is moved, whether to a standard position or not, the software examines the step count and sensor values, and attempts to determine if the mechanism is in a known position.

The “Position Tool”

The “Position Tool” is a program that aids in this part of the calibration procedure. It moves the system to each of the standard positions, and then records the values, which may then be written to a file. The file is read whenever the system is booted.

Warning: The “Cycle Tool” does not change the in-memory version of the table that it produces. To update the in-memory version of the table

1. Bring up the “Parameters” window (right-click the “Related Screens” button);
2. Install the updated table with gmake;
3. Press the “Load LUT” and “Load ILUT” buttons.

- 6.3 Warning: If the “Cycle Tool” is run while the mechanism is not working correctly, and if the values are written to disk, then the automatic verification of a move command will be suspect.

The screenshot shows a window titled "et_wish #2" with a menu bar (File, Mechanism, Tools, Help). Below the menu is a table with five columns: "count", "count tol", "val tol", "sens (p)", and "sens (b)". The table has 15 empty rows. Below the table are five input fields labeled "Mech", ".lut base directory", ".lut Subdirectory", "Ident", and "Out File". The ".lut base directory" field contains the path "/home/polaris/yamada/prj/xiri/ENG/pv". At the bottom of the window are three buttons: "ReScan", "Write", and "Exit".

Each line of this display represents a single defined mechanism position. The fields in each line are as follows:

count. Nominal step count as this standard position.

count tol. How close must the step count be to the nominal step count for the mechanism to be considered to be at this standard position.

val tol. How close must the Hall-effect sensor voltages be to the expected voltages for the mechanism to be considered to be at this standard position.

sens (p). Voltage at this standard position on the primary sensor.

sens (b). Voltage at this standard position on the secondary sensor.

Furthermore, the following values are defined.

Mech. The EPICS prefix for all fields related to this mechanism

.lut base directory. Where output files are created.

.lut subdirectory. Subdirectory of .lut base directory where files are created

Ident. The name of the default output file.

Out File. Name of the output file (user editable).

ReScan. Collect data.

Write. Write data file.

Exit. Exit from the “Cycle Tool.”

6.4 Calibration Procedure (Phase 2)

1. Press the “ReScan” button to move the mechanism to all defined positions.
2. Examine the values. All values should be reasonable. (See Section 6.5.)
3. Examine the “Out File” field. Verify that the correct file will be created. Change the “Out File” field if you do not want to overwrite the normal file.
4. Press the “Write” button to write the data.
5. Press the “Exit” button to exit the “Cycle Tool.”
6. From a shell on the software server, change into the “.pv base directory” (from the “Cycle Tool”).
7. Run “gmake” in the “.pv base directory.”

Note: If you forget to press “Write” before pressing “Exit,” you will not be warned, but the file on disk will not be updated.

6.5 Expected Values (Phase 2)

The following tables describe the expected sensor values for each mechanism at the standard positions.

The window cover has no sensors, and never needs to be calibrated. All sensor values read as 1 or 2 at the limits and 0 between limits.

TABLE 5.

Expected values

Position	Significance	Primary Value (V)	Backup Value (V)
pos00	First Position	any ^a	any
...	...	...	...
posNN	Last Position	any	any
home	Datum Position	> 1	> 1
park	Parked Position	any ^b	any

- a. These values should change significantly from position to position. If the same pair of values (within a few hundred millivolts) appears repeatedly, then the mechanism is not moving. If too many value pairs are near zero, then the mechanism is not moving correctly. If more than one positive value pair appears (other than the home or park position), then the mechanism is stuck at the home position.
- b. The park position is arbitrary, and is placed where it is most convenient. Its value might be positive, negative, or zero.

6.6 Configuration file

All constants that are defined in the position tool are defined in ENG/pv/cfg.tcl. See the comments in that file for more details.

7 Low Level Calibration Procedure

Whenever a mechanism is modified, it will be necessary to recalibrate it. Note that this procedure should only be used under very unusual conditions. It is not a standard maintenance procedure.

This section will describe the calibration procedure which is required for each mechanism.

7.1 The Wheels and Wheel-like Mechanisms, the Pupil Viewer, and the Window Cover

Low-level calibration is never necessary for the filter wheels, for the pupil-mask wheel, for the entrance beam splitter, for the focal-plane mask wheel, for the pupil viewer, for the window cover, and for the steering mirrors. There are no constants which require adjustment.

7.2 The Focus Stages

1. Start the Engineering Screen for the Science Focus Stage or OIWFS Focus Stage.
 2. Adjust the “New Position” field until the value of sensor 2P is less than 0.005 V. Call this *position A*.
 3. Repeat for sensor 2B. Call this *position B*.
 4. While observing the hardware, adjust the “New Position” field until the sensor is adjacent to the magnet. Call this *position C*.
 5. Compute the value of *position C* - *position A*. This is the value of *DOFP*.
-

6. Compute the value of *position C - position B*. This is the value of *DOFB*.
7. Bring up the parameter window. In the *DOFP* field, enter the value of *DOFP*. In the *DOFB* field, enter the value of *DOFB*. Close the parameter window.
8. Press the Datum Button, and let the system datum.
9. Set the “New Position” field to 0. You should find that sensor 1P has a signal between 1 and 5 V. If this is not true, then there is a hardware problem or the calibration was performed incorrectly.
10. While observing the stage, adjust the “New Position” field until you know the safe limits of motion of the mechanism. It is better to be slightly conservative here. Call the lower limit *LOPR*. Call the upper limit *HOPR*.
11. Bring up the parameter window. In the *HOPR* field, enter the value of *HOPR*. In the *LOPR* field, enter the value of *LOPR*.
12. Edit *niri/ENG/pv/cfg.tcl*. Enter the value of *LOPR* in the *pos01* field for *ccFoc*. Enter the value of *HOPR* in the *pos02* field for *ccFoc*.
13. Run Cycle Tool.
14. Run Position Tool.
15. Bring up the parameter window. Enter 0 in the *DOF1* field.
16. Disable sensor 1P.
17. Press the Redatum button in the engineering window. You should eventually see “Peak at (some value)”. Multiply this value by -1 and enter it into the *DOF1* field.
18. Enter sensor 1P.
19. Calibration is complete. The values which were placed in the *DOF1*, *DOFP*, and *DOFB* fields should be entered into *ENG/pv/cc.pv* as the values for *focHallstep.DOF1*, *focHallstep.DOFP*, and *focHallstep.DOFB*.
20. Enter the lower limit in *ENG/pv/cc.pv* as the value of *focHallstep.LOPR* and *focHallstep.CDSB*. Enter the upper limit as *focHallstep.HOPR* and *focHallstep.CDSA*.
21. Run *gmake*, then reboot the IOC.

7.3 The Gimbal Mirror

This section has not been written yet.

8 Warnings

If the system needs to be recalibrated frequently, there is a problem with the mechanism. Recalibration can mask minor problems, particularly with mechanisms that are not moving freely.

ICD 1.6/1.10[ab]

“A&G to Niri/GNIRS On Instrument Wavefront Sensors”

niri_h ty_008

Hubert Yamada
2001-06-27

Version: 1.01

Issued By: Hubert Yamada

Approved By: _____ Date: _____
Group Manager
_____ Group Manager (if req) _____
_____ Systems Engineering _____

Revision Control

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Chapter 1 Introduction

1–1.0 Description

1–1.1 Purpose

This Interface Control Document (ICD) serves the following purposes:

- To describe the commands and parameters that will pass between the A&G system and the On-Instrument Wavefront Sensor (OIWFS) Components Controller (CC).
- To provide a prototype which may, with minimal modification, be used to describe the GNIRS OIWFS, and the OIWFS for any system which copies the NIRI OIWFS hardware and software.

This document only describes the ways in which the NIRI/GNIRS OIWFS differs from the prototypical OIWFS which is described in ICD 1.6/1.10 [19] and in [20].

At some time in the future, the NIRI/GNIRS OIWFS information may be merged with ICD 1.6/1.10.

In addition to the documents mentioned in the previous paragraphs, the specific interfaces described in this document are governed by the following parent Interface Control Documents, which describe the general properties of particular kinds of interface:

- ICD 1a [24], which describes how the OCS deals with commands and attributes and the behavior the OCS expects from the systems it commands.
- ICD 1b [25], which gives a general description of the way the Observatory Control System communicates with the other principal systems. It is also related to ICD 7a, [27], which describes the way an Instrument Control System communicates with its subsystems.
- ICD 2 [26], which describes the way in which an Instrument Control System makes information available to the outside world (and the OCS) by means of “Status Information Records”.
- ICD 7b [28], which describes the way that the Telescope Control System (TCS) interacts with its subsystems. (From a control standpoint, the OIWFS is a subsystem of the A&G, which is a subsystem of the TCS.)

The intended audience for this document is:

- The NIRI reviewers;
- The GNIRS reviewers;
- Any other groups which will be using the design of the NIRI/GNIRS OIWFS;
- The developers of the Gemini Acquisition and Guidance units;
- The developers of the GNIRS OIWFS components controller.

1–1.2 Scope

This document describes only aspects of the NIRI/GNIRS OIWFS software system which differ from, or are more specific than, the Gemini standard on-instrument wavefront sensor

(OIWFS), described in [19]. It is assumed that the reader is familiar with both of those documents.

This document does *not* describe the following topics:

- The OIWFS hardware and electronics (see [11] and [12], summarized in [3]);
- The OIWFS detector software, hardware, and electronics (see [12] and [13]);
- The GNIRS software, hardware, and electronics;
- Other instruments which will be using the NIRI/GNIRS OIWFS;
- The OIWFS Interlock system (see [6]);
- The names of the NIRI OIWFS mechanism positions (see [37]);
- The names of the GNIRS OIWFS mechanism positions;
- Interfacing an instrument to the NIRI/GNIRS OIWFS (see [6]);
- The software design (see [2], and [3]).

Unless otherwise stated, any references to the OIWFS refers only to the OIWFS mechanisms. The OIWFS detector is a completely separate system, and does not have an EPICS interface.

1–2.0 Related Documents and Drawings

1–2.1 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

Chapter 2 Details of the A&G to NIRI/GNIRS OIWFS Interface

2-1.0 Introduction

2-2.0 Physical System Interfaces

2-2.1 Mechanical Interface

Not Applicable for a software ICD.

2-2.2 Optical Interface

Not Applicable for a software ICD.

2-2.3 Electronic Interface

2-2.3.1 NIRI

The system hardware architecture for the NIRI OIWFS control software is the same as described in [16], with the following exception:

- The Instrument Sequencer, Components Controller, and OIWFS Components Controller will share the same IOC.

There will also be the following electronic VME modules:

- The XVME-240 Digital I/O Board:
 - To monitor the status of the stepper-motor control modules;
 - To reset the stepper-motor control modules;
- Four custom-built VME sensor-support boards:
 - To read out the NIRI OIWFS Hall-effect sensors;
- Two XVME-566 Analog Input Modules:
 - To read out the VME sensor-support boards for the NIRI OIWFS Hall-effect sensors;
- Two Oregon Microsystems VME8-8 Stepper-motor controller boards
 - To control the NIRI OIWFS stepper-motors.

Note that these boards will be shared between the NIRI OIWFS hardware and the NIRI hardware, and will perform other functions that are not listed above. In particular, the XVME-240 board is shared between the NIRI CC, the NIRI OIWFS, and is also used to interface to the interlock bus. See [3] for more details.

2-2.3.2 GNIRS / Standalone OIWFS

The system hardware architecture for the GNIRS OIWFS control software is the same as described in [16], with the following exception:

- The OIWFS Components Controller and its supporting hardware will be in the same VME crate as other GNIRS hardware, but will have a separate bus.

There will also be the following electronic VME modules:

- The XVME-240 Digital I/O Board;
 - To monitor the status of the stepper-motor control modules;
 - To reset the stepper-motor control modules;
- One custom-built VME sensor-support board:
 - To read out the GNIRS OIWFS Hall-effect sensors;
- One XVME-566 Analog Input Modules:
 - To read out the VME sensor-support boards for the GNIRS OIWFS Hall-effect sensors;
- One Oregon Microsystems VME8-8 Stepper-motor controller boards
 - To control the GNIRS OIWFS stepper-motors.

These boards will not be shared between the OIWFS hardware and any other system; however, the XVME-240 board will be used to interface to the interlock bus.

2-2.4 Mass/Balance

Not Applicable for a software ICD.

2-2.5 Thermal Interface

Not Applicable for a software ICD.

2-3.0 Software/Control Function Interface

2-3.1 Overview

2-3.1.1 Communications infrastructure

The communications infrastructure is the same as described in [16].

2-3.1.2 Software architecture

The software architecture is described in [2].

2-3.2 Behavior

The behavior of the NIRI/GNIRS OIWFS/A&G interface is the same as described in [20] and [19].

2-3.3 Implementation

This section contains tables summarizing the EPICS records used for the interface. The next sections (Section 2-3.4 on page 17 and Section 2-3.5 on page 21) contain the details of what those records do.

It is assumed in all cases that the record names given in this document will be prefixed by the string “niri:” or “nirs:”¹, as appropriate when the database is implemented, except for the SAD database which will be prefixed by “niri:”, “niri::”, “nirs:”, or “nirs::”².

2–3.3.1 Command (CAD/CAR) records

The columns of the command record tables are as follows:

Description. A brief description of the command.

Attribute. The name of the attribute which the Observatory Control System associates with a particular parameter of a particular command. Component position names are listed separately in Table 7.

CAD Record / Cad Record.FIELD. The name of the EPICS (minus the “niri:” or “nirs:” prefix) together with the field in that CAD record associated with each parameter of the command.

CAD ordering number. The recommended order in which this command should be executed. CAD records should be connected to their associated APPLY record in the order 1, 2, 3, 4, 5, etc... CAD records within the same ordering number set can be connected to the APPLY record in any order in that set.

CAR record. The CAR record associated with the command. Note that several CAD records may share the same CAR record.

2–3.3.1.1 CAD/CAR records—System Commands

These commands affect the overall behavior of NIRI and the NIRI OIWFS; they affect the GNIRS OIWFS, but not the other GNIRS components. Note, in particular, that it is not possible to reboot NIRI and the NIRI OIWFS separately. It may be possible to reboot the GNIRS OIWFS without rebooting GNIRS.

TABLE 1. NIRI/GNIRS system engineering command CAD/CAR records

Description	Attribute	CAD Record / CAD Record.FIELD (“niri:” or “nirs:” +)	CAR record
Reboot.		wfs:reboot	N/A
Select debug mode.	Debug mode	wfs:debug.A	wfs:debugC
Flush data from internal buffers.		wfs:flush	wfs:flushC

1. Or some other Gemini-assigned name, which must be at most four alpha-numeric characters, followed by a colon.
2. Or some other Gemini-assigned name, which must be at most four alpha-numeric characters, followed by one or two colons.

2-3.3.1.2 OIWFS command CAD/CAR records—OIWFS components

The commands in Table 2 are the only ones that are used during routine observation.

TABLE 2. OIWFS command CAD/CAR records

Description	Attribute ^a	CAD Record / CAD Record.FIELD ("niri:" or "nirs:" +)	CAR record ("niri:" or "nirs:" +)
Select a filter in the filter wheel ^b .	filter name	wfs:filtSel.A	wfs:filtC
Park the filter wheel.	(none)	wfs:filtPark	
Move gimbal mirrors, to given (x, y) position and leave it stationary. (x, y) is the position in the focal plane measured in mm in the instrument's frame of reference.	x y	wfs:prbMove.A wfs:prbMove.B	wfs:prbC
Park the gimbal mirror.	(none)	wfs:prbPark	
Select the focus-stage position by name.	position name	wfs:focSel.A	wfs:focC
Set the focus-stage position in mm of translation.	position	wfs:focMove.A	
Park the focus stage.	(none)	wfs:focPark	
The gimbal and focus stage are caused to follow a continuous stream of updated (x=gimbal x, y=gimbal y, z=focus) positions. These positions are not defined as CAD record arguments but in separate records (see ICD 1.6/1.10). ^c	(none)	wfs:folFollow	folC
"Follow" mode, initiated by the wfs:folFollow command, is terminated.	(none)	wfs:folStop	
Set the tolerance values for all three axes (x=gimbal x, y=gimbal y, z=focus) in mm. The tolerance is used to calculate the wfs:folInToler SIR record.	x y z	wfs:folSetTol.A wfs:folSetTol.B wfs:folSetTol.C	folSetC

a. "(none)" means that there are no arguments.

b. ICD 1.6/1.10 [19] has three separate components: a field-stop, a neutral-density filter, and a color filter. For the NIRI/GNIRS OIWFS, these are combined into a single wheel which may have multiple optical elements at each position.

c. The focus is not designed to be operated continuously, and it should not be necessary to do so. By default, the stream of focus offset values is ignored.

2-3.3.1.3 OIWFS Engineering commands CAD/CAR records—OIWFS components

The following commands are used for engineering purposes, and are not expected to be used during normal observing. Note that, although these commands are connected to the OIWFS APPLY record, they are normally activated directly by the engineering user interface.

The following commands affect the overall behavior of the OIWFS but do not affect the science detector.

TABLE 3.

NIRI/GNIRS OIWFS engineering command CAD/CAR records

Description	Attribute	CAD Record / CAD Record.FIELD ("niri:" or "nirs:" +)	CAR record ("niri:" or "nirs:" +)
Datum filter wheel.		wfs:filtDatm	wfs:filtC
Set filter wheel position in position numbers ^a .	position	wfs:filtMove.A	
Set filter wheel position in engineering units (motor microsteps) ^b .	motor position	wfs:filtEngMove.A	
Confirm that filter wheel has not lost its datum position.		wfs:filtRdtm	
Confirm that filter wheel and its Hall-effect sensors are operational		wfs:filtDiag	
Datum the gimbal mirror.		wfs:prbDatm	wfs:prbC
Datum the x-axis of the gimbal mirror.		wfs:prbDatm	wfs:prbxC
Tip gimbal mirror along its x-axis, given x position in engineering units (motor microsteps).	motor position	wfs:prbxEngMove.A	
Confirm that the OIWFS gimbal mirror x-axis motor has not lost its datum position.		wfs:prbxRdtm	
Confirm that the OIWFS gimbal mirror x-axis and its Hall-effect sensors are operational		wfs:prbxDiag	
Datum the y-axis of the gimbal mirror.		wfs:prbDatm	wfs:prbyC
Tip gimbal mirror along its y-axis, given y position in engineering units (motor microsteps).	motor position	wfs:prbyEngMove.A	
Confirm that OIWFS gimbal mirror y-axis motor has not lost its datum position.		wfs:prbyRdtm	
Confirm that OIWFS gimbal mirror y-axis motor and its Hall-effect sensors are operational		wfs:prbyDiag	
Datum focus mechanism.		wfs:focDatm	wfs:focC
Set focus stage translation in engineering units (motor microsteps).	motor position	wfs:focEngMove.A	
Confirm that focus stage has not lost its datum position.		wfs:focRdtm	
Confirm that focus stage and its Hall-effect sensors are operational		wfs:focDiag	

a. Only the standard positions (excluding datum and park) are fully repeatable.

b. Only the standard positions (excluding datum and park) are fully repeatable.

2-3.3.2 Status Information Records

The following tables show the status information provided by the OIWFS. The tables have the following columns:

SIR record. The part of the name of the EPICS SIR record which follows the "niri:" or "nirs:" prefix³.

Units. The units in which the quantity is stored. (Ustep refers to units of motor microsteps; “user units” or “real-world units” refer to the arbitrary units described by the Units SIR variable.)

Comments. A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.

2-3.3.2.1 WFS Status Information

Table 4 contains status information for all of the WFS.

TABLE 4. The WFS Overall Status Information

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS included	Type	Units	Comments
wfs:debugMode	WSDMODE	Never	string		Debugging mode [NOLOG NONE MIN FULL]
wfs:health		Never	string		WFS Health [GOOD WARNING BAD]
wfs:heartBeat	WSHRTB	Never	long	count	Heart beat record: Shows that WFS software is alive by returning a constantly increasing value.
wfs:historyLog	WSHLOG	Never	array of char ^a		History log record ^b
wfs:issPort	WSISSPT	Start	long	number	ISS port on which instrument is installed
wfs:label	WSLABEL	Never	string		The FDSC field describes the WFS Components Controller
wfs:name	WSNAME	Never	string		WFS Components Controller Name
wfs:simMode	WSSMODE	Start	string		Simulation mode [NONE FULL FAST VSM]
wfs:state	WSSTATE	Never	string		Inst. state [BOOTING INITIALIZING RUNNING CONFIGURING]
wfs:wfsBeam	WSWFSBM	Start	long	0/1	Is the WFS detector unobstructed? [YES=0 NO=1] ^c

- Should be “bigstring”
- This record is not useful, because of various EPICS limitations.
- The wfs:wfsBeam record only indicates whether the OIWFS components controller mechanisms are obstructing the beam; it does not indicate that light is actually getting to the detector. Furthermore, it does not account for the obstruction caused by the pick-off mirror. The instrument needs to combine the wfs:wfsBeam with information about any other components in order to provide an instrument-wide wfsBeam record.

- Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI/GNIRS database(s) (for example “niri:” or “nirs:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:” or “nirs:”.

2-3.3.2.2 The Wavefront Sensor Components Status Information

The following table summarizes the information for the various components.

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:filtDatumed	WSFLDTM	Never	long	0/1	Is the filter wheel datumed? [YES=0 NO=1]
wfs:filtEngCyclic	WSFLECY	Never	long	0/1	Ignore limits for the filter wheel? [YES=0 NO=1]
wfs:filtEngMax	WSFLEMX	Never	long	motor microsteps	Maximum position of the filter wheel in engineering units (motor microsteps)
wfs:filtEngMin	WSFLEMN	Never	long	motor microsteps	Minimum position of the filter wheel in engineering units (motor microsteps)
wfs:filtEngName	WSFLENM	Start	string		Engineering name of the filter wheel position
wfs:filtEngPos	WSFLEPS	Start	long	motor microsteps	Position of the filter wheel in engineering units (motor microsteps)
wfs:filtLabel	WSFLLBL	Never	string		The FDSC field describes the filter wheel
wfs:filtMax	WSFLMAX	Never	double	position number	Maximum position of the filter wheel in real-world units (position number)
wfs:filtMin	WSFLMIN	Never	double	position number	Minimum position of the filter wheel in real-world units (position number)
wfs:filtName	WSFLNAM	Start	string		User name of the filter wheel position
wfs:filtParked	WSFLPRK	Never	long	0/1	Is the filter wheel parked? [YES=0 NO=1]
wfs:filtPos	WSFLPOS	Start	double	position number	Position of the filter wheel position in real-world units (position number)
wfs:filtRdtmVal	WSFLRDT	Never	long	motor microsteps	Returned value from the filter wheel redatum command
wfs:filtReject	WSFLREJ	Never	long	0/-1	Combined the filter wheel CAR state [GOOD=0 REJECT=-1]
wfs:filtState	WSFLSTE	Never	string		the filter wheel SNL state
wfs:filtUnits	WSFLUNT	Start	string		Definition of real-world units ("user units") for the filter wheel ("pos num").
wfs:focAgDiff	WSFOCADF	Never	double	mm	The focus stage offset error in real-world units (mm of translation)
wfs:focAgEngDiff	WSFOCAEF	Never	double	motor microsteps	The focus stage offset error in engineering units (motor microsteps)
wfs:focDatumed	WSFOCDTM	Never	long	0/1	Is the focus stage datumed? [YES=0 NO=1]
wfs:focEngCyclic	WSFOCECY	Never	long	0/1	Ignore limits for the focus stage? [YES=0 NO=1]
wfs:focEngMax	WSFOCEMX	Never	long	motor microsteps	Maximum position of the focus stage in engineering units (motor microsteps)
wfs:focEngMin	WSFOCEMN	Never	long	motor microsteps	Minimum position of the focus stage in engineering units (motor microsteps)

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:focEngName	WSFOCENM	Start	string		Engineering name of the focus stage position ^a
wfs:focEngPos	WSFOCEPS	Start	long	motor microsteps	Position of the focus stage in engineering units (motor microsteps)
wfs:focInToler	WSFOCINT	Never	long	0/1	Is the focus stage offset in tolerance ^b ? [YES=0 NO=1]
wfs:focLabel	WSFOCLBL	Never	string		The FDSC field describes the focus stage
wfs:focMax	WSFOCMAX	Never	double	mm	Maximum position of the focus stage in real-world units (mm of translation)
wfs:focMin	WSFOCMIN	Never	double	mm	Minimum position of the focus stage in real-world units (mm of translation)
wfs:focName	WSFOCNAM	Start	string		User name of the focus stage position ^c
wfs:focParked	WSFOCPRK	Never	long	0/1	Is the focus stage parked? [YES=0 NO=1]
wfs:focPos	WSFOCPOS	Always	double	mm	Position of the focus stage position in real-world units (mm of translation)
wfs:focRdtmVal	WSFOCRDT	Never	long	motor microsteps	Returned value from the focus stage redatum command
wfs:focReject	WSFOCREJ	Never	long	0/-1	Combined focus stage CAR state [GOOD=0 REJECT=-1]
wfs:focState	WSFOCSTE	Never	string		the focus stage SNL state
wfs:focTol	WSFOCTOL	Never	double	mm	Tolerance of the focus stage offset in real-world units (mm of translation)
wfs:focUnits	WSFOCUNT	Always	string		Definition of real-world units ("user units") for the focus stage. ("mm").
wfs:folDatumed	WSFLDTM	Never	long	0/1	Are all dynamic mechanisms datumed? [YES=0 NO=1]
wfs:folInToler	WSFOLINT	Never	long	0/1	Are all dynamic mechanisms in tolerance ^d ? [YES=0 NO=1]
wfs:folIsFollow	WSFLFOL	Never	long	0/1	Are all dynamic components in follow mode? [YES=0 NO=1]
wfs:folIsIdle	WSFLIDL	Never	long	0/1	Are all dynamic components idle? [YES=0 NO=1]
wfs:folLabel	WSFLLBL	Never	string		The FDSC field describes the dynamic mechanisms
wfs:folName	WSFLNAM	Start	string		Dynamic components: name of current position
wfs:folParked	WSFLPRK	Never	long	0/1	Are all dynamic mechanisms parked? [YES=0 NO=1]
wfs:folPresent	WSFLPRS	Never	double	seconds	Current Gemini time
wfs:folReject	WSFLREJ	Never	long	0/1	Combined reject state for Follow Command
wfs:prbDatumed	WSPBDTM	Never	long	0/1	Is/Are Gimbal Mirror datumed? [YES=0 NO=1]

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:prbLabel	WSPBLBL	Never	string		The FDSC field describes the gimbal mirror
wfs:prbName	WSPBNAM	Start	string		the gimbal mirror name of current position ^e
wfs:prbParked	WSPBPRK	Never	long	0/1	Is the gimbal mirror parked? [YES=0 NO=1]
wfs:prbReject	WSPBREJ	Never	long	0/1	Combined reject state for the gimbal mirror
wfs:prbSky:xPos	WSPBSXPS	Never	double	sky units	Position of the gimbal mirror x in sky units
wfs:prbSky:yPos	WSPBSYPS	Never	double	sky units	Position of the gimbal mirror y in sky units
wfs:prbSkyLabel	WSPBSLBL	Never	string		The FDSC field describes the gimbal mirror
wfs:prbSkyUnits	WSPBSUNT	Start	string		Definition of "sky units"
wfs:prbxAgDiff	WSPBXADF	Never	double	mm	The gimbal-mirror x-axis offset error in real-world units (mm in the focal plane)
wfs:prbxAgEngDiff	WSPBXAEF	Never	double	motor microsteps	The gimbal-mirror x-axis offset error in engineering units (motor microsteps)
wfs:prbxDatumed	WSPBXDTM	Never	long	0/1	Is the gimbal-mirror x-axis datumed? [YES=0 NO=1]
wfs:prbxEngCyclic	WSPBXECY	Never	long	0/1	Ignore limits for the gimbal-mirror x-axis? [YES=0 NO=1]
wfs:prbxEngMax	WSPBXEMX	Never	long	motor microsteps	Maximum position of the gimbal-mirror x-axis in engineering units (motor microsteps)
wfs:prbxEngMin	WSPBXEMN	Never	long	motor microsteps	Minimum position of the gimbal-mirror x-axis in engineering units (motor microsteps)
wfs:prbxEngName	WSPBXENM	Start	string		Engineering name of the gimbal-mirror x-axis position ^f
wfs:prbxEngPos	WSPBXEPS	Start	long	motor microsteps	Position of the gimbal-mirror x-axis in engineering units (motor microsteps)
wfs:prbxInToler	WSPBXINT	Never	long	0/1	Is the gimbal-mirror x-axis offset in tolerance ^g ? [YES=0 NO=1]
wfs:prbxLabel	WSPBXLBL	Never	string		The FDSC field describes the gimbal-mirror x-axis
wfs:prbxMax	WSPBXMAX	Never	double	mm	Maximum position of the gimbal-mirror x-axis in real-world units (mm in the focal plane)
wfs:prbxMin	WSPBXMIN	Never	double	mm	Minimum position of the gimbal-mirror x-axis in real-world units (mm in the focal plane)
wfs:prbxName	WSPBXNAM	Start	string		User name of the gimbal-mirror x-axis position ^h
wfs:prbxParked	WSPBXPRK	Never	long	0/1	Is the gimbal-mirror x-axis parked? [YES=0 NO=1]

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:prbxPos	WSPBXPOS	Always	double	mm	Position of the gimbal-mirror x-axis position in real-world units (mm in the focal plane)
wfs:prbxRdtmVal	WSPBXRDT	Never	long	motor microsteps	Returned value from the gimbal-mirror x-axis redatum command
wfs:prbxReject	WSPBXREJ	Never	long	0/-1	Combined the gimbal-mirror x-axis CAR state [GOOD=0 REJECT=-1]
wfs:prbxState	WSPBXSTE	Never	string		the gimbal-mirror x-axis SNL state
wfs:prbxTol	WSPBXTOL	Never	double	mm	Tolerance of the gimbal-mirror x-axis offset in real-world units (mm in the focal plane)
wfs:prbxUnits	WSPBXUNT	Always	string		Definition of real-world units ("user units") for the gimbal-mirror x-axis. ("mm").
wfs:prbyAgDiff	WSPBYADF	Never	double	mm	The gimbal-mirror y-axis offset error in real-world units (mm in the focal plane)
wfs:prbyAgEngDiff	WSPBYAEF	Never	double	motor microsteps	The gimbal-mirror y-axis offset error in engineering units (motor microsteps)
wfs:prbyDatumed	WSPBYDTM	Never	long	0/1	Is the gimbal-mirror y-axis datumed? [YES=0 NO=1]
wfs:prbyEngCyclic	WSPBYECY	Never	long	0/1	Ignore limits for the gimbal-mirror y-axis? [YES=0 NO=1]
wfs:prbyEngMax	WSPBYEMX	Never	long	motor microsteps	Maximum position of the gimbal-mirror y-axis in engineering units (motor microsteps)
wfs:prbyEngMin	WSPBYEMN	Never	long	motor microsteps	Minimum position of the gimbal-mirror y-axis in engineering units (motor microsteps)
wfs:prbyEngName	WSPBYENM	Start	string		Engineering name of the gimbal-mirror y-axis position ⁱ
wfs:prbyEngPos	WSPBYEPS	Start	long	motor microsteps	Position of the gimbal-mirror y-axis in engineering units (motor microsteps)
wfs:prbyInToler	WSPBYINT	Never	long	0/1	Is the gimbal-mirror y-axis offset in tolerance? [YES=0 NO=1]
wfs:prbyLabel	WSPBYLBL	Never	string		The FDSC field describes the gimbal-mirror y-axis
wfs:prbyMax	WSPBYMAX	Never	double	mm	Maximum position of the gimbal-mirror y-axis in real-world units (mm in the focal plane)
wfs:prbyMin	WSPBYMIN	Never	double	mm	Minimum position of the gimbal-mirror y-axis in real-world units (mm in the focal plane)
wfs:prbyName	WSPBYNAM	Start	string		User name of the gimbal-mirror y-axis position ^k

TABLE 5. NIRI/GNIRS On-Instrument Wave-Front Sensor Status Records

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Include d	Type	Units	Comments
wfs:prbyParked	WSPBYPRK	Never	long	0/1	Is the gimbal-mirror y-axis parked? [YES=0 NO=1]
wfs:prbyPos	WSPBYPOS	Always	double	mm	Position of the gimbal-mirror y-axis position in real-world units (mm in the focal plane)
wfs:prbyRdtmVal	WSPBYRDT	Never	long	motor microsteps	Returned value from the gimbal-mirror y-axis redatum command
wfs:prbyReject	WSPBYREJ	Never	long	0/-1	Combined the gimbal-mirror y-axis CAR state [GOOD=0 REJECT=-1]
wfs:prbyState	WSPBYSTE	Never	string		the gimbal-mirror y-axis SNL state
wfs:prbyTol	WSPBYTOL	Never	double	mm	Tolerance of the gimbal-mirror y-axis offset in real-world units (mm in the focal plane)
wfs:prbyUnits	WSPBYUNT	Always	string		Definition of real-world units ("user units") for the gimbal-mirror y-axis ("mm").

- a. For the focus stage the position returned by wfs:focEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- b. Valid only when the system is in follow mode.
- c. For the focus stage the position returned by wfs:focName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- d. Valid only when the system is in follow mode.
- e. For the gimbal mirror, the position returned by wfs:prbName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- f. For the gimbal mirror x-axis, the position returned by wfs:prbxEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- g. Valid only when the system is in follow mode.
- h. For the gimbal mirror x-axis, the position returned by wfs:prbxName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- i. For the gimbal mirror y-axis, the position returned by wfs:prbyEngName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.
- j. Valid only when the system is in follow mode.
- k. For the gimbal mirror y-axis, the position returned by wfs:prbyName will often be "INVALID". This should not be treated as an error, merely as an indication that only a few positions (e.g., the PARK position) have names.

2-3.3.2.3 Component Health Information

The following table summarizes the health information records. All records return one of the strings: “GOOD”, “WARNING”, or “BAD”. Each record describes one aspect of the health of a component

TABLE 6. OIWFS Components Controller Component Health Information

SIR record ("niri:" or "nirs:" +)	FITS Keyword	FITS Included	Type	Units	Comments
wfs:fltDatumHealth		Never	string		The health of the filter wheel datum state
wfs:fltHallHealth		Never	string		The health of the filter wheel Hall-effect sensors
wfs:fltHealth		Never	string		The overall health of the filter wheel
wfs:fltModuleHealth		Never	string		The health of the filter wheel driver
wfs:fltPosErrHealth		Never	string		The health of the filter wheel backlash correction
wfs:fltSnIHealth		Never	string		The health of the filter wheel state code
wfs:focDatumHealth		Never	string		The health of the focus stage datum state
wfs:focHallHealth		Never	string		The health of the focus stage Hall-effect sensors
wfs:focHealth		Never	string		The overall health of the focus stage
wfs:focModuleHealth		Never	string		The health of the focus stage driver
wfs:focPosErrHealth		Never	string		The health of the focus stage backlash correction
wfs:focSnIHealth		Never	string		The health of the focus stage state code
wfs:folHealth		Never	string		The overall health of the dynamic mechanisms (the gimbal mirror and focus stage)
wfs:prbHealth		Never	string		The overall health of the gimbal mirror
wfs:prbxDatumHealth		Never	string		The health of the gimbal-mirror x-axis datum state
wfs:prbxHallHealth		Never	string		The health of the gimbal-mirror x-axis Hall-effect sensors
wfs:prbxHealth		Never	string		The overall health of the gimbal mirror x-axis
wfs:prbxModuleHealth		Never	string		The health of the gimbal-mirror x-axis driver
wfs:prbxPosErrHealth		Never	string		The health of the gimbal-mirror x-axis backlash correction
wfs:prbxSnIHealth		Never	string		The health of the gimbal-mirror x-axis state code
wfs:prbyDatumHealth		Never	string		The health of the gimbal-mirror y-axis datum state
wfs:prbyHallHealth		Never	string		The health of the gimbal-mirror y-axis Hall-effect sensors
wfs:prbyHealth		Never	string		The overall health of the gimbal mirror y-axis
wfs:prbyModuleHealth		Never	string		The health of the gimbal-mirror y-axis driver
wfs:prbyPosErrHealth		Never	string		The health of the gimbal-mirror y-axis backlash correction
wfs:prbySnIHealth		Never	string		The health of the gimbal-mirror y-axis state code

2-3.3.3 Position names

Table 7 contains a list of valid position names. Note that these names are case sensitive and must be entered exactly as presented here, including underscores. In addition to the listed

positions, all mechanisms respond to the special names “datum” and “park” which may or may not be distinct from the listed positions.

TABLE 7. Position Names (observing commands)

Component	Position Names	Engineering Position ^a	Comments
Filter Wheel	See [37]		
Gimbal Mirror	Neutral	(comp)	
	Minimum	(comp)	
	Maximum	(comp)	
Gimbal Mirror X-Axis	Neutral	pos03	Mirror centered
	Minimum	pos02	Minimum travel location
	Maximum	pos00	Maximum travel location
Gimbal Mirror Y-Axis	Neutral	pos03	Mirror centered
	Minimum	pos02	Minimum travel location
	Maximum	pos00	Maximum travel location
Focus Stage	Neutral	pos00	Center of travel
	Minimum	pos01	Minimum travel location
	Maximum	pos02	Maximum travel location

a. “(comp)” means that the positions are mapped into component positions, and not sent directly to a mechanism.

2–3.4 Detailed Command Description

A general table of the commands accepted by the NIRI/GNIRS OIWFS, together with their arguments, is given in Table 1 on page 7. This section contains a detailed description of each of the commands that will be sent between the A&G and the NIRI/GNIRS control software. It is based on the required behavior of the commands specified in [21] and [32].

If a directive is accepted it will cause the NIRI/GNIRS OIWFS to change its configuration. The “applyC” CAR record will go “BUSY” while the OIWFS control software is changing its configuration and go to “IDLE” if completed successfully or to “ERR” if the configuration change fails.

2–3.4.1 Sequence Commands

2–3.4.1.1 INIT

On receipt of this command, the OIWFS control software will execute its initialization sequence, in which it will: read its hardware setup files and lookup tables, and reset itself to the start-up state. The “initC” CAR record will go “BUSY” while the initialization is being carried out and go to “IDLE” if completed successfully or to “ERR” if the initialization fails. No mechanisms will be moved.

2–3.4.1.2 DATUM

The OIWFS will locate the datum (i.e. index, home or reference point) for all its mechanisms. The “datumC” CAR record will go “BUSY” while the datum search is being carried out and go to “IDLE” if completed successfully or to “ERR” if the datum searched fails.

2-3.4.1.3 PARK

On receipt of this command, the OIWFS will move its mechanisms to a configuration in which it can safely be switched off. It will utilize all the low-level CAD records dedicated to parking the individual mechanisms. The “parkC” CAR record will go “BUSY” while the mechanisms are being parked and go to “IDLE” if completed successfully or to “ERR” if any mechanism fails to park successfully.

2-3.4.1.4 DIAGNOSE

On receipt of this command, the OIWFS will perform self tests to ensure that its subsystems are healthy. The “diagnoseC” CAR record will go “BUSY” while the tests are being carried out and go to “IDLE” if completed successfully or to “ERR” if the tests fail.

Note: unlike the TEST command, the DIAGNOSE command will move mechanisms, as necessary to diagnose problems.

2-3.4.2 General Engineering Commands

2-3.4.2.1 REBOOT

On receipt of this command, the OIWFS control software will reboot its IOC and go through the same start-up procedure it would normally do when first switched on. The working system will reboot into the simulation mode NONE and debug mode NONE. The simulator will reboot into simulation mode FULL and debug mode NONE. There is no CAR record associated with this command because connection with the IOC will be lost while it reboots.

This command will also reboot the NIRI CC and IS. It will not reboot the GNIRS CC and IS.

2-3.4.2.2 DEBUG

On receipt of this command, the OIWFS control software will change to the given debug mode (NONE, MIN or FULL). This mode determines the amount and frequency of information logged by the NIRI control software, as described in the “Debugging Modes” section of ICD 1a, [24]. The “debugC” CAR record will go briefly “BUSY” and then back to “IDLE”.⁴

2-3.4.3 OIWFS Standard Mechanisms

The OIWFS will have a single filter wheel, a detector focusing stage, and a two-axis gimbal mirror. The filter wheel is considered a “discrete” mechanisms, because it normally is moved only to one of a small number of locations. The focus stage and both axes of the gimbal mirror are considered continuous mechanisms, because they can move to a wide range of locations.

The following commands will act on all of the above mentioned mechanisms; however, as noted below, the use of certain commands with certain mechanisms is discouraged.

2-3.4.3.1 Sel—Select Position by Name

This is the principal interface for all continuous mechanisms. It is provided as a convenience for discrete mechanisms. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

4. Note: This was the required behavior for this system. This requirement was dropped, so the debug command is non-functional.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Translate desired position name into an engineering position with the aid of a look-up table.
 Look up the expected value for the Hall-effect sensors at the desired position location.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism to the engineering position corresponding to the desired position.
 Verify that the expected sensor value matches the measured value.

2-3.4.3.2 Move—Move to specified Position

This is the principal interface for all discrete components. It is available for continuous components, but is of limited use for normal use of the instrument⁵. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the focus mechanism to the engineering position corresponding to the desired focus.
 For some mechanisms, compute the expected values for the Hall-effect sensors.
 For some mechanisms, verify that the final hall effect sensors match the expected values.

2-3.4.3.3 Park—Park Mechanism

This command is used to prepare a mechanism for instrument shutdown. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism to its parking position.
 Verify that the expected sensor value matches the measured value.

The park command is not needed for most OIWFS components, and is provided for uniformity. The single exception is the focus stage. The focus stage should be parked while the instrument being cooled down or warmed up.

2-3.4.3.4 Datm—Datum mechanism

This command is used to find the datum (home) position which defines the mechanism locations. Note that for multi-axis components (e.g., the steering mirrors), there are commands that act on the combined components and on the individual mechanisms.

5. For discrete mechanisms, the units are position numbers; i.e., the first position is at 1.0, the second position is at 2.0.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism and search for its datum.
 Set the hardware datum position.

2-3.4.3.5 MoveEng—Set position in engineering units (motor microsteps)

This command is used to move the components to locations using units that are convenient for working directly with the hardware. Note that for multi-axis components (e.g., the steering mirrors), the command acts only on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism to the specified engineering position.

The Hall-effect sensors will not be used for confirmation.

2-3.4.3.6 Rdtm—Redatum mechanism.

This command provides a quick test that will detect a loss of positioning. It will detect small positioning errors, but can underestimate large positioning errors. Note that for multi-axis components (e.g., the steering mirrors), the command acts only on the individual mechanisms. It returns an estimate of the difference between the actual datum position and the current datum position (which can occur, e.g., if a mechanism is sticking, being driven too quickly, or being driven with too high an acceleration).

On receipt of this command, the OIWFS control software will:

Ensure an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism and search for its datum.

The hardware datum position will be left unchanged. This command returns a value in the RdtmVal record for this mechanism.

This command is not supported by the pupil viewer or the window cover, because those mechanisms use a home-finding algorithm which is incompatible with this command.

2-3.4.3.7 Diag—Diagnose mechanism

This is a quick test that will detect a loss of positioning. It will detect arbitrarily large positioning errors, but will fail to detect small positioning errors. Note that for multi-axis components (e.g., the steering mirrors), command acts only on the individual mechanisms.

On receipt of this command, the OIWFS control software will:

Ensure that an observation is not taking place.
 Check to insure that protective interlocks are not set.
 Wait until the number of active mechanisms drops below the number of mechanisms that the power supply can support.
 Move the mechanism and attempt to verify that the motors and Hall-effect sensors are operating.

If an error is detected, the diagnose command will fail and will return an error code in its CAR record.

2-3.5 Detailed Status information

2-3.5.1 General records

2-3.5.1.1 name

This record contains a string constant describing the name of the instrument. For NIRI this record will contain “NIRI WFS”; for GNIRS the record will contain “NIRS WFS”.

2-3.5.1.2 state

This record will take the values BOOTING, INITIALISING, RUNNING and CONFIGURING and reflect the state of the OIWFS control software while it is initializing. The activities corresponding to these states are

- BOOTING — the state during IOC Init and while any database default values are being set.
- INITIALISING — the OIWFS control software will be checking its own hardware and reading its hardware configuration files during this phase.
- RUNNING — the OIWFS control software is ready to accept commands.
- CONFIGURING — the OIWFS control software is ready to accept commands but it is busy configuring and unable to make observations.

2-3.5.1.3 health

This record will reflect the overall health of the OIWFS. Expected values are GOOD, WARNING or BAD. If the overall health is BAD then the OIWFS will be unusable. If the Health is set to WARNING then the OIWFS will be able to execute some functions but not perhaps to specification.

2-3.5.1.4 historyLog

The record used to contain messages to be recorded in the history log.

2-3.5.2 Component Status Records

2-3.5.2.1 Datumed

This record indicates that a component has been datumed. For multi-axis components, this flag will only be set if all mechanisms have been datumed. This record takes on the following values:

- 0 = Datumed
- 1 = Not Datumed

2-3.5.2.2 EngCyclic

This record indicates that a mechanism may be set to a location beyond its physical limits (convenient for circular mechanisms like filter wheels). Note that this only affects the higher level commands; the low-level software may choose to ignore any invalid requests. This record takes on the following values:

- 0 = Limits can be exceeded
- 1 = Limits may not be exceeded

This record is for internal use by the OIWFS software. Its value should not be changed. It is documented here only for completeness.

2-3.5.2.3 EngMax

This record contains the maximum position that the mechanism can be moved to, in engineering units (motor microsteps).

2-3.5.2.4 EngMin

This record contains the minimum position that the mechanism can be moved to, in engineering units (motor microsteps).

2-3.5.2.5 EngName

This record contains the engineering name of the current mechanism position. This represents a physical configuration of a component, and will never change (unlike the “Name” record which may change, if filters or other optical elements are replaced or rearranged). The names of the positions are:

- park;
- home;
- pos00, pos01, pos02, ... , up to the maximum number of positions;
- INVALID (if not in a standard location).

2-3.5.2.6 EngPos

This record contains the current position of a mechanism, in engineering units (motor microsteps).

2-3.5.2.7 Label

The FDSC field of this record contains a text string which describes a component. This is convenient in providing a title string for edd/dm screens.

2-3.5.2.8 Max

This record contains the maximum position that the mechanism can be moved to, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.9 Min

This record contains the maximum position that the mechanism can be moved to, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.10 Name

This record contains the name of the current component position. This name typically describes a filter or other optical element and will be redefined if filters or other optical elements are replaced or rearranged (unlike the “EngName” record which describes a component’s physical configuration). The current names are listed in [36].

Note that for multi-axis components, there will be a Name record for each individual mechanism, as well as for the component as a whole.

2-3.5.2.11 Parked

This record indicates that a component is in a parked state. For multi-axis components, this record will be set only if all mechanisms are parked. This field takes the values:

- 0 = All mechanisms parked;
- 1 = Not all mechanisms are parked.

2-3.5.2.12 Pos

This record contains the current position of a mechanism, in real-world units. A text string describing the real-world units may be found in the Units record. For mechanisms such as a filter wheel in which there are no appropriate units, the position will typically be given as a position number, i.e., position 0 is 0.0, position 1 is 1.0.

2-3.5.2.13 RdtmVal

This record contains the result of the last redatum command, in engineering units (motor microsteps). (The redatum command returns an estimate of the how far the current mechanism datum position is from the actual datum position; it is useful in diagnosing hardware problems.)

2-3.5.2.14 Reject

This record contains a combination of the values of all CAD records for all mechanisms that make up a component. It contains the following values:

- -1 = At least one CAD record is in a reject state;
- 0 = No CAD records are in a reject state.

This record exists as a convenience for the user interface. There is no reason for the OCS to use it. It is documented here only for completeness.

2-3.5.2.15 State

This record contains a string which describes the current state of the mechanism-support code.

This record is for debugging the OIWFS software. The possible values that this record may take may change, without notice. It is documented here only for completeness.

2-3.5.2.16 Units

This record contains a text string which describes the real world units (e.g., “mm” or “arc sec”)

2-3.5.3 Component Health Records

All of these records return the standard health values: “GOOD”, “WARNING”, and “BAD”.

2-3.5.3.1 DatumHealth

This record reports “WARNING” if a mechanism (or if any mechanism of a multi-axis component) is not datumed. It is cleared by successful execution of a datum command.

2-3.5.3.2 HallHealth

This record reports “WARNING” if a Hall-effect sensor has been disabled. (The sensors can only be disabled from the engineering user interface, and are normally disabled only to deal with a failed sensor.)

This record reports “BAD” if both sensors of a Hall-effect sensor pair have been disabled. Note that a mechanism is still usable if both sensors are disabled, but cannot be datumed. (The sensors can only be disabled from the engineering user interface, and are normally disabled only to deal with a failed sensor.)

2-3.5.3.3 Health

This record reports the most severe error value of all other health records.

2-3.5.3.4 ModuleHealth

This record reports “BAD” if the stepper-motor driver module has entered a fault state. This normally indicates a serious hardware problem. It may be possible to clear the error state by issuing a driver-reset from the engineering user interface. If it is not possible to clear the error, then the stepper-motor driver module or other related hardware has failed, and must be repaired.

2-3.5.3.5 PosErrHealth

This record reports “WARNING” if a mechanism (or if any mechanism of a multi-axis component) has been moved in such a way that proper backlash correction has not been made. In this case, the reported mechanism position may not be the actual mechanism position. This situation can arise if a movement command has been canceled while a motion is in progress; if a movement command failed to complete correctly, if certain low-level engineering commands are executed, and immediately after the control software has been rebooted. It is cleared by successful execution of any motion command.

2-3.5.3.6 SnlHealth

This record reports “BAD” or “WARNING” if the underlying mechanism support code detects an error while a mechanism is in motion. The only situation which is likely to arise is “WARNING”, if an operation takes longer than expected.

2-3.6 Alarm conditions

The following situations will generate an alarm.

- Hall effect sensor disabled
- Motor-control module fault
- Mechanism not datumed
- Mechanism not backlash corrected
- Problem detected by SNL code (time-out)
- Temperature changing
- Temperature out of range

2-3.7 Debugging and Maintenance

This interface can be tested and debugged by means of an engineering user interface supplied as part of the OIWFS.

2-3.7.1 Debugging Modes

The OIWFS control software will have the standard debugging modes described in ICD 1a, [24].

2-3.7.2 Calibration

If a hall sensor fails or if a mechanism must be disassembled for maintenance, the sensors will need to be recalibrated. A separate program is be provided for this purpose. Calibration procedures are be described in a separate document [5].

2-3.8 Simulation

The Gemini simulation modes are described in ICD 1a, [24]. The OIWFS control software supports only the “NONE” mode. The simulator supports only the “FULL” and “FAST” modes.

The “FAST” simulation mode removes most of the simulated delays which are built into the “FULL” simulation mode, but does not fully meet the Gemini specification for the FAST simulation mode. The “VSM” mode is not available.

To switch from operating mode to simulation mode requires an alternate start-up script to be selected, and a reboot (see the release notes [7]). The command to switch between the “FAST” and “FULL” simulations modes is an engineering-only command and does not fall within the scope of this document (see [8]).

2-4.0 Safety Issues

It will be necessary to avoid moving OIWFS components while the mechanical components are not at thermal equilibrium. Records are be provided to provide a temperature interlock. Details are described in [6].

Gemini Near Infra-Red Imager and Wavefront Sensor



Near IR Imager (NIRI) and OIWFS — OIWFS Component Position Names

Hubert Yamada

niri_h ty_014 / 01 / 1999-11-23

This document describes the position names for the OIWFS mechanisms (i.e., filter names).

1 Introduction

1.1 Purpose

This document describes the position names for the OIWFS mechanisms. In general, information which logically belongs within an ICD but which is subject to change with relatively trivial hardware changes are presented here. In particular, the names of the various filters contained within the different OIWFS filter wheels are presented here.

The intended audience of this document is:

- The NIRI reviewers;
- The developers of the Gemini Observatory Control System (OCS);
- The maintainers of NIRI after it is delivered to Gemini;
- Any groups that will be using the NIRI OIWFS.

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Tables

2.1 Filter Wheel

Table 1 describes the named positions for the filter wheel.

TABLE 1.

Named Positions for Filter Wheel

Engineering Position Name	Position Name
pos00	open0
pos01	open1
pos02	J/53.8
pos03	J/537
pos04	H/26.9
pos05	H/214.8
pos06	K/107.4
pos07	K/1074
pos08	K'/107.4
pos09	K'/214.8
pos10	open/214.8
pos11	open11

ICD 1.9/1.10[ab]

“NIRI/GNIRS Science Instrument to On Instrument Wavefront Sensor”

niri_h ty_005

Hubert Yamada
2001-06-27

Version: 1.0

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Chapter 1 Introduction

1–1.0 Description

This document describes the NIRI and GNIRS instrument-to-wavefront-sensor-components interlock system (not to be confused with the Gemini Interlock Bus). These interlocks provide a way for the instrument to inform the wavefront sensor that it should not move its components at this time.

1–1.1 Purpose

This Interface Control Document (ICD) serves the following purposes:

- To describe the commands and parameters that will pass between the Near Infra-Red Imager (NIRI) and its on instrument wavefront sensor (OIWFS) and between the Gemini Near Infra-Red Spectrograph (GNIRS) and its OIWFS.

The intended audience for this document is:

- the NIRI reviewers;
- the GNIRS reviewers and developers.

1–1.2 Scope

This document describes commands which pass between NIRI/GNIRS and their respective OIWFS's.

This document does *not* describe the following topics:

- the NIRI detector and components, (see [18], [11], and [12], summarized in [3]);
- the GNIRS hardware and electronics, (see [18] and (???))
- the OIWFS hardware and electronics, (see [11] and [12], summarized in [3]);
- the NIRI/GNIRS OIWFS detector and components (see [11], [12], and [13], summarized in [3]);

The software design is not described here. For that, see [2] (currently under preparation), and [3] (currently under preparation).

1–1.3 Differences between NIRI and GNIRS

For GNIRS, the commands here affect only the OIWFS. For NIRI, the commands affect the entire instrument.

1–2.0 Related Documents and Drawings

1–2.1 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1–2.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

Chapter 2 Details of the NIRI/GNIRS to OIWFS Interface

2–1.0 Introduction

2–2.0 Physical System Interfaces

2–2.1 Mechanical Interface

Not Applicable for a software ICD.

2–2.2 Optical Interface

Not Applicable for a software ICD.

2–2.3 Electronic Interface

Not Applicable for this ICD.

2–2.4 Mass/Balance

Not Applicable for a software ICD.

2–2.5 Thermal Interface

Not Applicable for a software ICD.

2–3.0 Software/Control Function Interface (wavefront sensor)

2–3.1 Overview

2–3.1.1 Communications infrastructure

The communications infrastructure is the same as described in [16].

2–3.2 Implementation

This section contains tables summarizing the EPICS records used for the interface. The next sections (Section 2–3.2.1.1 on page 6) contain the details of what those records do.

It is assumed in all cases that the record names given in this document will be prefixed by the string “niri:” or “nirs:” when the database is implemented, except for the SAD database which might be prefixed by “niri::” or “nirs::”.

2–3.2.1 Command (CAD/CAR) records

The columns of the command record tables are as follows:

Description. A brief description of the command.

Attribute. The name of the attribute which the Observatory Control System associates with a particular parameter of a particular command.

CAD record/field. The name of the EPICS CAD record (minus the “niri:” or “nirs:” prefix) together with the field in that CAD record associated with each parameter of the command.

CAR record. The CAR record associated with the command. Note that several CAD records may share the same CAR record.

These commands are executed by asserting the top level WFS apply record.

TABLE 1. NIRI/GNIRS OIWFS CAD/CAR records

Description	Attribute	CAD record:field (“niri:wfs:” +)	CAR record (“niri:wfs:” +)
Configuration interlock	Lock for configuration (string, “LOCK” “UNLOCK”)	lockCfgSel.A	lockC
Observation interlock	Lock for observation (string, “LOCK” “UNLOCK”)	lockObsSel.A	
General interlock	Lock (string, “LOCK” “UNLOCK”)	lockGenSel.A	
Temperature interlock	Lock for cool down / warm up (string, “WARM” “COLD” “CHANGING”)	lockTmpSel.A	
Datum mechanism	Datum mechanisms	datum	datumC

2–3.2.1.1 Detailed Command Description

2–3.2.1.2 lockCfgSel

An instrument may have limitations on which mechanisms which may be moving simultaneously. This interlock is intended to prevent the wavefront sensor components from moving while the instrument is being reconfigured, if it is possible for wavefront sensor components to be damaged. There are two possible states:

“LOCK”. Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

“UNLOCK”. Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2–6.0, “Activity Completion,” p. 11, below.)

This command should be used with caution. Excessive use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding sub-system).

This command is not needed by NIRI.

2-3.2.1.3 lockObsSel

This command is provided to provide a low noise environment for sensitive observations. There are two possible states.

“LOCK”. Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

“UNLOCK”. Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2-6.0, “Activity Completion,” p. 11, below.)

This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).

This command is not needed by NIRI.

2-3.2.1.4 lockGenSel

This command is provided as a “general” command to lock all hardware for any unspecified reason.

There are two possible states:

“LOCK”. Hardware is prevented from moving. Any new commands which might cause the hardware to move will fail.

“UNLOCK”. Hardware is permitted to move (if no other locks are set).

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2-6.0, “Activity Completion,” p. 11, below.)

This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).

This command is not needed by NIRI.

2-3.2.1.5 lockTmpSel

The OIWFS motors (and, possibly, other components) can be damaged if a move command is issued before the system has reached a uniform temperature throughout. Furthermore, the physical parameters of the system (e.g., the datum position) may depend on the temperature. The lockTmpSel command informs the OIWFS about the current temperature of the system. There are three possible states:

“WARM”. Hardware is permitted to move (if no other locks are set). The system is at or near room temperature.

“COLD”. Hardware is permitted to move (if no other locks are set). The system is at or near its operating temperature.

“CHANGING”. Hardware is not permitted to move. The system is currently being cooled down or warmed up.

When the temperature setting is changed, the OIWFS will enter the not-datumed state. (The datum shifts slightly, depending on the temperature of the system.) Upon reboot, the system always begins in a “CHANGING” state, until explicitly set to another state.

The lock may be set at any time; however, it does not interrupt any currently executing commands. (See section 2–6.0, “Activity Completion,” p. 11, below.)

It is always better to be conservative and leave the state of the system “CHANGING”, until it can be guaranteed that the OIWFS mechanisms have reached their steady-state temperature.

This command also sets the value of various internal temperature-dependent parameters within the OIWFS. If this parameter is set incorrectly, mechanisms will move, but may incorrectly report that there is a hardware problem.

This command should be used with caution. Use of this lock will interfere with the proper functioning of the OIWFS (and NIRI) and will keep the OIWFS (and NIRI) from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).

2–3.2.1.6 DATUM

This command will cause all mechanisms to be datumed. Note that the OIWFS does *not* automatically datum itself when a move command is issued.

This command should be used with caution. Use of this lock may interfere with the proper functioning of the OIWFS and will keep the OIWFS from obeying commands that are issued by other systems (e.g., the Acquisition and Guiding subsystem).

2–3.2.2 Status Information Records

The following tables show the status information provided by the NIRI/GNIRS OIWFS components controller which are of importance to the instrument as a whole.

Unless otherwise noted, the SIR records described here are to be used for status information, and not to be used for real-time control purposes. It is possible that rapid changes in the underlying system may not be reflected in the SIR records.

The tables have the following columns:

SIR record. The part of the name of the EPICS SIR record which follows the “niri:” prefix¹.

Units. The units in which the quantity is stored

Comments. A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

1. Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI database (for example “niri:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:”.

Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.

TABLE 2. NIRI/GNIRS OIWFS status Information

SIR record ("niri:wfs:" +)	FITS keyword	FITS includ ed?	Type	Units	Comments
lockHealth		no	string		Health of subsystem ("GOOD" "WARNING" "BAD"). Intended to be combined with overall instrument Health. This can also be set if there is a hardware or configuration problem with the database. Diagnosis requires the use of the Engineering user interface, and is not considered a function of the Instrument to OIWFS interface.
lockCfg		no	long		Locked for configuration (non-zero = locked). If this is set, the system will be put into a "WARNING" state.
lockObs		no	long		Locked for observation (non-zero = locked). If this is set, the system will be put into a "WARNING" state.
lockGen ^a		no	long		Locked (generic) (non-zero = locked). If this is set, the system will be put into a "WARNING" state.
lockTmp		no	long		Locked for cool down / warm up (non-zero = locked. I.e., lockTmpVal is "CHANGING". If this is set, the system will be put into a "WARNING" state.
lockInit		no	long		Locked for initialization (non-zero = locked)? If this is set, the system will be put into a "WARNING" state.
lockLock		no	long		Combined value (<i>logical or</i>) of lockCfg, lockObs, lockGen, lockTmp, and lockInit.
lockTmpVal		no	string		Current temperature state ("COLD" "WARM" "CHANGING ^b ")
lockMaxMove		no	long		How many motors may be moved at the same time. ^c
lockActive		no	long		How many motors have been commanded to move and are waiting to move or are currently moving ^c .
lockMoving		no	long		How many motors are currently moving ^c .
wfsBeam		no	long		Flag to indicate that the WFS components are allowing light to fall on the on-instrument wavefront sensor. ^d

- The instrument should not trigger any special actions based on the state of the lockGen record, because it is intended that the lockGen should be usable at any time, for any reason, without side effects.
- For safety, when NIRI or the standalone OIWFS is rebooted, the system always restarts in the "CHANGING".
- For NIRI, this number is the total of the NIRI CC motors and NIRI OIWFS motors, *excluding the gimbal (prbx, prby) motors, which may be moved at any time*. For GNIRS, and any other non-NIRI systems, this number only represents the OIWFS motors. If the system attempts to move a motor when the maximum number of motors is already in motion, then the motion of that motor will silently be delayed until the requested action may be carried out; no error or warning will be issued.
- It is the responsibility of the instrument to combine the value of the wfs:wfsBeam with information from the components controller to produce the value of the global wfsBeam SIR record. This field indicates only that there is no obstruction, not that light is actually getting to the detector. Furthermore, it does not consider the obscuration caused by the pick-off mirror.

2-3.2.3 Detailed SIR record description

2-3.2.3.1 The lockMaxMove, lockActive, and lockMoving records

The lockMaxMove, lockActive, and lockMoving records provide a mechanism to restrict the number of mechanisms which may move simultaneously (necessary, because of power supply restrictions). The lockMaxMove record contains the maximum number of mechanisms which may be moved simultaneously. The lockMoving record contains the number of mechanisms which are currently moving. The lockActive record contains the number of mechanisms which are moving or which are waiting to move.

If an attempt is made to move more mechanisms than are permitted, then the system will silently postpone the motion, so that no more than the maximum number of mechanisms are in motion.

This restriction does not apply to the gimbal (prbx, prby) mechanisms, which are permitted to move at any time.

2-4.0 Software/Control Function Interface (Components Controller)

2-4.1 Overview

2-4.1.1 Communications Infrastructure

The communications infrastructure is the same as described in [16].

2-4.2 Implementation

The instrument may control the temperature interlock system by writing to the lock records which are described above. Alternatively, it may provide a status and information record² which will be used to automatically update the lock records.

2-4.2.1 Monitoring an external record.

The temperature interlock may be set by using a record in another IOC. This feature must be enabled as a configuration option (see [7], section 5.4, “Configuration,” p. 7 and section 6.4, “Configuration,” p. 10). *All three of the variables defined below must be set as described, or all three must be left unset.*

Set the configuration variable *NIRI_TMPVAR* to the name of a field of an EPICS record which contains a numeric value. If the field contains a 0, the value of the temperature interlock will be “CHANGING.”; a 1, the value is “COLD”; and a 2, the value is “WARM”.

The configuration variable *NIRI_TMPSCAN* should be set to the rate at which the field defined by *NIRI_TMPVAR* should be monitored. This should be a valid capfast scan rate, e.g., “1 second”. There is no reason to scan at a rate faster than “1 second.”

The configuration variable *NIRI_TMPHB* should be set to the name of a field which is changing at least as frequently as *NIRI_TMPSCAN*, and which resides within the same IOC as *NIRI_TMPVAR*. This is used to detect communications problems and IOC problems, and

2. Gemini requires a SIR record to be the interface between two systems. The software will work correctly with any record that can return a numeric value.

will automatically force the system into a “CHANGING” (locked) state if there is a problem.

2–4.2.2 Status Information Records

The following tables show the status information provided by the NIRI/GNIRS OIWFS components controller which are of importance to the instrument as a whole. The tables have the following columns:

SIR record. The part of the name of the EPICS SIR record which follows the “niri:” prefix³.

Units. The units in which the quantity is stored

Comments. A description of the item. The first paragraph of this will go into the SIR record description and also the comment field in the FITS header.

Note that only the type, units and comments are actually contained in the SIR records. The FITS keyword is provided by the DHS.

TABLE 3. NIRI/GNIRS OIWFS status Information

SIR record (“niri:wfs:” +)	FITS keyword	FITS includ ed?	Type	Units	Comments
lockTmpVal		no	string		Current temperature state (“COLD” “WARM” “CHANG- ING”)
lockTmpHB		no	long		A non-zero value indicates a communications error or a problem with the IOC containing $\{NIRI_TMPVAR\}$ and $\{NIRI_TMPHB\}$

2–5.0 Safety Issues

The instrument software must always call lockTmpSel when the system is expected to be at a non-equilibrium temperature. If the alternative interface is used (see section 2–4.2, “Implementation,” p. 10), the sir records which are being monitored must hold the appropriate temperature state. *Various components may not operate correctly (and could be damaged) if they are moved when there is a significant temperature gradient across the component (more than 50 degrees C).*

2–6.0 Activity Completion

The interlocks only prevent new actions from starting. They do not interrupt currently executing commands.

3. Note that the actual system has been designed so the Status Alarm Database can easily be given a different prefix from the rest of the NIRI database (for example “niri:”). It is assumed in this document that the Status Alarm Database prefix will be “niri:”.

In order to be sure that the filter wheel and the focusing stage have stopped, it is necessary to monitor their CAR records. These records are standard CAR records; when the mechanism is idle, these records will have the value “IDLE”.

The Gimbal Mirror is unusual; if it is in “follow” mode, it is possible for the mechanism to be in motion, even if the CAR record shows “IDLE”. For this reason, the CAR record should not be used to determine mechanism activity. Instead, use the IsIdle SIR record. When this SIR record has a zero value, the gimbals are guaranteed to be inactive.

TABLE 4. NIRI/GNIRS OIWFS Activity Records

Record ("niri:wfs:" +)	Record Type	Type	Units	Comments
IsIdle	SIR	long 0=TRUE, 1=FALSE		Both axes of gimbal mirror are inactive.
filtC	CAR	Standard CAR values		State of the filter wheel.
focC	CAR	Standard CAR values		State of the focusing stage.

Gemini Near Infra-Red Imager



Near IR Imager (NIRI)—The Engineering User Interface

Hubert Yamada

niri_h ty_007 / 05 / 1999-11-11

This document describes the operation of the engineering user interface for NIRI.

1 Introduction

The Engineering User Interface will be delivered with NIRI to Gemini. Its purposes are:

- to provide a user interface for testing and debugging NIRI;
- to provide a user interface before the standard NIRI user interface is available;
- to provide sample code which the designers of the NIRI standard user interface may inspect and reuse;
- to provide a user interface for the NIRI acceptance test procedure;
- to provide an interface to NIRI functions which the designers of the NIRI standard user interface may choose not to make available to the general users.

It is extremely important to read the warnings in Section 2.1, “Warnings (Very Important!),” on page 2!

1.1 Purpose

This document describes the operation of the NIRI engineering user interface.

The intended audience of this document is:

- the NIRI reviewers;
- the developers of the Gemini Observatory Control System (OCS) and the NIRI standard user interface;
- the developers of the GNIRS OIWFS components controller and other groups which will be using the design of the NIRI OIWFS
- the scientists, engineers, and technicians who will be designing and testing NIRI;

- the committee which will be reviewing the NIRI acceptance test procedure and who will be verifying that NIRI passes its acceptance tests;
- the early users of NIRI.

1.2 Definitions, Acronyms and Abbreviations

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

1.3 Stylistic Conventions

References to documents in Section 1.5, “References,” on page 2 of “Near IR Imager (NIRI)—Definitions, Abbreviations, and References” are given like this; [1].

1.4 References

See “Near IR Imager (NIRI)—Definitions, Abbreviations, and References”.

2 Warnings and General Information

2.1 Warnings (Very Important!)

There are a number of screens which contain functions which may damage NIRI or which may leave it in a state from which no software recovery is possible. These screens are easily distinguished from the standard NIRI screens, because they have a bright yellow title bar on a black background. The functions on these screens are intended for use by the qualified personnel only, and are not intended for general use. They include several functions which are capable of bypassing various safety features.

2.2 A Few Notes About Edd/Dm Interfaces

3 General User Commands

3.1 Starting the User Interface

3.2 The Introduction Screen

3.3 The Components Controller Main Screen

3.4 The On-Instrument Wavefront Sensor Main Screen

4 Restarting After the IOC Has Been Rebooted

4.1 Wait for initialization to complete

4.2 Restart screens

4.3 Look for RED

4.3.1 Set the Temperature Interlock

4.3.2 Start the Closed Cycle Coolers (if needed)

4.4 Look for YELLOW

4.4.1 Datum Mechanisms

5 Advanced User commands

The commands which are described in this section can damage the instrument or leave it in an unusable state. Use with extreme caution. *It is extremely important to read the warnings in Section 2.1, “Warnings (Very Important!),” on page 2!*

Notice: The screens which are described in this section do not meet the Gemini interface standards. They do not interact with the EPICS database using the mandatory CAD/CAR/APPLY/SIR interface. This is explicitly permitted in [33].

5.1 Starting the Low-Level Engineering Interface



5.2 The Main Engineering Screens

FIGURE 1. The Top Level CC Screen

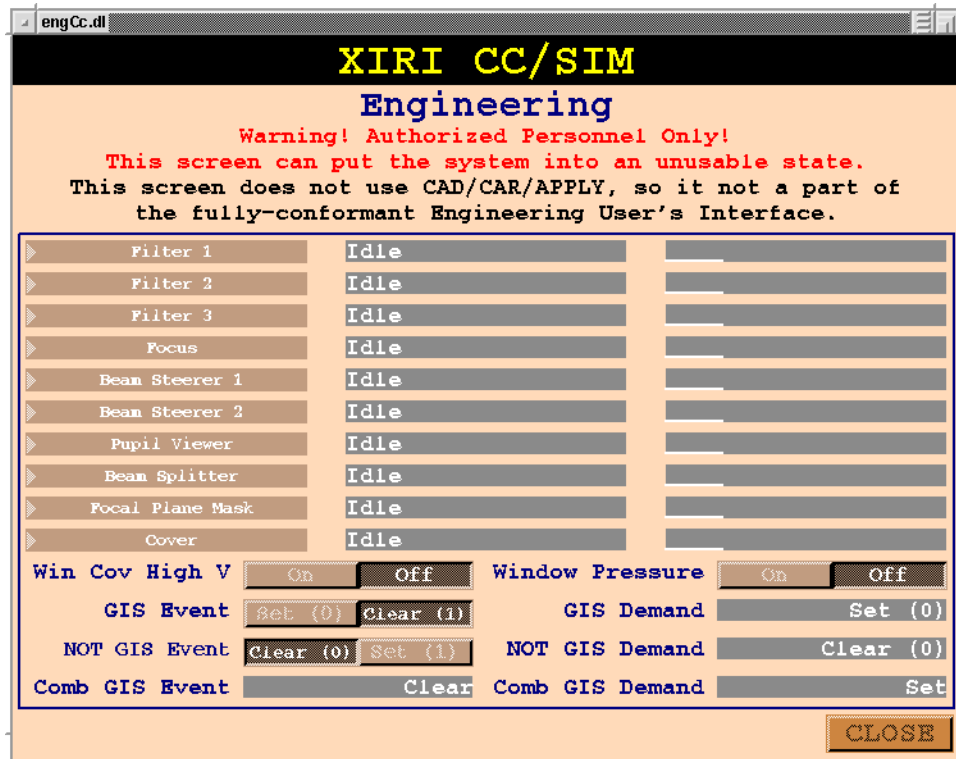
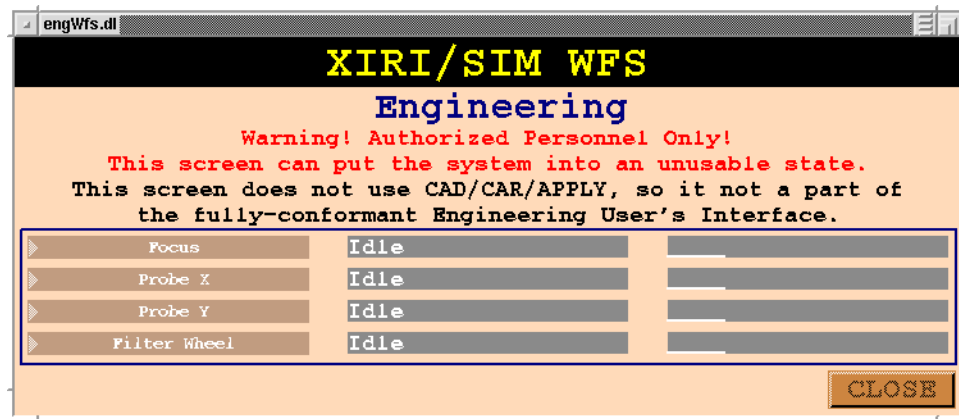


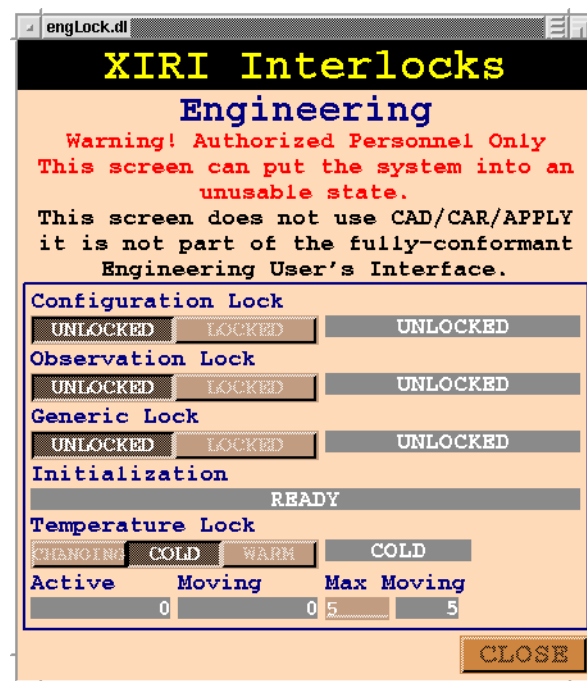
FIGURE 2. The Top Level OIWFS Screen



5.3 The Global Parameters Screen



5.4 The Engineering Interlocks Screen



5.5 The Low Level Mechanism Control Screen

engMech.dl

XIRI CC/SIM Filter 1

Engineering

Warning! Authorized Personnel Only!
 This screen can put the system into an unusable state.
 This screen does not use CAD/CAR/APPLY so it is not a
 part of the fully-conformant Engineering User's
 Interface.

Status Idle Error/Info 0			Sensors 1P (V) 2P (V) 3.997 0.000  -5.000 5.000 -5.000 5.000 Disable Disable Enable Enable		
Motor User Raw Stop New Pos. Cur Pos Pos Err 0 0 1 Low High Sens Tol 0 288000 0.300 Speed Acc T(s) Datumed 3000.0 1.0 0 Datum Redatum Datum/Diag Cycle End Cycle Diagnose Set ctr SW Count HW Count Use Set 0.0 0.0			1B (V) 2B (V) 3.998 0.000  -5.000 5.000 -5.000 5.000 Disable Disable Enable Enable		
Select home home 0.000 Ex Md Ps Ex Sen P Ex Sen B 0 0.000 0.000 Mod Pos Rd Sen P Rd Sen B 0 3.997 3.998 Update			Stripchart Related Screens External Tools CLOSE		

5.6 The Parameter Screen

engParams.dll

XIRI CC/SIM Filter 1

Engineering Parameters

Warning! Authorized Personnel Only!
This screen can put the system into an unusable state.

This screen does not use CAD/CAR/APPLY, so it is not a part of the fully-conformant Engineering User's Interface.

Cycle Parameters

	Toler	Set Exp Value				
	0.300	NO	YES			
Pos.	Exp1	Val1	Exp2	Val2		
A	13400	-0.779	0.000	0.000	0.000	P
		-0.779	0.000	0.000	0.000	B
B	-10600	-3.119	0.000	0.000	0.000	P
		-2.339	0.000	0.000	0.000	B
C	0	3.997	0.000	0.000	0.000	P
		3.998	0.000	0.000	0.000	B

Motor Control

Auto Off	Mot Pwr	Module	Reset Module
No	Yes	Auto On	Off
		Good	Reset

Motion / Homing Parameters

Backlash	DSLP	DSLB	THRP	WTHP
1000	0	0	0.000	400
Timeout	DOFP	DOFB	THRB	WTHB
120	0	0	2.000	400
DOF1	LOPR	HOPR	ZERO	
-30	0	288000	0	

Reload LUTs:

5.7 The Temperature Control Screen

engTmp.dll

XIRI/SIM Temperature control Engineering

Warning! Authorized Personnel Only!
 This screen can put the system into an unusable state.
 Center the science focusing stage before cooling or warming.
 This screen does not use CAD/CAR/APPLY, so it not a part of
 the fully-conformant Engineering User's Interface.

Temperature Sensors		Temperature Controllers			
		OIWFS	Img Buf Mass	Cold Pl Center	
Serial I/O		Serial I/O			
Top Strap	300.00	Temperature	300.00	300.00	300.00
Bottom Strap	300.00	Heater %	0.00	0.00	0.00
Cryopump	300.00	Set Point	0.00	0.00	0.00
Cold Pl Edge	300.00	Gain	0.00	0.00	0.00
CC Cabinet	300.00	Rate	0.00	0.00	0.00
WFS Cabinet	300.00	Reset	0.00	0.00	0.00
7	300.00	Range	Off	Off	Off
8	300.00	Tune	Manual	Manual	Manual
Units	K	Units	K	K	K

Closed Cycle Coolers					
Cooling	Off	Low	High	State	Off
Low	30000			Pos	0
High	55000			Vel	1000
Enable 1	Auto	On	Off	Jog	0
Enable 2	Auto	On	Off	Dmov	1
				Abort	
Timer	Cancel			Heater Mode	Normal

CLOSE

